DR64 Primary HSI B2

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1 Introduction

This document describes a technical detail of the Primary Board B2. This document describes both production intent hardware and Debug hardware, for the production vs Debug changes please refer Chapter 9.

2 B2 Board Connectivity

HW Architecture block diagram

Pri_SV62 + TDA4+A-PHY Master System Arch Diagram B2 V1.0 250624.pdf

3 MCU Block Diagram

Detailed view of MCU connectivity

Pri_SV62 + TDA4+A-PHY Master System Arch Diagram B2 V1.0 250624.pdf

4 S32G3

4.1 S32G3 Power tree diagram

Attached the power tree of Primary B2. (needs to be updated to B2)

Power tree_B1

4.2 S32G3 has two main power supplies:

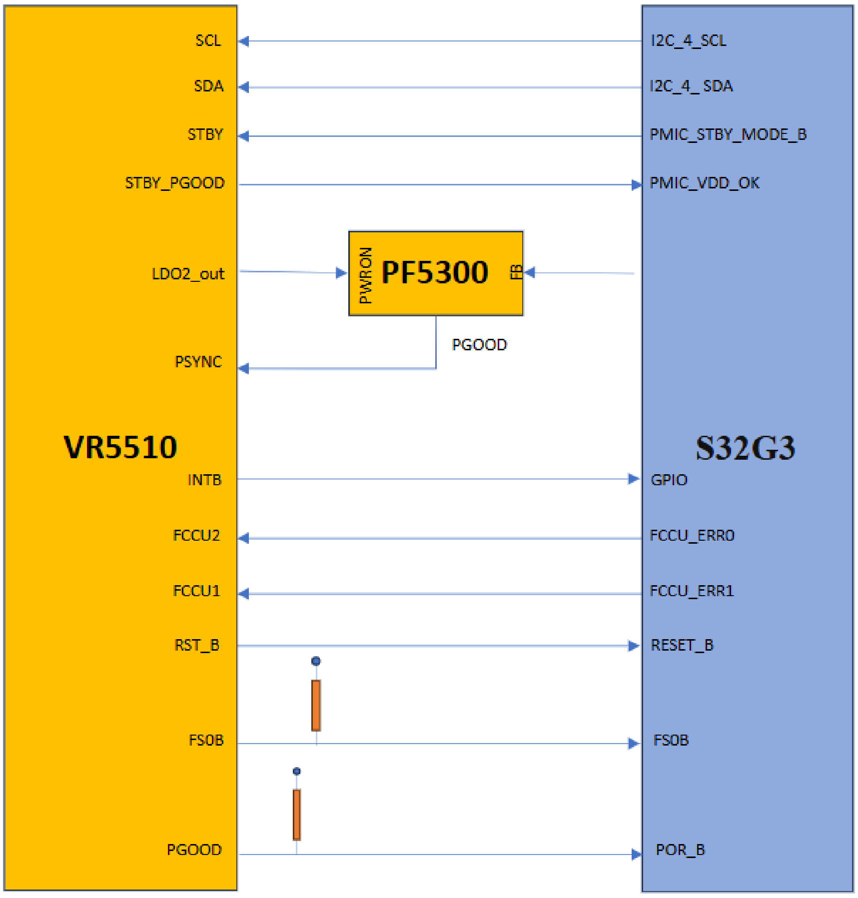
Power supplies

- PMIC VR5510 with several outputs, used for I/Os & peripherals.
- Buck regulator PF5300 used for 0.8V MCU core voltage

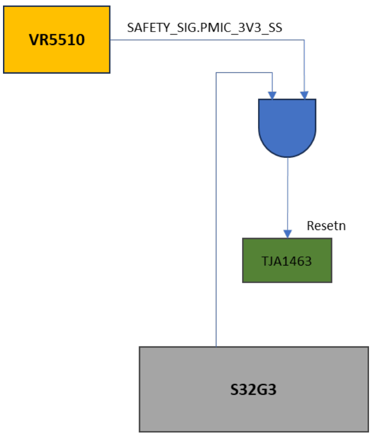
4.3 S32G3 & Peripherals Power Rails

Device	QTY	MCU_3V3	MCU_0V8_STBY	MCU_1V8	MCU_1V1	MCU_0V8_VDD_CORE	MCU_1V8_ANA	5V_CAN
MCU	x 1	3.3V	0.8V	1.8V	1.1V	0.8V	1.8V	
LPDDR4	x 1			1.8V	1.1V			
NOR Flash	x 1			1.8V				
eMMC	x 1	3.3V		1.8V				
CAN Trans.	x 7	3.3V						5V
Misc.	x 10	3.3V						

4.4 S32G3 PMIC Connectivity - HW Safety



4.5 PMIC FS0B Connectivity



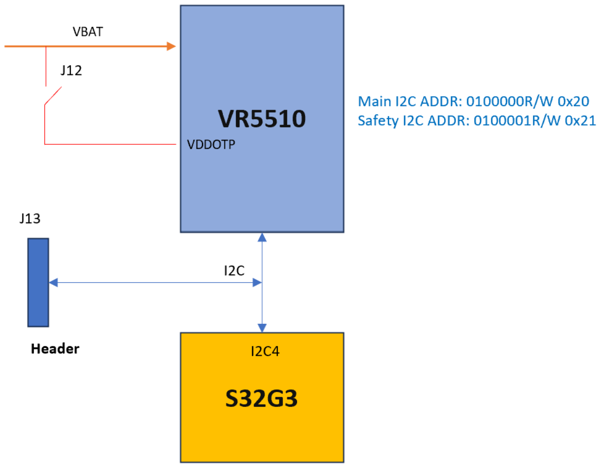
4.6 PMIC

4.6.1 Configuration

MCU PMIC is a Pre-programmed device, P.N: MVR5510AMDALES. Below is the pre-configured data

R-MVR5510AMDALES.pdf

4.6.2 Debug Mode



The VR5510 enters in Debug mode with the following sequence:

- Apply VDDOTP pin = VDDOTP.
- Apply VSUP1/2 > V_{SUP_UVH} and PWRON1 > $PWRON1_{VIH}$ or PWRON2 > $PWRON2_{VIH}$.
- The device now starts in Debug mode, ready for debugging or OTP programming.
- Apply VDDOTP = 0 V to turn on the device with the modified configuration.

4.7 S32G Hardware Configuration (BOOTMOD) pins

FUSE_SEL	BMODE1	BMODE2	Boot mode	Available Boot interface	Use case
0	0	0	Serial boot + XOSC in Differential mode	Serial Communication interfaces: UART and CAN	Production configuration for a unprogrammed chip
0	0	1	Serial boot + XOSC in Crystal mode or Bypass mode	Serial Communication interfaces: UART and CAN	Production configuration for a unprogrammed chip
0	1	0	Boot from RCON	Memory Interfaces: QSPI, SD card, MMC, eMMC	Development configuration
1	0	X	Boot from fuses	Memory Interfaces: QSPI, SD card, MMC, eMMC	Production configuration for a programmed chip
1	1	0	Serial boot	Serial Communication interfaces: UART and CAN	Debug of a programmed chip
X	1	1	Reserved		Reserved

4.7.1 BOOTMOD Pins – SW1 Debug

Peripheral	Signal	Pin/Port#	Net Name
BOOT	mode, 0	[W10] PA_02	BOOTMOD0
BOOT	mode, 1	[W11] PA_03	BOOTMOD1

NOTE: If the switch is closed the value of the BOOTMODE will be 1.

4.7.2 MCU Development Config.

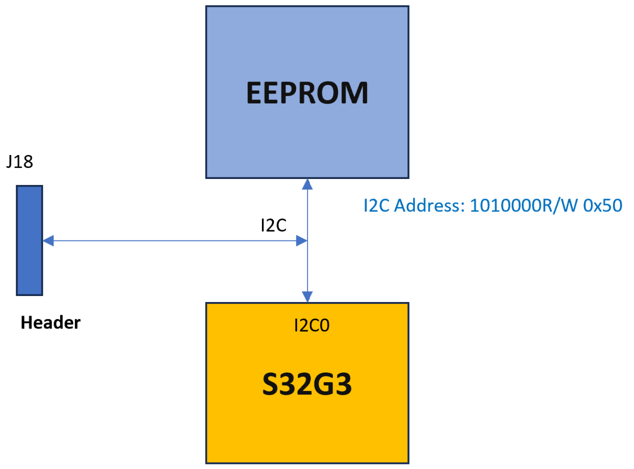
4.7.2.1 BOOT STRAP

BOOT_CFG2 [4]	PA_02	PA_03	Mode		RCON8/PB_01	
FUSE_SEL	BOOTMOD1	BOOTMOD2			eePROM/Serial	Pins/Parallel
0	1	0	Boot from RCON	→	1	0

4.8 Serial BOOT

Peripheral	Signal	Pin/Port#	Net Name
I2C_0	sda, 0	[W12] PB_00	MCU_IF.3v3_I2C0_SDA
I2C_0	scl, 0	[E7] PB_01	MCU_IF.3v3_I2C0_SCL

4.8.1 I2C eePROM



4.8.2 Description

- eePROM P.N: AT24C01C
- I2C tool like Aardvark shall be used to download the eePROM.
- J18 is connected to I2C0

SM-39428 - The first 4 bytes must be set:

- NOR Flash - 0x0000
- eMMC – 0x0060
- This eePROM can be downloaded by JTAG too.
- Board type details can be accessed from the eePROM
- Board Revision details can be accessed from the eePROM

[Draft **]**

4.9 MI information

Reference: MCU MI Software Requirements SV62, MI information

SM-39431 - MCU MI data shall be written to NOR flash with an allocated size of at least 1024 Bytes. NOR Flash 0x0FFE0000 - 0x0FFFFFFF (128 KB) address range is allocated for MI block **[** Draft **]**

SM-39423 - Board Name
The board name is a text describing the board. String is zero terminated.

- Parameter name: board_name
- Type: str
- size in bytes: 64
- Min value: 1 char
- Max value: 63 chars

[Draft **]**

SM-39426 - Board_ID
This board_id value should be the dec value of the VW_ECU_HW_version 3 first bytes value.

- Parameter name: board_id
- Type: u32, dec value
- size in bytes: 4
- Min value: 0
- Max value: 32 bits

[Draft **]**

SM-39412 - Board revision
The MCU MI shall contain the ECU board revision. which describes the board **revision** for the board type indicated in the board_id field (e.g. B sample rev1, B sample rev2).

- Parameter name: board_rev
- Type: u32, dec value
- size in bytes: 4
- Min value: 0
- Max value: 32 bits

[Draft **]**

SM-44467 - Below are MI parameters

parameter	Type	Size in bytes	Min value	Max value	DR64_B2_primary	DR64 B2_secondary
starting address					0x0FFE0000	0x0FFE0000
Ending address					0x0FFFFFFF	0x0FFFFFFF

parameter	Type	Size in bytes	Min value	Max value	DR64_B2_primary	DR64 B2_secondary
magic_id	u32	4	0x0C051D19	0x0C051D19	0x0C051D19	0x0C051D19
ver	u32	4	0	0	1	1
size	u32	4	313	1024	801	801
reserved_header	str	32				
board_name	str	64	1 char	63 chars	DR64-PRM-B2	DR64-SEC-B2
board_id	u32	4	0	32 bits	20	21
board_rev	u32	4	0	32 bits	6	4
board_sku	str	16	1 char		2	0
MCU_vendor	u32	4	0	0	1	1
MCU_model	u32	4	0	5	4	4
EQ gen	u8	1	6	6	6	6
EQ_type	u8	1	0	1	1	1
EQ_ver	u16	2	0x000	0x0200	0x0200	0x0200
EQ_num	u8	1	0	3	2	2
MCU_chip_idx	u8	1	0	1	1	1
TDA_vendor	u32	4	0	0	0	0
TDA_model	u32	4	0	3	1	0
UART_Asclin_Module	u8	1			NA	NA
Debug_level	u8	1	0	0	Refer to debug sheet	Refer to debug sheet
UART_RX_pin	u16	2			NA	NA
UART_TX_pin	u16	2			NA	NA
MCU_HC_idx	u16	2	0	N/A	0	0
eth_mac_add_2.5G_0	byte[6]	6			0x027DFA00B300	0x027DFA00B200
eth_mac_add_2.5G_1	byte[6]	6			0x027DFA00B301	0x027DFA00B201
eth_mac_add_TDA	byte[6]	6				
BRD_PN		8			Refer to BRD sheet	Refer to BRD sheet
reserved_ME	str	125				
Vendor Serial Number	str	32				
production_date	str	16				
reserved_CM	str	128				
VW FAZIT Identification String	str	23				
ECU Serial Number	str	20				
VW ECU Hardware Number	str	11			1T6907018A	1T6907018A

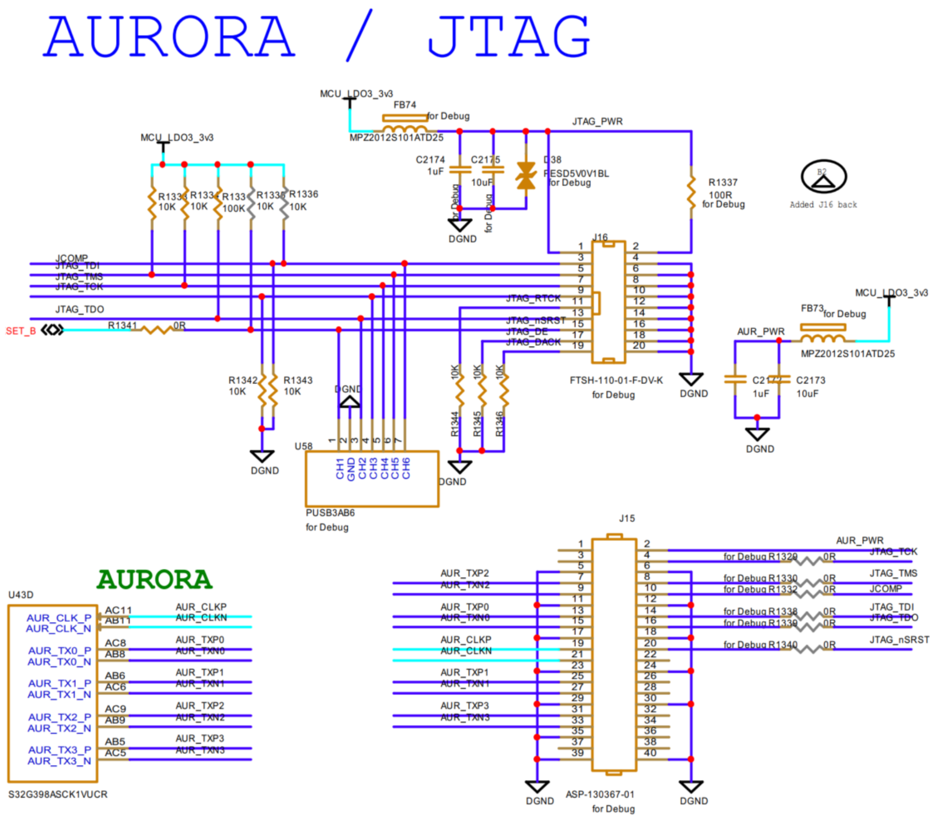
parameter	Type	Size in bytes	Min value	Max value	DR64_B2_primary	DR64 B2_secondary
VW ECU Hardware Version Number	str	3			Liquid cooled HO5	Liquid cooled HO5
reserved_OEM	str	256				
CRC	u32	4				

[Draft]

4.10 Debug S32G

4.10.1 Lauterbach/S32DBGPROBE - JTAG programmer device

For B2 board, both J15 and J16 exists and can support JTAG like following:



J16:

Connector Pin	Signal	Pin/Port#	Signal Name
5	JTAG test data input	[V7] PA_00	JTAG_TDI
13	JTAG test data output	[AA7] PA_01	JTAG_TDO
9	JTAG test clock	[W9] PA_04	JTAG_TCK
7	JTAG test mode selection	[U7] PA_05	JTAG_TMS
3	jcomp Pin	[W7] JCOMP	JCOMP
15	JTAG reset pin	RESET_B	JTAG_nSRST

J15:

Connector Pin	Signal	Pin/Port#	Signal Name
14	JTAG test data input	[V7] PA_00	JTAG_TDI
16	JTAG test data output	[AA7] PA_01	JTAG_TDO
4	JTAG test clock	[W9] PA_04	JTAG_TCK
8	JTAG test mode selection	[U7] PA_05	JTAG_TMS
10	jcomp Pin	[W7] JCOMP	JCOMP
20	JTAG reset pin	RESET_B	JTAG_nSRST
7	Aurora Transmit lane 2	[AC9] AUR_TXP2	AUR_TXP2
9	Aurora Transmit lane 2	[AB9] AUR_TXN2	AUR_TXN2
13	Aurora Transmit lane 0	[AC8] AUR_TXP0	AUR_TXP0
15	Aurora Transmit Lane 0	[AB8] AUR_TXN0	AUR_TXN0
19	Aurora reference Clock input	[AC11] AUR_CLKP	AUR_CLKP
21	Aurora reference Clock input	[AB11] AUR_CLKN	AUR_CLKN
25	Aurora Transmit Lane 1	[AC6] AUR_TXP1	AUR_TXP1
27	Aurora Transmit Lane 1	[AB6] AUR_TXN1	AUR_TXN1
31	Aurora Transmit Lane 3	[AB5] AUR_TXP3	AUR_TXP3
33	Aurora Transmit Lane 3	[AC5] AUR_TXN3	AUR_TXN3

4.10.2 Debug UART

#	Peripheral	Signal	Routed pin/signal	Label
U11	LINFlex_0	txd	[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX
Y12	LINFlex_0	rxd	[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX
A9	LINFlex_2	txd	[A9] PK_11	DEBUG.3v3_MCU_UART_TX
C14	LINFlex_2	rxd	[C14] PK_12	DEBUG.3v3_MCU_UART_RX

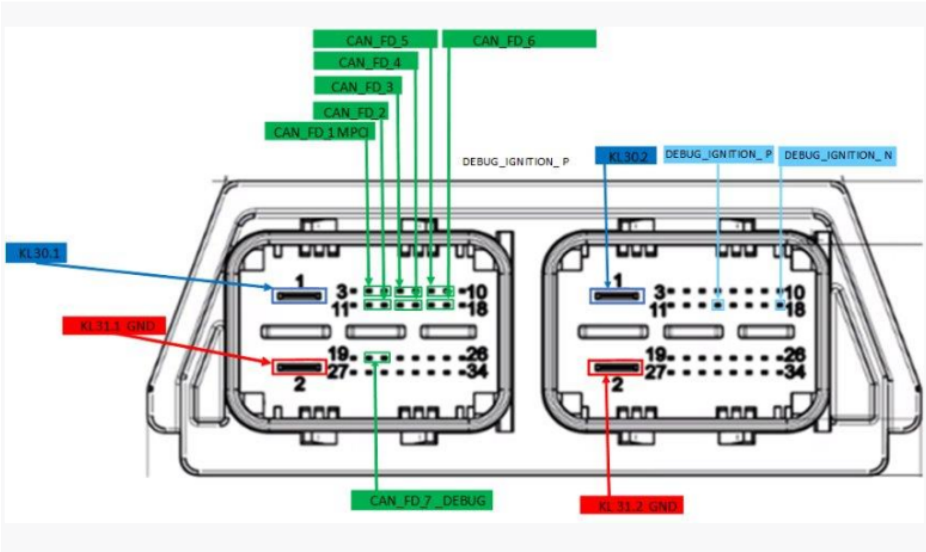
4.10.3 Debug CAN

#	Peripheral	Signal	Routed pin/signal	Label
E12	LLCE_CAN_6	txd	[E12] PJ_11	CAN.3v3_DEBUG_TX
E16	LLCE_CAN_6	rxd	[E16] PJ_12	CAN.3v3_DEBUG_RX

4.10.4 Debug GPIO

SM-39323 - Test point of PB_11[G8] pin of S32G3 is used for the 1 PPS signal generation purpose, so that a high-impedance measuring instrument can be connected to it permanently. **[** Draft **]**

4.11 Front Panel – DR64 Main Connector



4.12 U104 Configuration

Control Register

Following the successful acknowledgment of the address byte, the bus controller sends a command byte shown in

0	0	0	0	0	B2	B1	B0

CONTROL REGISTER BITS

B2	B1	B0	Command Byte (Hex)	Register	Protocol	Power-Up Default
CONTROL REGISTER BITS						
0	0	0	0x00	Input Port 0	Read byte	xxxx xxxx
0	0	1	0x01	Input Port 1	Read byte	xxxx xxxx
0	1	0	0x02	Output Port 0	Read-write byte	1111 1111
0	1	1	0x03	Output Port 1	Read-write byte	1111 1111
1	0	0	0x04	Polarity Inversion Port 0	Read-write byte	0000 0000
1	0	1	0x05	Polarity Inversion Port 1	Read-write byte	0000 0000
1	1	0	0x06	Configuration Port 0	Read-write byte	1111 1111
1	1	1	0x07	Configuration Port 1	Read-write byte	1111 1111

Configuration registers

The Configuration registers (registers 6 and 7) shown below configure the directions of the I/O pins. If a bit in this register is set to 1, the corresponding port pin is enabled as an input with a high-impedance output driver. If a bit in this register is cleared to 0, the corresponding port pin is enabled as an output.

Registers 6 And 7 (Configuration Registers)

C0	C0.7	C0.6	C0.5	C0.4	C0.3	C0.2	C0.1	C0.0
Default	1	1	1	1	1	1	1	1
C1	C1.7	C1.6	C1.5	C1.4	C1.3	C1.2	C1.1	C1.0

Default	1	1	1	1	1	1	1	1

Output Port registers

The Output Port registers (registers 2 and 3) shown below show the outgoing logic levels of the pins defined as outputs by the Configuration register. Bit values in this register have no effect on pins defined as inputs. In turn, reads from this register reflect the value that is in the flip-flop controlling the output selection, not the actual pin value.

Registers 2 And 3 (Output Port Registers)

O0	O0.7	O0.6	O0.5	O0.4	O0.3	O0.2	O0.1	O0.0
Default	1	1	1	1	1	1	1	1
O1	O1.7	O1.6	O1.5	O1.4	O1.3	O1.2	O1.1	O1.0
Default	1	1	1	1	1	1	1	1

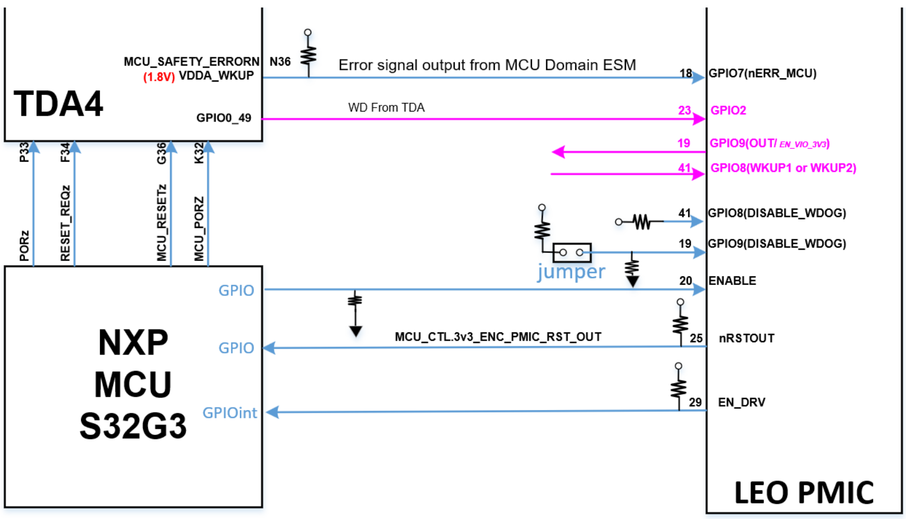
Input Port registers

The Input Port registers (registers 0 and 1) shown below reflect the incoming logic levels of the pins, regardless of whether the pin is defined as an input or an output by the Configuration register. It only acts on read operation. Writes to these registers have no effect. The default value, X, is determined by the externally applied logic level

Registers 0 And 1 (Input Port Registers)

I0	I0.7	I0.6	I0.5	I0.4	I0.3	I0.2	I0.1	I0.0
Default	X	X	X	X	X	X	X	X
I1	I1.7	I1.6	I1.5	I1.4	I1.3	I1.2	I1.1	I1.0
Default	X	X	X	X	X	X	X	X

4.13 TDA4 RESET Diagram



TDA4 Reset IF					
Device/Pin		Dir. S32G3	Net Name	Description	Notes
U43.A8	INT GPIO	I	MCU_CTL.3v3_ENC_PMIC_RST_OUT	LEO PMIC	PMIC Output

TDA4 Reset IF					
U103.10	I2C#4 Add. 0x74	O	ENC.3v3_RESET_REQZ	Main Domain external Warm Reset request	TDA4 Input
U43.P23	GPIO	O	ENC.1v8_PORZ	SoC PORz Reset Signal	TDA4 Input
U43.L21	GPIO	O	ENC.1v8_MCU_PORZ	MCU Domain Cold Reset	TDA4 Input
U103.11	I2C#4 Add. 0x74	O	ENC.3v3_MCU_RESETZ	MCU Domain Warm Reset	TDA4 Input

TDA4 Modes

TDA4 UART_MODE:

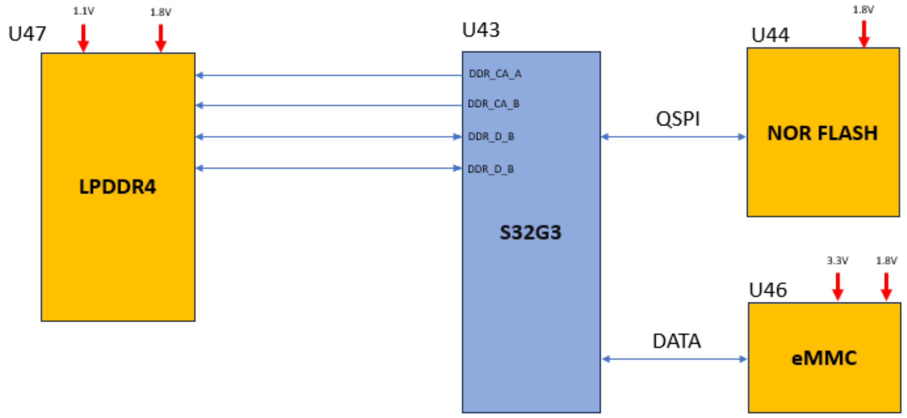
- ENC.3v3_MCU_BOOTMODE3- High
- ENC.3v3_MCU_BOOTMODE4- High
- ENC.3v3_MCU_BOOTMODE5- High
- ENC.1v8_SoC_BOOTMODE0- Low
- ENC.3v3_SoC_BOOTMODE4- Low
- ENC.3v3_SoC_BOOTMODE5- Low
- ENC.3v3_SoC_BOOTMODE6- Low

TDA4 NOR_MODE:

- ENC.3v3_MCU_BOOTMODE3- High
- ENC.3v3_MCU_BOOTMODE4- High
- ENC.3v3_SoC_BOOTMODE4- High
- ENC.3v3_SoC_BOOTMODE6- High
- ENC.3v3_MCU_BOOTMODE5- Low
- ENC.1v8_SoC_BOOTMODE0- Low
- ENC.3v3_SoC_BOOTMODE5- Low

4.14 MCU Memories

4.14.1 Block Diagram



4.14.2 MCU & Peripherals memories

SM-39243 -

Device	P.N	Manuf.	Ref. Des.	Description	Variants
MCU	S32G398ASCK1VUCR	NXP	U43	15MB RAM, 8 Cores	
LPDDR4	MT53E1G32D2FW-046 AAT:C	Micron	U47	4GB	
NOR Flash	MX66UM2G45GXRR00	Macronix	U44	256MB	
eMMC	KLMCG4JEUD-B04Q063	Samsung	U46	64GB	SV62
	MTFC128GAZAQJP-AAT/ Change in variant	Micron		128GB	CH63
	MTFC128GAZAQJP-AAT/ Change in variant	Micron		128GB	DR64

[Draft]

4.14.3 Information of Memories

SM-39246 -

			Device ID [h]				Variants
Device	P.N	MNF ID [h]	Byte2	Byte3	Sector Size	no. of Sectors	
NOR Flash	MX66UM2G45GXRR00	C2	80	3C	4KB	64K	
eMMC	KLMCG4JEUD-B04Q063	15	57		512B	125M	SV62
	MTFC128GAZAQJP-AAT	15	57		512B	250M	CH63/DR64

[Draft]

SM-39255 - NOR memory frequency shall be set to 100MHz due to stability issues at low temperature. [Draft]

SM-39217 -

Device	P.N	MNF ID [h]	Rev ID [h]	Page Size [bytes]
LPDDR4	MT53E1G32D2FW-046 AAT:C	FF	3	2048

[Draft]

SM-39220 - pSLC mode should be used in eMMC for the following reasons

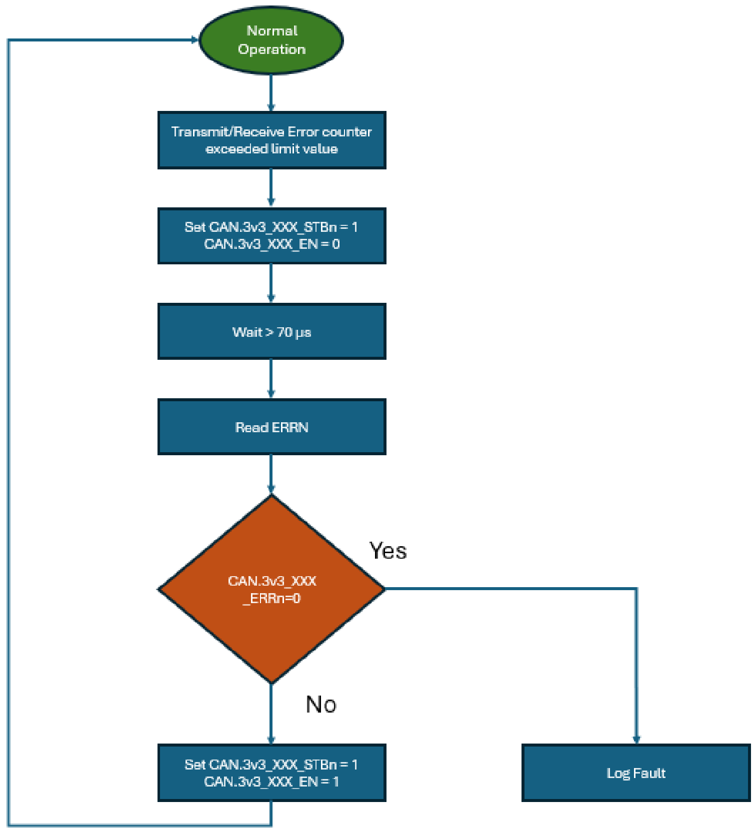
- avoid data corruption, even with power loss.
- for better data retention.

The DAF partition and partitions identified with potential risk of crossing the supported write/erase cycles (when used in MLC mode) during vehicle life time needs to be in pSLC mode. [Draft]

4.15 MCU CANs

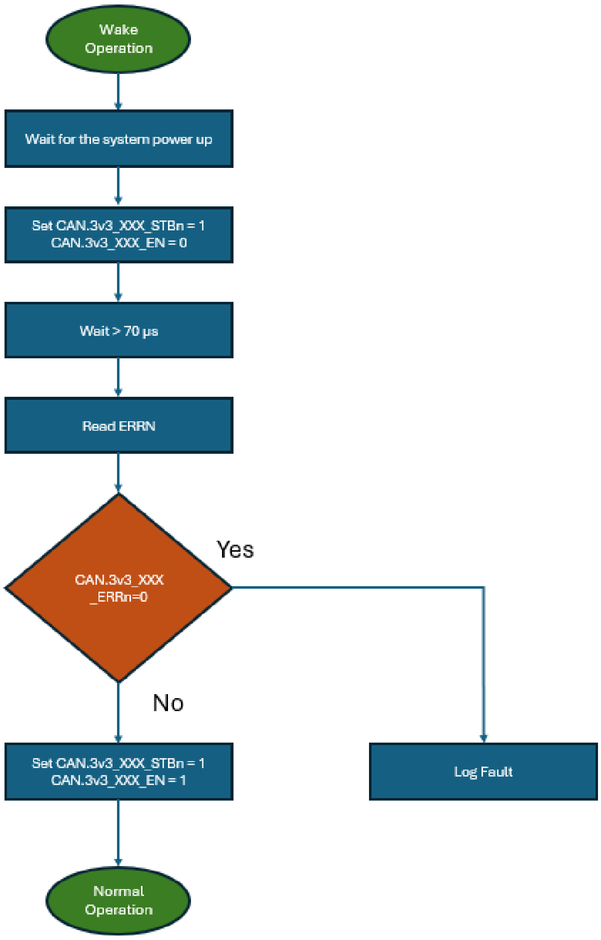
SM-39235 - For HW Diagnostics of CAN please refer Chapter 4.1 *SuppliersManagement/HW_SW Interface/HWDIAG* [Draft]

SM-39239 - Flow chart for the CAN error checking during the Normal operation



[Draft **]**

SM-39227 - Flow chart for the CAN error checking after power up. (Self test)



[Draft]

4.15.1 Flex_CAN

SM-39231 -

Peripheral	Routed pin/signal	Label	Connector Name	VOBES Name	Description
CAN_1	[AA10] PB_09	CAN.3v3_B2B_TX			B2B
CAN_1	[E6] PB_14	CAN.3v3_B2B_RX			
CAN_0	[F15] PC_11	CAN.3v3_HCP5_RX	B4: CAN.HCP5_H B5: CAN.HCP5_L	CAN_FD_1_MCPI	HCP5
CAN_0	[G11] PC_12	CAN.3v3_HCP5_TX			

[Draft]

4.15.2 LLCE_CAN

4.15.2.1 RADARs

SM-39293 -

Peripheral	Routed pin/signal	Label	Connector Name	VOBES Name
LLCE_CAN_1	[B11] PJ_01	CAN.3v3_RADAR2_TX	B6: CAN.RADAR2_H B7: CAN.RADAR2_L	NA for DR64
LLCE_CAN_1	[D17] PJ_02	CAN.3v3_RADAR2_RX		
LLCE_CAN_2	[D10] PJ_03	CAN.3v3_RADAR3_TX	B16: CAN.RADAR3_H B17: CAN.RADAR3_L	NA for DR64
LLCE_CAN_2	[C15] PJ_04	CAN.3v3_RADAR3_RX		
LLCE_CAN_3	[C11] PJ_05	CAN.3v3_RADAR4_TX	B12: CAN.RADAR4_H B13: CAN.RADAR4_L	NA for DR64
LLCE_CAN_3	[D15] PJ_06	CAN.3v3_RADAR4_RX		
LLCE_CAN_7	[A11] PJ_13	CAN.3v3_RADAR5_TX	B14: CAN.RADAR5_H B15: CAN.RADAR5_L	NA for DR64
LLCE_CAN_7	[C16] PJ_14	CAN.3v3_RADAR5_RX		
LLCE_CAN_15	[C9] PK_13	CAN.3v3_RADAR1_TX	B8: CAN.RADAR1_H B9: CAN.RADAR1_L	NA for DR64
LLCE_CAN_15	[B13] PK_14	CAN.3v3_RADAR1_RX		

[Draft]

4.15.2.2 EyeQs

SM-39296 -

Pin#	Peripheral	Signal	Pin/Port#	Net Name	
E10	LLCE_CAN_13	txd	[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	EyeQ 0
E14	LLCE_CAN_13	rxd	[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	
F11	LLCE_CAN_9	txd	[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	EyeQ1
D14	LLCE_CAN_9	rxd	[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	

[Draft]

4.16 MCU UARTs

SM-39288 -

#	Peripheral	Signal	Routed pin/signal	Label	
G10	LLCE_LIN_2	txd	[G10] PC_05	ENC.3v3_UART_RX_R	TDA4
A14	LLCE_LIN_2	rxd	[A14] PC_06	ENC.3v3_UART_TX_R	
F10	LLCE_LIN_0	txd	[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	EyeQ_0 (A)
A12	LLCE_LIN_0	rxd	[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	EyeQ_0 (A)
B9	LLCE_LIN_3	txd	[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	EyeQ_1 (B)
G13	LLCE_LIN_3	rxd	[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	EyeQ_1 (B)

[Draft]

4.17 MCU ADC Monitor

4.17.1 Configuration

SM-39291 - Configuration for the S32G can be defined by below method

- Base address of ADC are:
  - ADC_0 base address: 401F_8000h
  - ADC_1 base address: 402E_8000h
- MCR configuration register with Offset: 0h is used for the main configuration
- Analog Watchdog Threshold (THRHLR**n**) register is used to define the upper thresholds [16:27] and lower thresholds [0:11]
  - THRHLR0 Offset: 60h
  - THRHLR1 Offset: 64h
  - THRHLR2 Offset: 68h
  - THRHLR3 Offset: 6Ch
  - THRHLR4 Offset: 280h
  - THRHLR5 Offset: 284h
  - THRHLR6 Offset: 288h
  - THRHLR7 Offset: 28Ch
- Channel Interrupt Mask 0 (CIMR0) with Offset: 24h and Channel Interrupt Mask 1 (CIMR1) with Offset: 28h is used to enable and disable the register
- Watchdog Threshold Interrupt Status (WTISR) with Offset: 30h corresponds to the interrupt generated on the converted value being higher than the programmed higher threshold or lower than the programmed lower threshold.

[Draft]

4.17.2 ADC Monitor Voltages

SM-39269 -

#	Signal	Routed pin/signal	Label	voltage [V]			Notes	HWDIAG
				Min.	Typ.	Max.		
F23	adc0_ch, 0	[F23] ADC_CH_00	USS.LSS_MON	0 0X000		0.56 0X4FA	NA for DR64	☐ PLATFORM-HW-DEV-15239
E22	adc0_ch, 1	[E22] ADC_CH_01	SAFETY_SIG.RED_GND_IMON	0 0X000	0.4 0X38E	1.4 0XC71	NA for SV62	☐ PLATFORM-HW-DEV-12716
E23	adc0_ch, 2	[E23] ADC_CH_02	VBAT_INPUT_STAGE.MAIN_HSS_VMON	0.244 0X22B	0.8 0X71C	1.8 0XFFF		☐ PLATFORM-HW-DEV-9363
D22	adc0_ch, 3	[D22] ADC_CH_03	VBAT_INPUT_STAGE.MAIN_HSS_IMON	0 0X000	0.4 0X38E	1.4 0XC71		☐ PLATFORM-HW-DEV-9366
B21	adc0_ch, 4	[B21] ADC_CH_04	VR5510_AMUX	32 Analog Channels				☐ PLATFORM-HW-DEV-9373
B22	adc0_ch, 5	[B22] ADC_CH_05	SAFETY_SIG.MAIN_GND_IMON	0 0X000	0.4 0X38E	1.4 0XC71		☐ PLATFORM-HW-DEV-12715

#	Signal	Routed pin/signal	Label	voltage [V]			Notes	HWDIAG
D23	adc0_ch, 6	[D23] ADC_CH_ 06	HEATER.HSS_CURRENT	0.099 0X0E1	na	1.49 0XD3E	NA for DR64	☐ PLATFORM-HW-DEV-12264
C22	adc0_ch, 7	[C22] ADC_CH_ 07	VBAT_INPUT_STAGE.MAIN_IN_HSS_ VMON	0.244 0X22B	0.61 0X56C	1.4 0XC71		☐ PLATFORM-HW-DEV-9364
A19	adc1_ch, 0	[A19] ADC_CH_ 08	HEATER.HSS_MON	0.89 0X399	na	1.58 0XE0B	NA for DR64	☐ PLATFORM-HW-DEV-12893
B20	adc1_ch, 1	[B20] ADC_CH_ 09	HEATER.LSS_MON	0 0X000	na	1.58 0XE0B	NA for DR64	☐ PLATFORM-HW-DEV-9329
A20	adc1_ch, 2	[A20] ADC_CH_ 10	VBAT_INPUT_STAGE.RED_IN_HSS_ VMON	0.244 0X22B	0.61 0X56C	1.4 0XC71	NA for SV62	☐ PLATFORM-HW-DEV-9365
C21	adc1_ch, 3	[C21] ADC_CH_ 11	VBAT_INPUT_STAGE.MCU_HSS_IMON	0 0X000	0.3 0X2AA	1 0X8E3		☐ PLATFORM-HW-DEV-9367

Equation for Analog to digital conversion is:

(read value)*4096/1.8 to hex

Cyan * the heater current considered here is 0.2A to 3A [Draft]

4.17.3 ADC Monitor Calculations

SM-41349 - For DR64

Net Name	Real value
VBAT_INPUT_STAGE.MAIN_HSS_VMON	(read value)*15
VBAT_INPUT_STAGE.MAIN_HSS_IMON	(read value)*40
HEATER.HSS_CURRENT	(read value)*2.01
HEATER.HSS_MON	(read value)*10.1
HEATER.LSS_MON	(read value)*10.1
VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON	[(read value)*(19.1)] +0.43
VBAT_INPUT_STAGE.RED_IN_HSS_VMON	[(read value)*(19.1)] +0.43
VBAT_INPUT_STAGE.MCU_HSS_IMON	(read value)*8
USS.LSS_MON	(read value)*22.234
SAFETY_SIG.MAIN_GND_IMON	(read value-0.25)/0.025
SAFETY_SIG.RED_GND_IMON	(read value-0.25)/0.025

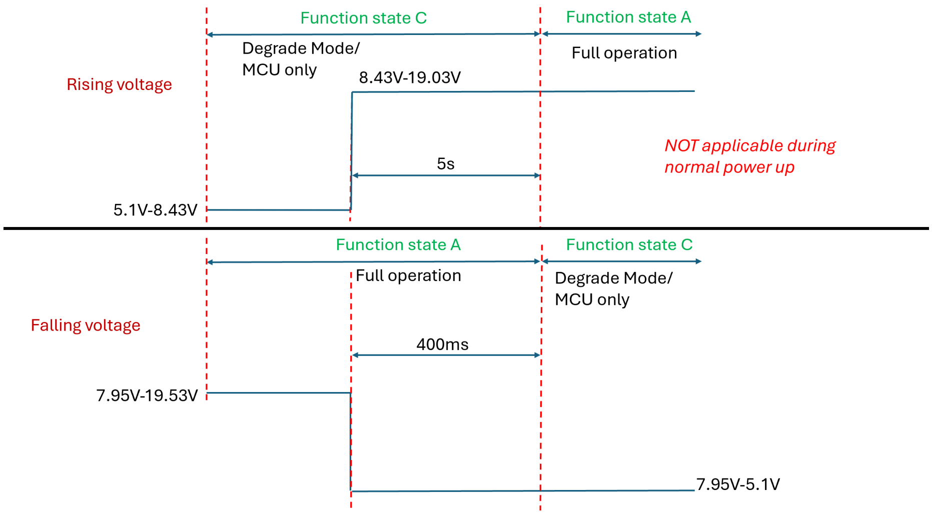
[Draft]

SM-39280 - HW Modes

HWA-1220 - MCU Only state shall be engaged in the following scenarios:

MCU only mode to Full operation: **Low** to high input voltage

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON OR VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is greater than or equal to **0.41984V (3BBh)** for more than 5s ,device goes to full operation mode. Fu ll operation to MCU only mode : High to **Low** input voltage VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON AND VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is equal to or less than **0.39419V (381h)** for more than 400ms ,device goes to MCU only mode.

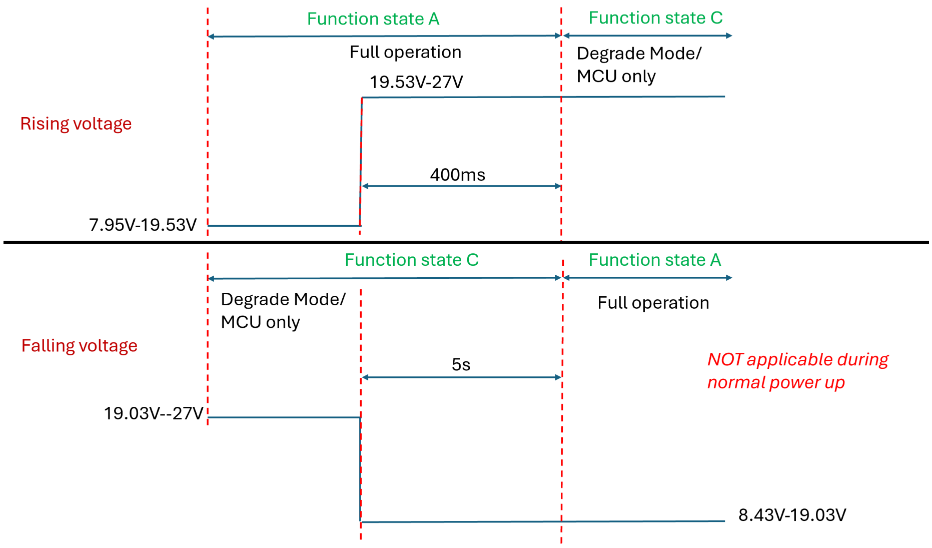


Full operation MCU only mode : Low to **High** input voltage

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON AND VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is greater than or equal to**1.00081V (8E5h)** for more than 400ms ,device goes to MCU only mode.

MCU only mode to Full operation : **High** to low input voltage

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON OR VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is equal to or less than **0.974118V (8A8h)** for more than 5s ,device goes to full operation mode.



[Draft]

4.17.4 VR5510_AMUX Values: HW Safety

SM-39284 - Please refer PLATFORM-HW-DEV-9373 for details **[** Draft **]**

SM-39272 -

AMUX [4:0]	Signal Select	P.S Out	Nom.	UV	OV	Remarks
00000	GND	NA	NA	NA	NA	
00001	VDDIO voltage divided by 2	3.3V	1.65	1.595	1.705	
00010	AMUX Temperature Sensor		0.69 (25°C)	0.495 (125°C)	0.816 (-40°C)	AMUX temp. T(°C) = [(VAMUX - VTEMP25) /VTEMP_COEFF] + 25 > VAMUX=VSUP/20 > VTEMP25=Temp. sensor voltage at 25°C (min. 0.67; typ. 0.69; max. 0.71) > VTEMP_COEFF (min. -2; max. -1.9)[mV/°C]
00011	Bandgap Main	NA	NA	NA	NA	
00100	Bandgap F.S	NA	NA	NA	NA	
00101	BUCK1 voltage	MCU_1v8	1.8	1.764	NA	
00110	BUCK2 voltage	1.1V	1.1	1.064	1.136	
00111	BUCK3 voltage divided by 2.5	MCU_1v1	0.44	0.425	0.454	
01000	VPRE voltage divided by 4	MCU_3v3	0.825	0.799	0.851	
01001	BOOST Voltage divided by 4	5V_CAN	1.25	1.19	1.31	
01010	LDO1 voltage divided by 4	MCU_1v8_ANA	0.45	0.435	0.465	
01011	LDO2 voltage divided by 4	LDO2_1V8	0.45	0.435	0.465	
01100	BOS voltage divided by 4	5V	1.25	1.19	1.31	
01101	Reserved					
01110	VSUP1 voltage divided by 20	Vbat_FILT_MCU	0.6	0.265	1.35	0.6 is for 12V nominal
01111	PWRON1 voltage divided by 20	PWRON1	0.6	0.265	1.35	0.6 is for 12V nominal
10000	PWRON2 voltage divided by 4	NA	NA	NA	NA	
10001	HVLDO voltage divided by 2	MCU_0v8_STBY	0.4	0.341	0.446	
10010	LDO3 voltage divided by 4	MCU_LDO3_3v3	0.8	0.795	0.855	
Others	GND	NA	NA	NA	NA	

[Draft **]**

4.18 Voltage Monitoring Validation - HW Safety

4.18.1 TPS389008-Q1 BIST

SM-39276 - TPS389008-Q1 BIST

When the TPS38900x-Q1 is powered ON, BIST is optionally executed (depending on TEST_CFG.AT_POR register bit)
Built-In Self Test (BIST) is performed:

- At Power On Reset (POR), if TEST_CFG.AT_POR=1
- When exiting ACTIVE state due to ACT transitioning from 1→0, if TEST_CFG.AT_SHDN=1

Configuration load from OTP is assisted by ECC (supporting SEC-DED). This is to protect against data integrity issues and to maximize system availability.

TEST_CFG Register (Address = 0x12)

Bit	Field	Type	Default	Description
7:4	RSVD		X	RSVD
3	AT_SHDN		X	Run BIST when exiting ACTIVE state due to ACT transitioning 1 to 0. Device ready after tCFG_WB. This bit cannot be set in OTP/NVM. Always defaults to 0 when loading configuration from OTP/NVM.
2	RESERVED	R	X	
1:0	AT_POR		X	Run BIST at POR. Device ready after tCFG_WB. 00b = Valid OTP configuration, skip BIST at POR 01b = Corrupt OTP configuration, run BIST at POR 10b = Corrupt OTP configuration, run BIST at POR 11b = Valid OTP configuration, run BIST at POR

During BIST, NIRQ is de-asserted (asserted in case of failure), input pins are ignored, and the I2C block is inactive with SDA and SCL de-asserted. The BIST includes device testing to meet the Technical Safety Requirements. Once BIST is completed without failure, I2C is immediately active and the device enters the IDL sate after loading the configuration data from OTP. If BIST fails and/or ECC reports Double-Error Detection (DED), NIRQ is asserted, the device enters FAILSAFE state, and a best effort attempt is made to active I2C. TEST_INFO register may provide additional information on the test results.

The detailed behaviour upon success/failure of the BIST is controlled by INT_TEST and IEN_TEST registers. Reporting of the BIST results is carried out through:

- NIRQ pin: pulled low depending on the test result and BIST_C and BIST bits in IEN_TEST
- I_BIST_C and BIST bits in INT_TEST register depending on IEN_TEST settings
- VMON_STAT.ST_BIST_C register bit
- TEST_INFO[3:0] register bits

TEST_INFO Register (Address = 0x31)

Bit	Field	Type	Default	Description
7:6	RSVD	R	X	RSVD
5	ECC_SEC	R	X	Status of ECC single-error correction on OTP configuration load. 0 = no error correction applied 1 = single-error correction applied

Bit	Field	Type	Default	Description
4	ECC_DED	R	X	Status of ECC double-error detection on OTP configuration load. 0 = no double-error detected 1 = double-error detected
3	BIST_VM	R	X	Status of Volatile Memory test output from BIST. 0 = Volatile Memory test pass 1 = Volatile Memory test fail
2	BIST_NVM	R	X	Status of Non-Volatile Memory test output from BIST. 0 = Non-Volatile Memory test pass 1 = Non-Volatile Memory test fail
1	BIST_L	R	X	Status of Logic test output from BIST. 0 = Logic test pass 1 = Logic test fail
0	BIST_A	R	X	Status of Analog test output from BIST. 0 = Analog test pass 1 = Analog test fail

IEN_TEST Register (Address = 0x1C)

Bit	Field	Type	Default	Description
07:04	RSVD		X	RSVD
3	ECC_SEC		0b	ECC single-error correction fault (on OTP load) interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled
2	RSVD		X	RSVD
1	BIST_Complete_INT		0b	Built-In Self-Test complete interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled
0	BIST_Fail_INT		0b	Built-In Self-Test fault interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled Although expected to be always enabled, it is desirable to have the option to disable it.

INT_TEST Register (Address = 0x23)

Bit	Field	Type	Default	Description
7:4	RSVD	R/W1C	X	RSVD
3	ECC_SEC	R/W1C	X	ECC single-error corrected on OTP configuration load: Write- 1-to-clear will clear the bit. The bit will be set again during next OTP configuration load if the same fault is detected. 0b = No single-error corrected (or IEN_TEST.ECC_SEC is disabled) 1b = Single-error corrected

Bit	Field	Type	Default	Description
2	ECC_DED	R/W1C	X	ECC double-error detected on OTP configuration load: The fault bit is always enabled (there is no associated interrupt enable bit). Write- 1-to-clear will clear the bit. The bit will be set again during next OTP configuration load if the same fault is detected. 0b = No double-error detected on OTP load 1b = Double-error detected on OTP load
1	I_BIST_C	R/W1C	X	Indication of Built-In Self-Test complete: Write- 1-to-clear will clear the bit. The bit will be set again on completion of next BIST execution.Write- 1-to-clear will clear the bit. The bit will be set again on completion of next BIST execution. 0b = BIST not complete (or IEN_TEST.BIST_C is disabled) 1b = BIST complete
0	BIST	R/W1C	X	Built-In Self-Test fault: Write- 1-to-clear will clear the bit. The bit will be set again during next BIST execution if the same fault is detected. 0b = No BIST fault detected (or IEN_TEST.BIST is disabled) 1b = BIST fault detected

[Draft **]**

SM-39176 - TPS389008-Q1 can be used to set register with over voltage and under voltage value. The TPS389008-Q1 is designed to assert active low output signals (NIRQ) when the monitored voltage is outside the safe window.

- Overvoltage threshold can be set using OV_LF[i], INT_OVLF Register (Address = 0x18) detect OV fault has been occurred.
- Undervoltage threshold can be set using UV_LF[i], INT_UVLF Register (Address = 0x14) detect UV fault has been occurred.
- i = 1 to 8

TPS389008-Q1 can be used to set register with over voltage and under voltage value. The TPS389008-Q1 is designed to assert active low output signals (NIRQ) when the monitored voltage is outside the safe window.

- Overvoltage threshold can be set using OV_HF[i], INT_OVHF Register (Address = 0x16) detect OV fault has been occurred.
- Undervoltage threshold can be set using UV_HF[i], INT_UVHF Register (Address = 0x12) detect UV fault has been occurred.
- i = 1 to 8

Details can found in PLATFORM-HW-DEV-9413 and PLATFORM-HW-DEV-14930 **[** Draft **]**

SM-39179 - Equation for Analog to digital conversion is:

- Max Voltage below 1.475: [((voltage) -0.2V)/0.005] to hex
- Max Voltage above 1.475: [((voltage) -0.8V)/0.020] to hex

[Draft **]**

Configuration files of the voltage monitors is in the below link
VM config files

4.18.2 VM#1

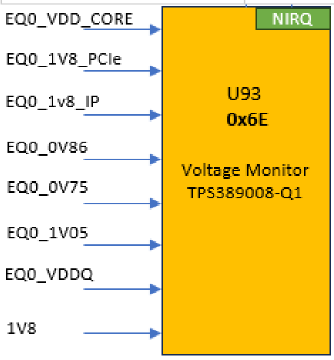
4.18.2.1 U93

For configuration file for U93 TPS389008M6HRTERQ1, Check VM config files

SM-39158 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)
U93 TPS389008M6HRTERQ1	EyeQ6H Main	0	0011 0111b	0x37



[Draft]

4.18.2.2 U93 rails list

SM-39150 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.875V (core) +2.5% / -3%	EQ0_VDD_CORE	0.84875	0.875	0.896875	☐ PLATFORM-HW-DEV-9386
Chan 2: 1.8V (PCIE) +/-5%	EQ0_1V8_PCIe	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9388
Chan 3: 1.8V (IP) +/-3%	EQ0_1v8_IP	1.746	1.8	1.854	☐ PLATFORM-HW-DEV-9391
Chan 4: 0.86V (PCIE) +/-5%	EQ0_0V86	0.817	0.86	0.903	☐ PLATFORM-HW-DEV-9393
Chan 5: 0.75V (MIPI) +/-5%	EQ0_0V75	0.7125	0.75	0.7875	☐ PLATFORM-HW-DEV-9395
Chan 6: 1.05V (DDR) +/-3%	EQ0_1V05	1.0185	1.05	1.0815	☐ PLATFORM-HW-DEV-9374
Chan 7: 0.5V (DDR) +/-5%	EQ0_VDDQ	0.475	0.5	0.525	☐ PLATFORM-HW-DEV-9375
Chan 8: 1.8V (Main) +/-5%	1V8	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9376

[Draft]

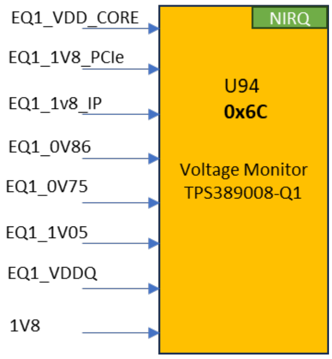
4.18.3 VM#2

4.18.3.1 U94

For configuration file for U94 TPS389008M6HRTERQ1, Check VM config files

SM-39167 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)
U94 TPS389008M6HRTERQ1	EyeQ6H Surround	0	0011 0110b	0x36



[Draft]

4.18.3.2 U94 rails list

SM-39170 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.875V (core) +2.5% / -3%	EQ1_VDD_CORE	0.84875	0.875	0.896875	☐ PLATFORM-HW-DEV-9416
Chan 2: 1.8V (PCIE) +/-5%	EQ1_1V8_PCIe	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9417
Chan 3: 1.8V (IP) +/-3%	EQ1_1v8_IP	1.746	1.8	1.854	☐ PLATFORM-HW-DEV-9418
Chan 4: 0.86V (PCIE) +/-5%	EQ1_0V86	0.817	0.86	0.903	☐ PLATFORM-HW-DEV-9419
Chan 5: 0.75V (MIPI) +/-5%	EQ1_0V75	0.7125	0.75	0.7875	☐ PLATFORM-HW-DEV-9420
Chan 6: 1.05V (DDR) +/-3%	EQ1_1V05	1.0185	1.05	1.0815	☐ PLATFORM-HW-DEV-9421
Chan 7: 0.5V (DDR) +/-5%	EQ1_VDDQ	0.475	0.5	0.525	☐ PLATFORM-HW-DEV-9414
Chan 8: 1.8V (Main) +/-5%	1V8	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9415

[Draft]

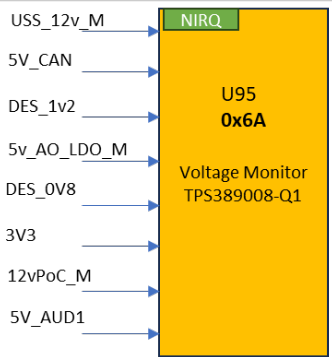
4.18.4 VM#3

4.18.4.1 U95

For configuration file for U95 TPS389008M66RTERQ1, Check VM config files

SM-39164 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit)
U95 TPS389008M66RTERQ1	Deserializer, CAN, POC, USS & General	0	0011 0101b	0x35



[Draft]

4.18.4.2 U95 rails list

SM-39210 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG	Notes
Chan 1: 1.1V (USS_12v_M) +/-5%	USS_12v_M	1.03835	1.1	1.14765	☐ PLATFORM-HW-DEV-9405	1. NA for DR64. 2. Formula for actual result: Vin x 11
Chan 2: 5V (5V_CAN) +/-5%	5V_CAN	4.75	5	5.25	☐ PLATFORM-HW-DEV-9406	
Chan 3: 1.2V (DES_1v2) +/-5%	DES_1V2	1.14	1.2	1.26	☐ PLATFORM-HW-DEV-9407	
Chan 4: 1.2V (5V_AOLDO_M) +/-5%	5v_AO_LDO_M	1.12575	1.185	1.24425	☐ PLATFORM-HW-DEV-9408	Formula for actual result: Vin x 4.22
Chan 5: 0.8V (DES_0V8) +/-5%	DES_0V8	0.76	0.9	0.84	☐ PLATFORM-HW-DEV-9409	
Chan 6: 3.3V (3V3) +/-5%	3V3	3.135	3.3	3.465	☐ PLATFORM-HW-DEV-9410	
Chan 7: 1.1V (12vPoC_M) +/-5%	12vPoC_M	1.03835	1.1	1.14765	☐ PLATFORM-HW-DEV-9411	Formula for actual result: Vin x 11
Chan 8: 5V	5V_AUD1	4.75	5	5.25	☐ PLATFORM-HW-DEV-12886	NA for SV62.

[Draft]

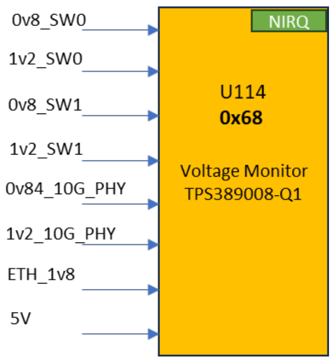
4.18.5 VM#4

4.18.5.1 U114

For configuration file for U114 TPS389008M67RTERQ1, Check VM config files

SM-39183 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)
U114 TPS389008M67RTERQ1	Switch, Phy& General	0	0011 0100b	0x34



[Draft]

4.18.5.2 U114 rails list

SM-39191 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG	Notes
Chan 1: 0.8V () +/-5%	0v8_SW0	0.76	0.8	0.84	☐PLATFORM-HW-DEV-9380	
Chan 2: 1.2V () +/-5%	1v2_SW0	1.14	1.2	1.26	☐PLATFORM-HW-DEV-9378	
Chan 3: 0.8V () +/-5%	0v8_SW1	0.76	0.8	0.84	☐PLATFORM-HW-DEV-9384	
Chan 4: 1.2V () +/-5%	1v2_SW1	1.14	1.2	1.26	☐PLATFORM-HW-DEV-9382	
Chan 5: 0.84V () +/-5%	0v84_10G_PHY	0.798	0.84	0.882	☐PLATFORM-HW-DEV-9389	NA for production
Chan 6: 1.2V () +/-5%	1v2_10G_PHY	1.14	1.2	1.26	☐PLATFORM-HW-DEV-14228	NA for production
Chan 7: 1.8V (ETH_1v8) +/-5%	ETH_1v8	1.71	1.8	1.89	☐PLATFORM-HW-DEV-9392	
Chan 8: 5V (Main 5V) +/-5%	5V	4.75	5	5.25	☐PLATFORM-HW-DEV-9390	

[Draft]

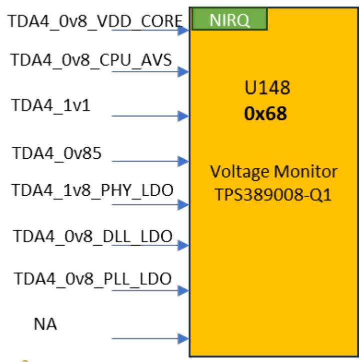
4.18.6 VM#5

4.18.6.1 U148

For configuration file for U148 TPS389008M64RTERQ1, Check VM config files

SM-39185 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)
U148 TPS389008M64RTERQ1	Encoder	4	0011 0100b	0x34



[Draft]

4.18.6.2 U148 rails list

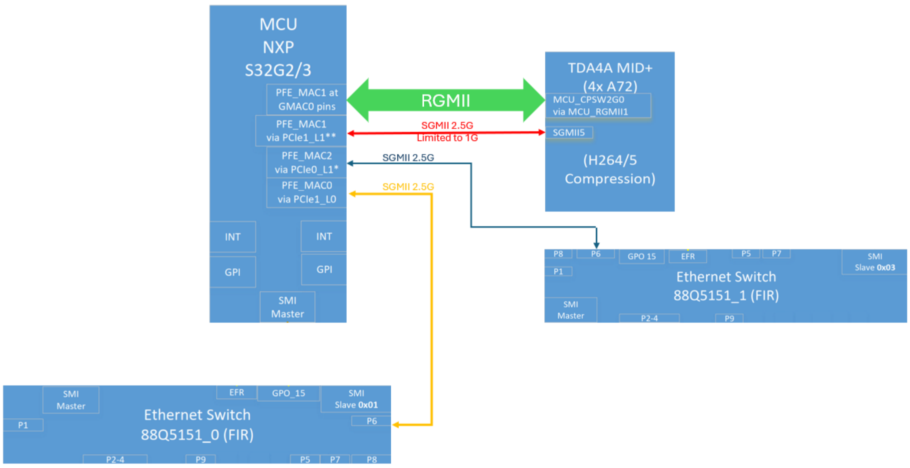
SM-39188 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.8V (TDA4_0v8_VDD_CORE) +/-5%	TDA4_0v8_VDD_CORE	0.76	0.8	0.84	☐PLATFORM-HW-DEV-9396
Chan 2: 0.8V (TDA4_0v8_CPU_AVS) +/-5%	TDA4_0v8_CPU_AVS	0.76	0.8	0.84	☐PLATFORM-HW-DEV-9394
Chan 3: 1.1V (TDA4_1v1) +/-5%	TDA4_1v1	1.045	1.1	1.155	☐PLATFORM-HW-DEV-9377
Chan 4: 0.85V (TDA4_0v85) +/-5%	TDA4_0v85	0.8075	0.85	0.8925	☐PLATFORM-HW-DEV-9381

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 5: 1.8V (TDA4_1v8_PHY_LDO) +/-5%	TDA4_1v8_PHY_LDO	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9379
Chan 6: 0.8V (TDA4_0v8_DLL_LDO) +/-5%	TDA4_0v8_DLL_LDO	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9385
Chan 7: 1.8V (TDA4_1v8_PLL_LDO) +/-5%	TDA4_1v8_PLL_LDO	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9383
Chan 8: 0V disabled	P.D		0		

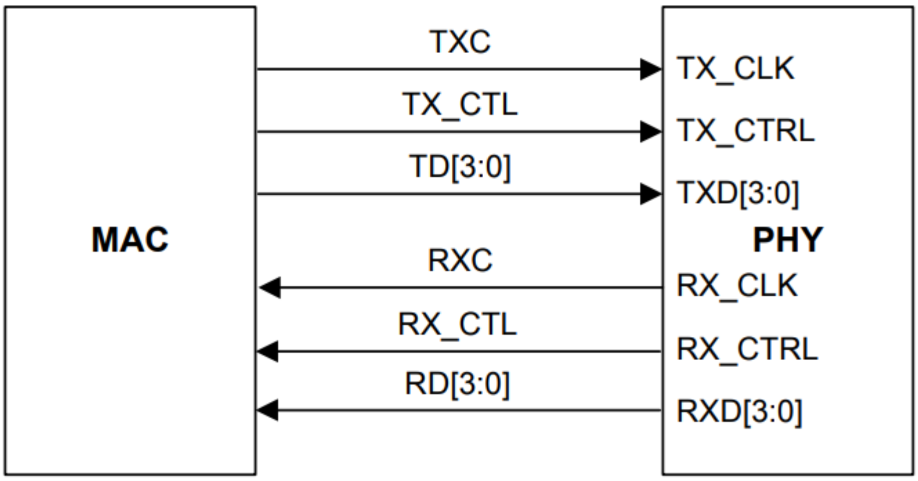
[Draft]

4.19 MCU Ethernet Connectivity



SM-39203 - [Draft]

4.20 MCU – TDA4 Main Link: RGMII



SM-39206 -

MCU – PFE_MAC1 **TDA4 (CPSW2G0)** [Draft]

4.20.1 RGMII signal connectivity

SM-39197 - Data signals are working at 150MHz

Pin#	Peripheral	Pin/Port#	Net Name
P20	PFE_MAC1	[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK
U23	PFE_MAC1	[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL
T23	PFE_MAC1	[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0
T22	PFE_MAC1	[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1
U22	PFE_MAC1	[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2
T21	PFE_MAC1	[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3
R21	PFE_MAC1	[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK
R23	PFE_MAC1	[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL
P19	PFE_MAC1	[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0
P18	PFE_MAC1	[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1
N18	PFE_MAC1	[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2
U21	PFE_MAC1	[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3

[Draft **]**

4.21 MCU SGMII

4.21.1 Chosen working modes

SM-39200 - ETH.MCU_2G5_SGMII should be configured as below
serdes0:mode=xpcs0&xpcs1,clock=int,fmhz=100;xpcs0_0:speed=2G5;xpcs0_1:speed=2G5 **[** Draft **]**

SM-39125 - ETH.MCU_2G5_SGMII_1 should be configured as below
serdes1:mode=xpcs0&xpcs1,clock=int,fmhz=125;xpcs1_0:speed=2G5;xpcs1_1:speed=2G5 **[** Draft **]**

4.21.2 SGMII Connectivity

SM-39106 -

SGMII#	Mode	PHY Lane 0	PHY Lane 1	Connect to	ext. Ref. Clock
0	SGMII Only		PFE_MAC2	ETH. SWITCH_0	NA
1	SGMII 2.5G	PFE_MAC0		ETH. SWITCH_1	NA

[Draft **]**

SM-39107 - * Note: MCU - TDA4 SGMII link is not used **[** Draft **]**

4.22 I2C

4.22.1 Buses

SM-39113 -

#	Peripheral	Signal	Routed pin/signal	Label
W12	I2C_0	sda, 0	PB_00	MCU_IF.3v3_I2C0_SDA
E7	I2C_0	scl, 0	PB_01	MCU_IF.3v3_I2C0_SCL
A6	I2C_2	scl, 2	PB_05	MCU_IF.3v3_I2C2_SCL

#	Peripheral	Signal	Routed pin/signal	Label
G9	I2C_2	sda, 2	PB_06	MCU_IF.3v3_I2C2_SDA
Y8	I2C_4	sda, 4	PC_01	MCU_IF.3v3_I2C4_SDA
W8	I2C_4	scl, 4	PC_02	MCU_IF.3v3_I2C4_SCL
C6	I2C_1	scl, 1	PB_03	MCU_IF.3v3_I2C1_SCL
E8	I2C_1	sda, 1	PB_04	MCU_IF.3v3_I2C1_SDA

[Draft]

4.22.2 Addresses

SM-39115 -

Device	Ref. Des.	I2C Bus #	Add. (7 bit)
PMIC BUCK	U50	4	0x28
PMIC	U54	4	Main = 0x20
			Safety = 0x21
GPIO Ex.	U103	4	0x74
Temp. Sensor	U61	4	0x4C
V.M	U148	4	0x34
P.S	U32	2	0x30
eePROM	U59	0	0x50
P.S	B_U100	2	0x46
Temp. Sensor	U60	4	0x48
Temp. Sensor	U62	4	0x49
P.S	B_U26	2	0x40
Temp. Sensor	U64	4	0x4A
P.S	B_U18	2	0x43
P.S	A_U26	1	0x40
P.S	A_U18	1	0x43
P.S	A_U100	1	0x46
Clk Gen.	U88	1	0x6B
Clk Gen.	U87	1	0x6A
P.S	U33	1	0x33
V.M	U93	0	0x37
V.M	U94	0	0x36
V.M	U95	0	0x35
V.M	U114	0	0x34
GPIO Ex.	U104	0	0x75
GPIO Ex.	U146	0	0x74
P.S	U34	2	0x43

Device	Ref. Des.	I2C Bus #	Add. (7 bit)
Temp. Sensor	U127	0	0x49
USS P.S	U91	1	0x75
Debug IO expander	U1	0	0x77

[Draft]

4.23 Temperature Sensors - HW Safety

SM-39109 - The thermal reaction is defined below

1. Critical Temperature high (low) Temperature (per SYS.3):
   a. Operating beyond this temperature shall be avoided
   b. there's time left to do graceful shutdown considering possible temperature gradients
   c. Perform graceful shutdown
2. Non-Critical high (Low) Temperature (SYS.3):
   a. operating outside specified temperature range
   b. just for logging purposes / quality analysis in the field etc

[Draft]

4.23.1 TMP75B-Q1

4.23.1.1 Addresses & Location

SM-39111 -

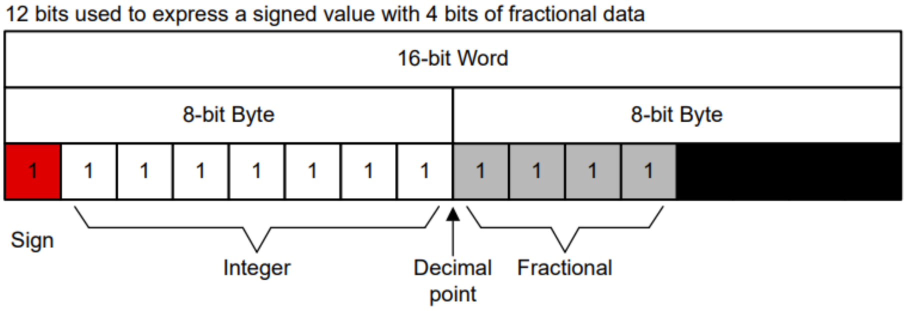
Temp. Sense #	Location	I2C Bus #	Add. (hex, 7bit)
U60	Near EyeQ_1 DDR	4	0x48
U61	Near ENCODER DDR	4	0x4C
U62	Between EyeQ_0 & EyeQ_1 DDRs	4	0x49
U64	Between EyeQ_0 & MCU DDRs	4	0x4A
U127	Near Valens SerDes	0	0x49

[Draft]

4.23.1.2 Temperature Sensors Read Conversion

SM-39121 - Negative numbers are represented in binary twos complement format. Following power-up or reset, the Temperature register reads 0°C until the first conversion is complete. The user can obtain 9, 10, 11, or 12 bits of resolution by addressing the Configuration register and setting the resolution bits accordingly. For 9-, 10-, or 11-bit resolution, the most significant bits (MSBs) in the Temperature register are used with the unused least significant bits (LSBs) set to zero

12 bits used to express a signed value with 4 bits of fractional data



[Draft]

SM-39123 - The Configuration register is an 16-bit read/write register used to store bits that control the operational modes of the temperature sensor. Read and write operations are performed MSB first. The power-up or reset value of the Configuration register are all bits equal to 0.

BYTE	D15	D14	D13	D12	D11	D10	D9	D8
1	OS	CR		FQ		POL	TM	SD

BYTE	D7	D6	D5	D4	D3	D2	D1	D0
0	Reserved	Read only -FFh						

Below Configuration can be used for configuring the TMP75B-Q1

D15: D8 - 0b 00000010

Number of consecutive faults: 0

Polarity: 0, Alert pin will be active low

Conversion rate : 37Hz (continuous conversion)

TM:1 Interrupt mode is used

Shutdown mode disabled

[Draft **]**

SM-39117 - For the temperature monitoring registers please refer

Chapter 1.5 of *SuppliersManagement/Primary Board HW Diagnostics/HWDIAG* **[** Draft **]**

4.23.1.3 Temperature limits

SM-39119 -

THIGH register is used to set the higher threshold and **T**LOW register used to set the lower threshold for TMP75-Q1.

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature[c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy	HWDIAG
Temp_sense1 (U60)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐ PLATFORM-HW-DEV-12630
Temp_sense5 (U61)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐ PLATFORM-HW-DEV-12637
Temp_sense3 (U62)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐ PLATFORM-HW-DEV-12633
Temp_sense2 (U127)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐ PLATFORM-HW-DEV-12632
Temp_sense4 (U64)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐ PLATFORM-HW-DEV-12635

Cooling TH: Temperature above which cooling is requested **[** Draft **]**

4.23.2 S32G internal temperature sensor

4.23.2.1 Configuration

SM-39132 -

- TMU base address: 400A_8000h
- Mode (TMR) register with Offset: 0h allows the software to control the thermal monitoring operation: Mode, enable/disable, Low pass filter
- Status (TSR) register with Offset: 4h reports the monitoring and calibration status during operation.
- Monitor Site (TMSR) register with Offset: 8h indicates which remote site sensor is actively being monitored by TMU when TMR[MODE]=10b
- Monitor Temperature Measurement Interval (TMTMIR) register with Offset: Ch allows you to control at what frequency the module reads the temperature sensors.
- Report Immediate Temperature at Site (TRITSR0 - TRITSR2) registers indicates Last Temperature Reading in Kelvin at the Site
  - TRITSR0 with Offset: 100h
  - TRITSR1 with Offset: 110h
  - TRITSR2 with Offset: 120h

- TMU interrupt: NCF designation is FCCU NCF [46] and NVIC interrupt ID is 120. Following interrupts provide FCCU interrupt
  - Immediate High Temperature Threshold Interrupt Enable
  - Immediate Low Temperature Threshold Interrupt Enable
  - Average High Temperature Threshold Interrupt Enable
  - Average Low Temperature Threshold Interrupt Enable
- Threshold for the temperature can be set using following registers
  - Monitor High Temperature Immediate Threshold (TMHTITR) with Offset: 50h
  - Monitor High Temperature Average Threshold (TMHTATR) with Offset: 54h
  - Monitor Low Temperature Immediate Threshold (TMHTITR) with Offset: 60h
  - Monitor Low Temperature Average Threshold (TMHTATR) with Offset: 64h
- Temperature Configuration (TTCFGR) with Offset: 80h register, in conjunction with Sensor Configuration (TSCFGR), is used to initialize the internal sensor translation table used during monitoring.
- Sensor Configuration (TSCFGR) with Offset: 84h register, in conjunction with Sensor Configuration (TSCFGR), is used to initialize the internal sensor translation table used during monitoring.
- Interrupt Enable (TIER) register with Offset: 20h determines if a detected status condition causes a system interrupt.
- Interrupt Detect (TIDR) register with Offset: 24h indicates if a status condition is detected that could generate an interrupt.
- Interrupt Average Site Capture (TIASCR) register with Offset: 34h indicates which remote sensor sites are associated with the detection of an interrupt event related to average reported temperatures.
- Interrupt Immediate Site Capture (TIISCR) register with Offset: 30h indicates which remote sensor sites are associated with the detection of an interrupt event related to current reported temperatures.

The procedure to enable the monitoring mode is as follows:

1. Clear the interrupt detect register, Interrupt Detect (TIDR)
2. Clear the interrupt capture registers: Interrupt Immediate Site Capture (TIISCR) and Interrupt Average Site Capture (TIASCR)
3. Enable interrupt handling by writing 1 to the appropriate fields in the interrupt enable register, Interrupt Enable (TIER).
4. Write the appropriate values to the following temperature threshold registers:
   • Monitor High Temperature Immediate Threshold (TMHTITR)
   • Monitor Low Temperature Immediate Threshold (TMLTITR)
   • Monitor High Temperature Average Threshold (TMHTATR)
   • Monitor Low Temperature Average Threshold (TMLTATR)
5. Write the appropriate value to the monitoring interval register, Monitor Temperature Measurement Interval (TMTMIR).

6. Select the ALPF coefficient in the Mode (TMR). The ALPF coefficient is used in the average temperature calculations.
7. Enable the monitor mode by setting TMR[MODE] to 10b. You control the monitored sites by writing an appropriate value to TMSR[SITE]. You must enable at least one active site, and set the other mode control fields as needed.
8. If the monitoring interval is too short, as indicated by TSR[MIE], you may need to increase the interval or reduce the number of sites.

[Draft]

SM-39128 - S32G threshold configuration

The below registers are configured with respect to temperature limit table

Monitor High Temperature Immediate Threshold (TMHTITR): Critical Temperature High [c]

Monitor Low Temperature Immediate Threshold (TMLTITR): Critical Temperature Low [c]

Monitor High Temperature Average Threshold (TMHTATR): Non-Critical High Temperature[c]

Monitor Low Temperature Average Threshold (TMLTATR): Non-Critical Low Temperature[c] [Draft]

4.23.2.2 Temperature limits

SM-39130 -

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature[c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy
MCU_Temp_sens_0	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	
MCU_Temp_sens_1	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	
MCU_Temp_sens_2	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	

NOTE: temperature conversion is done by converting temperature to Kelvin to a 9bit hex [Draft]

4.23.3 EyeQ internal temperature sensor

4.23.3.1 Configuration

SM-39137 - EyeQ6H Thermal Application Note [Draft]

4.23.3.2 Temperature limits

SM-39139 - EyeQ and TDA4 Limits

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature [c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy
EQ_0_Temp_sens_Combined	120	-35	118	-30	NA	3.5
EQ_1_Temp_sens_Combined	120	-35	118	-30	NA	3.5
TDA4_Temp_sens_Combined	125	-35	118	-30	NA	

[Draft]

4.24 SMI

4.24.1 MDC/IO port

SM-39133 -

Peripheral	Signal	Pin/Port	Label
PFE_MAC0	mdc	[M19] PF_00	MCU_IF.1v8_MDC
PFE_MAC0	md	[L22] PF_01	MCU_IF.1v8_MDIO

[Draft]

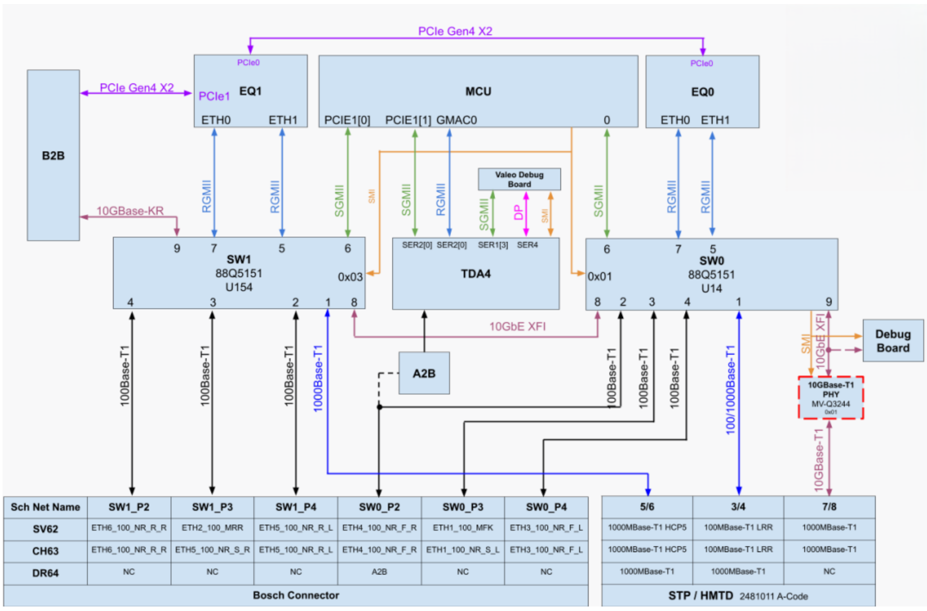
4.24.2 SMI Address List

SM-39135 -

Device	P.N	Ref. Des.	Address	Notes
ETH. SW_0	88Q5151	U14	0x01	
ETH. SW_1	88Q5151	U154	0x03	
10G PHY	MVQ3244	U150	0x09	U154 Slave, NA DR64

[Draft]

4.25 Ethernet Connectivity

SM-39144 - 

[Draft]

4.25.1 DR64 Ethernet Porting List

4.25.1.1 SW0 – U14

SM-39440 -

Port	Interface	Connect to
1	1000BASE-T1	HMTD Conn.
2	N.C	N.C

Port	Interface	Connect to
3	N.C	N.C
4	N.C	N.C
5	RGMII	EQ6H_0 Eth1
6	2.5G	MCU PCIe1_L0
7	RGMII	EQ6H_0 Eth0
8	10G	SW1 P8
9	N.C	N.C

[Draft]

4.25.1.2 SW1– U154

SM-39456 -

Port	Interface	Connect to
1	1000BASE-T1	HMTD Conn.
2	N.C	N.C
3	N.C	N.C
4	N.C	N.C
5	RGMII	EQ6H_1 Eth0
6	2.5G	MCU PCIe0_L1
7	RGMII	EQ6H_1 Eth1
8	10G	SW0 P8
9	10G	B2B

[Draft]

4.26 Interrupts - HW Safety

SM-39457 -

Sources	EIRQ	Port	Trigger Type	Net Name	SV62	CH63	DR64
ETH. SW0	EIRQ[20]	PL_12	Falling	SAFETY_SIG.SW0_1V8_EFR	No	Yes	Yes
	EIRQ[22]	PL_14		SAFETY_SIG.SW0_1V8_HARTBEAT	No	Yes	Yes
ETH. SW1	EIRQ[21]	PL_13	Falling	SAFETY_SIG.SW1_1V8_EFR	No	Yes	Yes
	EIRQ[23]	PH_01		SAFETY_SIG.SW1_3V3_HARTBEAT	No	Yes	Yes
EyeQ main	EIRQ[4]	PL_03	Rising+Falling	SAFETY_SIG.EQ0_3v3_CRERR	Yes	Yes	Yes
	EIRQ[5]	PB_08	Rising+Falling	SAFETY_SIG.EQ0_3v3_NCRERR	Yes	Yes	Yes
CAN	EIRQ[16]	PL_08	Falling	CAN.3v3_RADAR4_ERR	NA	Yes	NA
	EIRQ[17]	PL_09	Falling	CAN.3v3_RADAR5_ERR	NA	Yes	NA
EyeQ Surround	EIRQ[12]	PB_15	Rising+Falling	SAFETY_SIG.EQ1_3v3_CRERR	Yes	Yes	Yes
	EIRQ[15]	PC_03	Rising+Falling	SAFETY_SIG.EQ1_3v3_NCRERR	Yes	Yes	Yes

Sources	EIRQ	Port	Trigger Type	Net Name	SV62	CH63	DR64
Temp. Sensor	EIRQ[27]	PC_04	Rising+Falling	TEMP_SENS_ALERT	Yes	Yes	Yes
Voltage Monitors	EIRQ[11]	PK_08	Rising+Falling	SAFETY_SIG.VMs_3v3_INT	Yes	Yes	Yes
VBAT Input	EIRQ[1]	PK_05	Rising+Falling	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	NA	Yes	Yes
	EIRQ[26]	PH_05	Rising+Falling	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	Yes	Yes	Yes
	EIRQ[24]	PH_02	Rising+Falling	SAFETY_SIG.3v3_Main_BAT_Fault_INT	Yes	Yes	Yes
	EIRQ[7]	PB_10	Rising+Falling	SAFETY_SIG.3v3_Red_BAT_Fault_INT	NA	Yes	Yes
	EIRQ[25]	PH_03	Rising+Falling	VBAT_INPUT_STAGE.3v3_DIODE_PG	Yes	Yes	Yes
MCU PMIC	EIRQ[31]	PC_08	Falling	PMIC_INTB	Yes	Yes	Yes
TDA4	EIRQ[19]	PL_11		ENC.1v8_MCU_TIMESYNC1	No	Yes	NA
	EIRQ[2]	PL_01	Rising+Falling	SAFETY_SIG.3v3_ENC_PMIC_SS	Yes	Yes	Yes
	EIRQ[24]	PD_15	Falling	ENC.1v8_MCU_PORZ	No	Yes	Yes
	EIRQ[3]	PL_02	Rising+Falling	MCU_CTL.3v3_ENC_PMIC_RST_OUT	No	Yes	Yes

No: Polling is used

Yes: Interrupt is used

NA: Will not be monitored **[** Draft **]**

4.27 GPIO init state

SM-39454 - Production vs Debug GPIO init

- When debug level =0, then the production GPIO init configuration needs to be applied
- When debug level =any other value, then the debug GPIO init configuration needs to be applied

SM-3778 - debug level

[Draft **]**

4.27.1 MCU GPIOs init state

4.27.1.1 DR64 GPIOs init state

4.27.1.1.1 DR64 Debug

SM-39464 - MCU GPIOs init state DR64 Debug

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[W10] PA_02	BOOTMOD0	BOOT	Input	<NA>	PD			50MHz 3.3V
[W11] PA_03	BOOTMOD1	BOOT	Input	<NA>	PD			50MHz 3.3V
[Y11] PA_07	MCU_CTL.FR_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[U10] PA_08	EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[B8] PA_09	EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[E13] PA_10	DEBUG.3v3_DETECT	GPIO	Input	<NA>				50MHz 3.3V
[D8] PA_11	MCU_CTL.FR_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C7] PA_12	CAN.3v3_B2B_EN	GPIO	Output	Low				50MHz 3.3V
[U12] PA_13	MCU_CTL.FR_STBN	GPIO	High-Z	<NA>	PD			1MHz 3.3V
[W13] PA_15	MCU_CTL.FR_BGE	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[W12] PB_00	MCU_IF.3v3_I2C0_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[E7] PB_01	MCU_IF.3v3_I2C0_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[D7] PB_02	WAKE.3v3_MCU_KEEPALIVE	GPIO	Output	Low				50MHz 3.3V
[AA10] PB_09	CAN.3v3_B2B_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E6] PB_14	CAN.3v3_B2B_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A6] PB_05	MCU_IF.3v3_I2C2_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[G9] PB_06	MCU_IF.3v3_I2C2_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[F7] PB_08	SAFETY_SIG.EQ0_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[U18] PH_02	SAFETY_SIG.3v3_Main_BAT_Fault_INT	EIRQ	Input	<NA>	PD			83MHz 3.3V
[V11] PB_10	SAFETY_SIG.3v3_Red_BAT_Fault_INT	EIRQ	Input	<NA>	PD			50MHz 3.3V
[G8] PB_11	MCU_IF.FR_TXEN	GPIO	Output	Low				50MHz 3.3V
[F8] PB_12	MCU_IF.FR_TX	GPIO	High-Z	<NA>	PD	OD		50MHz 3.3V
[G7] PB_13	MCU_IF.FR_RX	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y9] PA_06	MCU_CTL.SW1_RST	GPIO	High-Z	High				50MHz 3.3V
[B5] PB_15	SAFETY_SIG.EQ1_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[V9] PC_00	DISCHARGE.ETH_MCU	GPIO	Output	Low				50MHz 3.3V
[Y8] PC_01	MCU_IF.3v3_I2C4_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[W8] PC_02	MCU_IF.3v3_I2C4_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[U9] PC_03	SAFETY_SIG.EQ1_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[C13] PC_04	TEMP_SENS_ALERT	EIRQ	Input	<NA>				50MHz 3.3V
[G10] PC_05	ENC.3v3_UART_RX_R	GPIO	Input	<NA>			115Kbps	50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[A14] PC_06	ENC.3v3_UART_TX_R	GPIO	Input	<NA>				50MHz 3.3V
[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[B6] PC_08	PMIC_INTB	EIRQ	Input	<NA>				50MHz 3.3V
[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[F15] PC_11	CAN.3v3_HCP5_RX	CAN	Input	<NA>			2Mbps	50MHz 3.3V
[G11] PC_12	CAN.3v3_HCP5_TX	CAN	Output	<NA>			2Mbps	50MHz 3.3V
[Y7] PC_13	CAN.3v3_HCP5_EN	GPIO	Input Output	Low				50MHz 3.3V
[Y10] PD_11	EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[V23] PD_12	EQ_IOs.EQ0_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[R19] PD_13	EQ_IOs.EQ1_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[N23] PH_00	EQ_IOs.EQ0_1_1v8_GPIO_B14_SYNC	GPIO	Output	Low				150MHz 1.8V
[P23] PD_14	ENC.1v8_PORZ	GPIO	Output	Low				150MHz 1.8V
[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK	GPIO	Output	Low				100MHz 3.3V
[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL	GPIO	Output	Low				150MHz 3.3V
[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0	GPIO	Output	Low				150MHz 3.3V
[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1	GPIO	Output	Low				150MHz 3.3V
[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2	GPIO	Output	Low				150MHz 3.3V
[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3	GPIO	Output	Low				150MHz 3.3V
[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK	RGMII CLK	Input	<NA>				150MHz 3.3V
[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL	RGMII CTL	Input	<NA>				150MHz 3.3V
[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0	RGMII DATA	Input	<NA>				150MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1	RGMII DATA	Input	<NA>				150MHz 3.3V
[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2	RGMII DATA	Input	<NA>				150MHz 3.3V
[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3	RGMII DATA	Input	<NA>				150MHz 3.3V
[T18] PE_14	CAN.3v3_RADAR1_STB	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[L22] PF_01	MCU_IF.1v8_MDIO	MDIO	Input Output	<NA>				150MHz 1.8V
[V17] PE_15	CAN.3v3_RADAR5_STB	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[M19] PF_00	MCU_IF.1v8_MDC	MDC	Output	<NA>				150MHz 1.8V
[Y21] PF_02	CAN.3v3_RADAR5_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[AA12] PA_14	SAFETY_SIG.DC_EQs_IN	GPIO	Output	High				50MHz 3.3V
[V14] PF_03	TDA4_EFUSE_EN	GPIO	Output	Low				150MHz 1.8V
[V13] PF_04	ENC.1v8_SoC_BOOTMODE0	GPIO	Output	Low				150MHz 1.8V
[U8] PF_15	MCU_CTL.power_EN_n	GPIO	Output	Low				50MHz 3.3V
[T19] PH_01	SAFETY_SIG.SW1_3v3_HARTBEAT	GPIO	Input	<NA>				150MHz 3.3V
[L21] PD_15	ENC.1v8_MCU_PORZ	GPIO	Output	Low				150MHz 1.8V
[U20] PH_03	VBAT_INPUT_STAGE.3V3_DIODE_PG	EIRQ	Input	<NA>	PU			150MHz 3V3
[P21] PE_00	EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[V21] PH_04	CAN.3v3_RADAR3_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W23] PH_05	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	EIRQ	Input	<NA>	PU			150MHz 3.3V
[T20] PH_06	CAN.3v3_RADAR4_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y23] PH_07	CAN.3v3_RADAR4_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W21] PH_08	CAN.3v3_RADAR2_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[M23]PE_01	EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[W22] PH_09	CAN.3v3_RADAR2_STBn	GPIO	High-Z	<NA>	PD			50MHZ 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[V20] PH_10	CAN.3v3_RADAR1_EN	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[U19] PJ_00	CAN.3v3_RADAR3_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[B11] PJ_01	CAN.3v3_RADAR2_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D17] PJ_02	CAN.3v3_RADAR2_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D10] PJ_03	CAN.3v3_RADAR3_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C15] PJ_04	CAN.3v3_RADAR3_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C11] PJ_05	CAN.3v3_RADAR4_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D15] PJ_06	CAN.3v3_RADAR4_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B15] PK_06	SAFETY_SIG.PMIC_3v3_SS	GPIO	Input	<NA>				50MHz 3.3V
[D12] PJ_07	CAN.3v3_RADAR1_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[D11]PK_05	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	EIRQ	Input	<NA>	PU			50MHz 3.3V
[F16] PJ_08	CAN.3v3_RADAR3_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[B12] PJ_09	MCU_CTL.SW0_RST	GPIO	High-Z	High				50MHz 3.3V
[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	CAN	Output	<NA>				50MHz 3.3V
[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	CAN	Input	<NA>				50MHz 3.3V
[E12] PJ_11	CAN.3v3_DEBUG_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E16] PJ_12	CAN.3v3_DEBUG_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A11] PJ_13	CAN.3v3_RADAR5_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C16] PJ_14	CAN.3v3_RADAR5_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A7] PL_04	CAN.3v3_B2B_STBn	GPIO	Output	Low				50MHz 3.3V
[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	CAN	Output	<NA>				50MHz 3.3V
[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	CAN	Input	<NA>				50MHz 3.3V
[C6] PB_03	MCU_IF.3v3_I2C1_SCL	I2C SCL	Output	<NA>		OD	400KHz	50MHz 3.3V
[F14] PK_04	IGN_SENSE	GPIO	Input	<NA>	PU			50MHz 3.3V
[E8] PB_04	MCU_IF.3v3_I2C1_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[A15] PK_00	CAN.3v3_HCP5_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[C10] PK_07	EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[F13] PK_08	SAFETY_SIG.VMs_3v3_INT	GPIO	Input	<NA>				50MHz 3.3V
[A10] PJ_15	CAN.3v3_HCP5_STBn	GPIO	Input Output	Low				50MHz 3.3V
[E11] PK_03	HCP5_INH	GPIO	Input	<NA>				50MHz 3.3V
[A9] PK_11	DEBUG.3v3_MCU_UART_TX	UART	Output	<NA>			115Kbps	50MHz 3.3V
[C14] PK_12	DEBUG.3v3_MCU_UART_RX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[C9] PK_13	CAN.3v3_RADAR1_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B13] PK_14	CAN.3v3_RADAR1_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[E9] PL_01	SAFETY_SIG.3v3_ENC_PMIC_SS	EIRQ	Input	<NA>				50MHz 3.3V
[A8] PL_02	MCU_CTL.3v3_ENC_PMIC_RST_OUT	GPIO	Input	<NA>				50MHz 3.3V
[A13] PL_03	SAFETY_SIG.EQ0_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[D16] PJ_10	CAN.3v3_RADAR2_ERRn	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C8] PL_05	CAN.3v3_B2B_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[D13] PL_06	B2B_CAN_INH	GPIO	Input	<NA>				50MHz 3.3V
[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[N19] PL_08	CAN.1v8_RADAR4_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[A5] PB_07	SAFETY_SIG.DC_EQ1_OUT	GPIO	Input	<NA>				50MHz 3.3V
[L23] PL_09	CAN.1v8_RADAR5_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[P22] PL_10	ENC.1v8_MCU_TIMESYNC0	GPIO	Input	<NA>	PD			166MHz 1.8V
[N22] PL_11	ENC.1v8_MCU_TIMESYNC1	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[N20] PL_12	SAFETY_SIG.SW0_1V8_EFR	GPIO	Input	<NA>				150MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[N21] PL_13	SAFETY_SIG.SW1_1V8_EFR	GPIO	Input	<NA>				150MHz 1.8V
[M21] PL_14	SAFETY_SIG.SW0_1V8_HARTBEAT	GPIO	Input	<NA>				166MHz 1.8V
[K20] PG_00	QSPI_A_SCK	QSPI CK	Output	<NA>			100MHz	133MHz 1.8V
[K21] PG_01	N.C	QSPI_A_ CK	High-Z	<NA>	PD			150MHz 1.8V
[J21] PG_02	N.C	QSPI_A_C K2	High-Z	<NA>	PD			150MHz 1.8V
[H21] PG_03	N.C	QSPI_A_C K2	High-Z	<NA>	PD			150MHz 1.8V
[G21] PG_04	QSPI_A_CS0	QSPI_A_C S0	Output	Low				133MHz 1.8V
[H22] PG_05	TEST_POINT	QSPI_A_C S1	Output	Low				150MHz 1.8V
[L18] PF_05	QSPI_A_DATA0	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L19] PF_06	QSPI_A_DATA1	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L20] PF_07	QSPI_A_DATA2	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K22] PF_08	QSPI_A_DATA3	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K19] PF_09	QSPI_A_DATA4	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[J23] PF_10	QSPI_A_DATA5	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[H23] PF_11	QSPI_A_DATA6	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K18] PF_12	QSPI_A_DATA7	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K23] PF_13	QSPI_A_DQS	QSPI DQS	Input	<NA>	PD			133MHz 1.8V
[J19] PF_14	QSPI_A_INT	QSPI INT	Input	<NA>				133MHz 1.8V
[E19] PC_14	eMMC_CK	USDHC CK	Output	<NA>				166MHz 1.8V
[F21] PC_15	eMMC_CMD	USDHC CMD	Input Output	<NA>				166MHz 1.8V
[G22] PD_00	eMMC_DATA0	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[E20] PD_01	eMMC_DATA1	USDHC DATA	Input Output	<NA>				166MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[H19] PD_02	eMMC_DATA2	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H20] PD_03	eMMC_DATA3	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H18] PD_04	eMMC_DATA4	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D18] PD_05	eMMC_DATA5	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F19] PD_06	eMMC_DATA6	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F20] PD_07	eMMC_DATA7	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D20] PD_08	eMMC_RST	USDHC RST	Output	<NA>				166MHz 1.8V
[G19] PD_09	N.C	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[D19] PD_10	eMMC_DQS	USDHC DQS	Input	<NA>	No			166MHz 1.8V

Not applicable for DR64 [Draft]

4.27.1.1.2 DR64 Production

SM-39459 - MCU GPIOs init state DR64 Production

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[W10] PA_02	BOOTMOD0	BOOT	Input	<NA>	PD			50MHz 3.3V
[W11] PA_03	BOOTMOD1	BOOT	Input	<NA>	PD			50MHz 3.3V
[Y11] PA_07	MCU_CTL.FR_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[U10] PA_08	EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[B8] PA_09	EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[E13] PA_10	DEBUG.3v3_DETECT	GPIO	Input	<NA>				50MHz 3.3V
[D8] PA_11	MCU_CTL.FR_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C7] PA_12	CAN.3v3_B2B_EN	GPIO	Output	Low				50MHz 3.3V
[U12] PA_13	MCU_CTL.FR_STBN	GPIO	High-Z	<NA>	PD			1MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[W13] PA_15	MCU_CTL.FR_BGE	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[W12] PB_00	MCU_IF.3v3_I2C0_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[E7] PB_01	MCU_IF.3v3_I2C0_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[D7] PB_02	WAKE.3v3_MCU_KEEPALIVE	GPIO	Output	Low				50MHz 3.3V
[AA10] PB_09	CAN.3v3_B2B_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E6] PB_14	CAN.3v3_B2B_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A6] PB_05	MCU_IF.3v3_I2C2_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[G9] PB_06	MCU_IF.3v3_I2C2_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[F7] PB_08	SAFETY_SIG.EQ0_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[U18] PH_02	SAFETY_SIG.3v3_Main_BAT_Fault_INT	EIRQ	Input	<NA>	PD			83MHz 3.3V
[V11] PB_10	SAFETY_SIG.3v3_Red_BAT_Fault_INT	EIRQ	Input	<NA>	PD			50MHz 3.3V
[G8] PB_11	MCU_IF.FR_TXEN	GPIO	Output	Low				50MHz 3.3V
[F8] PB_12	MCU_IF.FR_TX	GPIO	High-Z	<NA>	PD	OD		50MHz 3.3V
[G7] PB_13	MCU_IF.FR_RX	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y9] PA_06	MCU_CTL.SW1_RST	GPIO	High-Z	High				50MHz 3.3V
[B5] PB_15	SAFETY_SIG.EQ1_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[V9] PC_00	DISCHARGE.ETH_MCU	GPIO	Output	Low				50MHz 3.3V
[Y8] PC_01	MCU_IF.3v3_I2C4_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[W8] PC_02	MCU_IF.3v3_I2C4_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[U9] PC_03	SAFETY_SIG.EQ1_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[C13] PC_04	TEMP_SENS_ALERT	EIRQ	Input	<NA>				50MHz 3.3V
[G10] PC_05	ENC.3v3_UART_RX_R	GPIO	Input	<NA>			115Kbps	50MHz 3.3V
[A14] PC_06	ENC.3v3_UART_TX_R	GPIO	Input	<NA>				50MHz 3.3V
[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[B6] PC_08	PMIC_INTB	EIRQ	Input	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX	GPIO	High-Z	<NA>	External PU			50MHz 3.3V
[F15] PC_11	CAN.3v3_HCP5_RX	CAN	Input	<NA>			2Mbps	50MHz 3.3V
[G11] PC_12	CAN.3v3_HCP5_TX	CAN	Output	<NA>			2Mbps	50MHz 3.3V
[Y7] PC_13	CAN.3v3_HCP5_EN	GPIO	Input Output	Low				50MHz 3.3V
[Y10] PD_11	EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[V23] PD_12	EQ_IOs.EQ0_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[R19] PD_13	EQ_IOs.EQ1_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[N23] PH_00	EQ_IOs.EQ0_1_1v8_GPIO_B14_SYNC	GPIO	Output	Low				150MHz 1.8V
[P23] PD_14	ENC.1v8_PORZ	GPIO	Output	Low				150MHz 1.8V
[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK	GPIO	Output	Low				100MHz 3.3V
[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL	GPIO	Output	Low				150MHz 3.3V
[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0	GPIO	Output	Low				150MHz 3.3V
[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1	GPIO	Output	Low				150MHz 3.3V
[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2	GPIO	Output	Low				150MHz 3.3V
[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3	GPIO	Output	Low				150MHz 3.3V
[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK	RGMII CLK	Input	<NA>				150MHz 3.3V
[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL	RGMII CTL	Input	<NA>				150MHz 3.3V
[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0	RGMII DATA	Input	<NA>				150MHz 3.3V
[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1	RGMII DATA	Input	<NA>				150MHz 3.3V
[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2	RGMII DATA	Input	<NA>				150MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3	RGMII DATA	Input	<NA>				150MHz 3.3V
[T18] PE_14	CAN.3v3_RADAR1_STB	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[L22] PF_01	MCU_IF.1v8_MDIO	MDIO	Input Output	<NA>				150MHz 1.8V
[V17] PE_15	CAN.3v3_RADAR5_STB	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[M19] PF_00	MCU_IF.1v8_MDC	MDC	Output	<NA>				150MHz 1.8V
[Y21] PF_02	CAN.3v3_RADAR5_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[AA12] PA_14	SAFETY_SIG.DC_EQs_IN	GPIO	Output	High				50MHz 3.3V
[V14] PF_03	TDA4_EFUSE_EN	GPIO	Output	Low				150MHz 1.8V
[V13] PF_04	ENC.1v8_SoC_BOOTMODE0	GPIO	Output	Low				150MHz 1.8V
[U8] PF_15	MCU_CTL.power_EN_n	GPIO	Output	Low				50MHz 3.3V
[T19] PH_01	SAFETY_SIG.SW1_3v3_HARTBEAT	GPIO	Input	<NA>				150MHz 3.3V
[L21] PD_15	ENC.1v8_MCU_PORZ	GPIO	Output	Low				150MHz 1.8V
[U20] PH_03	VBAT_INPUT_STAGE.3V3_DIODE_PG	EIRQ	Input	<NA>	PU			150MHz 3V3
[P21] PE_00	EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[V21] PH_04	CAN.3v3_RADAR3_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W23] PH_05	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	EIRQ	Input	<NA>	PU			150MHz 3.3V
[T20] PH_06	CAN.3v3_RADAR4_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y23] PH_07	CAN.3v3_RADAR4_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W21] PH_08	CAN.3v3_RADAR2_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[M23]PE_01	EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[W22] PH_09	CAN.3v3_RADAR2_STBn	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[V20] PH_10	CAN.3v3_RADAR1_EN	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[U19] PJ_00	CAN.3v3_RADAR3_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[B11] PJ_01	CAN.3v3_RADAR2_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D17] PJ_02	CAN.3v3_RADAR2_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D10] PJ_03	CAN.3v3_RADAR3_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C15] PJ_04	CAN.3v3_RADAR3_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C11] PJ_05	CAN.3v3_RADAR4_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D15] PJ_06	CAN.3v3_RADAR4_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B15] PK_06	SAFETY_SIG.PMIC_3v3_SS	GPIO	Input	<NA>				50MHz 3.3V
[D12] PJ_07	CAN.3v3_RADAR1_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[D11]PK_05	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	EIRQ	Input	<NA>	PU			50MHz 3.3V
[F16] PJ_08	CAN.3v3_RADAR3_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[B12] PJ_09	MCU_CTL.SW0_RST	GPIO	High-Z	High				50MHz 3.3V
[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	CAN	Output	<NA>				50MHz 3.3V
[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	CAN	Input	<NA>				50MHz 3.3V
[E12] PJ_11	CAN.3v3_DEBUG_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[E16] PJ_12	CAN.3v3_DEBUG_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A11] PJ_13	CAN.3v3_RADAR5_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C16] PJ_14	CAN.3v3_RADAR5_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A7] PL_04	CAN.3v3_B2B_STBn	GPIO	Output	Low				50MHz 3.3V
[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	CAN	Output	<NA>				50MHz 3.3V
[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	CAN	Input	<NA>				50MHz 3.3V
[C6] PB_03	MCU_IF.3v3_I2C1_SCL	I2C SCL	Output	<NA>		OD	400KHz	50MHz 3.3V
[F14] PK_04	IGN_SENSE	GPIO	Input	<NA>	PU			50MHz 3.3V
[E8] PB_04	MCU_IF.3v3_I2C1_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[A15] PK_00	CAN.3v3_HCP5_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[C10] PK_07	EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[F13] PK_08	SAFETY_SIG.VMs_3v3_INT	GPIO	Input	<NA>				50MHz 3.3V
[A10] PJ_15	CAN.3v3_HCP5_STBn	GPIO	Input Output	Low				50MHz 3.3V
[E11] PK_03	HCP5_INH	GPIO	Input	<NA>				50MHz 3.3V
[A9] PK_11	DEBUG.3v3_MCU_UART_TX	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C14] PK_12	DEBUG.3v3_MCU_UART_RX	GPIO	High-Z	<NA>	Externa l PU			50MHz 3.3V
[C9] PK_13	CAN.3v3_RADAR1_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B13] PK_14	CAN.3v3_RADAR1_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[E9] PL_01	SAFETY_SIG.3v3_ENC_PMIC_SS	EIRQ	Input	<NA>				50MHz 3.3V
[A8] PL_02	MCU_CTL.3v3_ENC_PMIC_RST_OUT	GPIO	Input	<NA>				50MHz 3.3V
[A13] PL_03	SAFETY_SIG.EQ0_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[D16] PJ_10	CAN.3v3_RADAR2_ERRn	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C8] PL_05	CAN.3v3_B2B_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[D13] PL_06	B2B_CAN_INH	GPIO	Input	<NA>				50MHz 3.3V
[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[N19] PL_08	CAN.1v8_RADAR4_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[A5] PB_07	SAFETY_SIG.DC_EQ1_OUT	GPIO	Input	<NA>				50MHz 3.3V
[L23] PL_09	CAN.1v8_RADAR5_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[P22] PL_10	ENC.1v8_MCU_TIMESYNC0	GPIO	Input	<NA>	PD			166MHz 1.8V
[N22] PL_11	ENC.1v8_MCU_TIMESYNC1	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[N20] PL_12	SAFETY_SIG.SW0_1V8_EFR	GPIO	Input	<NA>				150MHz 1.8V
[N21] PL_13	SAFETY_SIG.SW1_1V8_EFR	GPIO	Input	<NA>				150MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[M21] PL_14	SAFETY_SIG.SW0_1V8_HARTBEAT	GPIO	Input	<NA>				166MHz 1.8V
[K20] PG_00	QSPI_A_SCK	QSPI CK	Output	<NA>			100MHz	133MHz 1.8V
[K21] PG_01	N.C	QSPI_A_CK	High-Z	<NA>	PD			150MHz 1.8V
[J21] PG_02	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[H21] PG_03	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[G21] PG_04	QSPI_A_CS0	QSPI_A_CS0	Output	Low				133MHz 1.8V
[H22] PG_05	TEST_POINT	QSPI_A_CS1	Output	Low				150MHz 1.8V
[L18] PF_05	QSPI_A_DATA0	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L19] PF_06	QSPI_A_DATA1	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L20] PF_07	QSPI_A_DATA2	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K22] PF_08	QSPI_A_DATA3	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K19] PF_09	QSPI_A_DATA4	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[J23] PF_10	QSPI_A_DATA5	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[H23] PF_11	QSPI_A_DATA6	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K18] PF_12	QSPI_A_DATA7	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K23] PF_13	QSPI_A_DQS	QSPI DQS	Input	<NA>	PD			133MHz 1.8V
[J19] PF_14	QSPI_A_INT	QSPI INT	Input	<NA>				133MHz 1.8V
[E19] PC_14	eMMC_CK	USDHC CK	Output	<NA>				166MHz 1.8V
[F21] PC_15	eMMC_CMD	USDHC CMD	Input Output	<NA>				166MHz 1.8V
[G22] PD_00	eMMC_DATA0	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[E20] PD_01	eMMC_DATA1	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H19] PD_02	eMMC_DATA2	USDHC DATA	Input Output	<NA>				166MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[H20] PD_03	eMMC_DATA3	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H18] PD_04	eMMC_DATA4	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D18] PD_05	eMMC_DATA5	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F19] PD_06	eMMC_DATA6	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F20] PD_07	eMMC_DATA7	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D20] PD_08	eMMC_RST	USDHC RST	Output	<NA>				166MHz 1.8V
[G19] PD_09	N.C	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[D19] PD_10	eMMC_DQS	USDHC DQS	Input	<NA>	No			166MHz 1.8V

Not applicable for DR64

Not applicable for production [Draft]

4.27.2 GPIOs Expanders Init State

At power on, all ports are configured as input.

SM-39460 - Before setting the pins to output, the pin value required value to be set to Init Value. [Draft]

4.27.2.1 DR64 GPIO expander

SM-39392 - DR64 GPIO expander

IC: U103

I2C: MCU_IF.3v3_I2C4

Address: 0x74

Pin/Port#	Net Name	Init State	Init Value	Notes
P00	ENC.3v3_MCU_BOOTMODE3	Input		
P01	ENC.3v3_MCU_BOOTMODE4	Input		
P02	ENC.3v3_MCU_BOOTMODE5	Input		
P03	ENC.3v3_SoC_BOOTMODE4	Input		
P04	ENC.3v3_SoC_BOOTMODE5	Input		
P05	MCU_CTL.3v3_ENC_PMIC_EN	Output	Low	
P06	ENC.3v3_RESET_REQZ	Output	Low	
P07	ENC.3v3_MCU_RESETZ	Output	Low	
P10	EN_RESET	Output	High	
P11	EQ_IOs.EQ1_3v3_GPIO_A7_MCU_GPIO0	Output	Low	

Pin/Port#	Net Name	Init State	Init Value	Notes
P12	EQ_IOs.EQ0_3v3_GPIO_A7_MCU_GPIO0	Output	Low	
P13	RADAR2_CAN_INH	Output	Low	
P14	ENC.3v3_SoC_BOOTMODE6	Input		
P15	EQ_IOs.EQ0_3v3_GPIO_A8_UBOOT	Output	Low	
P16	EQ_IOs.EQ1_3v3_GPIO_A8_UBOOT	Output	Low	
P17	FR_INH	Output	Low	

Not applicable for DR64 [Draft]

SM-39393 - DR64 GPIO expander
IC: U104
I2C: MCU_IF.3v3_I2C0
Address: 0x75

Pin/Port#	Net Name	Init State	Init Value
P00	DEBUG.10G_PHY_RST	Output	Low
P01	EQ_IOs.EQ0_3v3_FCMU_RST	Output	Low
P02	EQs_IOs.EQ0_3v3_GPIO_A14-TIMER0_EXT_ICAP1	Output	Low
P03	EQs_IOs.EQ1_3v3_GPIO_A14-TIMER0_EXT_ICAP1	Output	Low
P04	HEATER.3v3_LSS_EN	Output	Low
P05	MCU_CTL.I2B_5V_EN	Output	Low
P06	MCU_CTL.3v3_VDIs_CLKs_EN	Input	
P07	MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN	Input	
P10	HEATER.3v3_LSS_STATUS	Output	Low
P11	MCU_EEPROM_WP	Input	
P12	EQ_IOs.EQ1_3v3_FCMU_RST	Output	Low
P13	USS.3v3_LSS_EN	Output	Low
P14	USS.3v3_LSS_STATUS	Output	Low
P15	HEATER.3v3_HSS_EN	Output	Low
P16	HEATER.3v3_HSS_DIAG_EN	Output	Low
P17	B2B_GPIO	Input	

Not applicable for DR64 [Draft]

SM-39390 - DR64 GPIO expander
IC: U146
I2C: MCU_IF.3v3_I2C0
Address: 0x74

Pin/Port#	Net Name	Init State	Init Value
P00	EQ_IOs.EQ0_3V3_GPIO_A12_MCU_GPIO1	Output	Low

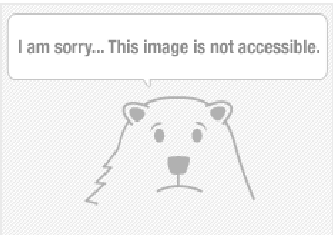
Pin/Port#	Net Name	Init State	Init Value
P01	EQ_IOs.EQ0_3V3_GPIO_A15_MCU_INCAP	Output	Low
P02	EQ_IOs.EQ0_3V3_GPIO_A16_MCU_OUTCMP	Output	Low
P03	EQ_IOs.EQ0_3V3_GPIO_A17_MCU_GPIO2	Output	Low
P04	EQ_IOs.EQ1_3V3_GPIO_A12_MCU_GPIO1	Output	Low
P05	EQ_IOs.EQ1_3V3_GPIO_A15_MCU_INCAP	Output	Low
P06	EQ_IOs.EQ0_3v3_XINT	Input	
P07	EQ_IOs.EQ1_3v3_XINT	Input	
P10	EQ_IOs.EQ1_3V3_GPIO_A16_MCU_OUTCMP	Output	Low
P11	EQ_IOs.EQ1_3V3_GPIO_A17_MCU_GPIO2	Output	Low
P12	MCU_CTL.3v3_USS_CLK_EN	Input	
P13	MCU_CTL.3v3_EQ1_CLKs_EN	Input	
P14	MCU_CTL.3v3_EQ0_CLKs_EN	Input	
P15	MCU_CTL.12v_PoC_EN	Output	Low
P16	EFUSE_EN	Output	Low
P17	VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN	Output	Low

[Draft]

SM-39391 -

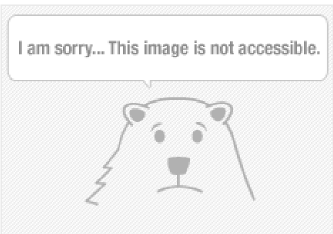
Control Register

Following the successful acknowledgment of the address byte, the bus controller sends a command byte shown in



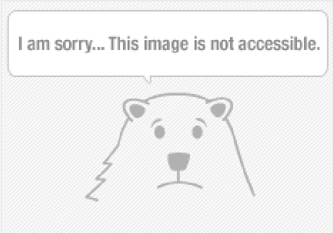
Configuration registers

The Configuration registers (registers 6 and 7) shown below configure the directions of the I/O pins. If a bit in this register is set to 1, the corresponding port pin is enabled as an input with a high-impedance output driver. If a bit in this register is cleared to 0, the corresponding port pin is enabled as an output.



Output Port registers

The Output Port registers (registers 2 and 3) shown below show the outgoing logic levels of the pins defined as outputs by the Configuration register. Bit values in this register have no effect on pins defined as inputs. In turn, reads from this register reflect the value that is in the flip-flop controlling the output selection, not the actual pin value.



[Draft]

SM-39399 -
Setting the Pins as output: Following the successful acknowledgment of the address byte, a command byte of 0x06(PORT0)/0x07(PORT1) is send. By this you can configure the PORT0/PORT1. Then set the bits of register 6 or 7 as 0 to configure as output and 1 to configure it as 0. [Draft]

4.28 Power supplies

4.28.1 Power supply configuration

4.28.1.1 USS PS #U91

SM-39401 - Not applicable for DR64 [Draft]

SM-39395 - USS power supply

	Ref. Des	I2C bus	Device address
USS P.S	U91	1	0x75

[Draft]

SM-39397 - REF Register (Address = 0h, 1h) [reset = 10100100b, 00000001b]:
REF sets the internal reference voltage of the TPS55287-Q1. The 01h register is the high byte and the 00h register is the low byte. One LSB of register 00h stands for 0.5645mV of the internal reference voltage. The default register value is 00000001 10100100b of 282mV. When the register value is 00000000 00000000b, the reference voltage is 45mV. When the register value is 00000111 10000000b, the reference voltage is 1.129V. The output voltage of the TPS55287-Q1 also depends on the output feedback ratio, which is either set in register 04h or set by an external resistor divider.
The REF register can be configured by an I2C controller before setting the OE bit in register 06h. For 5V output voltage, set the REF register value to 00000001 10100100b. To set the internal reference voltage, write the register 00h first, then write the register 01h.

Bit(s)	Field	Type	Reset Value	Description
15–11	Reserved	R	00000b	Reserved bits (read-only)
10–0	VREF	R/W	00110100100b (282 mV)	Internal reference voltage setting 00000000000b → 45 mV 00000000001b → 45.5645 mV 00000000010b → 46.129 mV .... 00110100100b → 282 mV (default) .... 01100110100b → 508 mV 10110001100b → 846 mV 11110000000b → 1129 mV 11111111110b → 1200 mV

Configuration:

REG	REG Address	Reference voltage	Hex	Binary
REF	0h, 01h	1.1V	**0x74Dh**	00000111 01001101

[Draft]

SM-39389 - VOUT_FS Register (Address = 4h) [reset = 00000011b]:

Register 04h sets the selection for the output feedback voltage, either by an internal resistor divider or external resistor divider, and sets the internal feedback ratio when using internal feedback resistor divider

Bit(s)	Field	Type	Reset Value	Description
7	FB	R/W	0b	Output feedback voltage control: • 0b – Use internal output voltage feedback (default) • 1b – Use external feedback; FB/INT becomes voltage input
6–2	RESERVED	R	00000b	Reserved
1–0	INTFB	R/W	11b	Internal feedback ratio: • 00b – Ratio = 0.2256 • 01b – Ratio = 0.1128 • 10b – Ratio = 0.0752 • 11b – Ratio = 0.0564 (default)

Configuration:

REG	REG Address	FB Internal or external	INTFB	Hex	Binary
VOUT_FS	4h	External ->1b	Default-> 11b	**0x83h**	1000 0011

[Draft]

SM-39385 - IOUT_LIMIT Register (Address = 2h) [reset = 11100100b]

IOUT_LIMIT sets the current limit target voltage between the ISP pin and the ISN pin. The default value in the current limit register is 11100100b standing for 50mV. 1 LSB stands for 0.5mV. The bit7 enables the current limit or disables the current limit.

Bit(s)	Field	Type	Reset Value	Description
7	Current_Limit_EN	R/W	1b	Enable/Disable current limit: • 0b – Disabled • 1b – Enabled (default)
6–0	Current_Limit_Setting	R/W	1100100b (50.0 mV)	Sets current limit threshold between ISP and ISN pins: • 0000000b – 0.0 mV • 0000001b – 0.5 mV • 0000010b – 1.0 mV • 0000011b – 1.5 mV • 0000100b – 2.0 mV … • 1100100b – 50.0 mV (default) … • 1111111b – 63.5 mV

Configuration:

REG	REG Address	Current limit EN/Disable	Current limit setting	Hex	Binary
IOUT_LIMIT	2h	Disable -> 0b	Default -> 1100100b	**0x64h**	0110 0100

[Draft]

SM-39388 - CDC Register (Address = 5h) [reset = 11100000b]

Register 05h sets masks for SC bit, OCP bit, and OVP bit in register 07h. In addition, register 05h sets the voltage rise added to the setting output voltage with respect to the sensed differential voltage between the ISP pin and the ISN pin.

Bit(s)	Field	Type	Reset Value	Description
7	SC_MASK	R/W	1b	Short Circuit Mask • 0b – Disabled SC indication • 1b – Enable SC indication (default)

Bit(s)	Field	Type	Reset Value	Description
6	OCP_MASK	R/W	1b	Overcurrent Mask • 0b – Disabled OCP indication • 1b – Enable OCP indication (default)
5	OVP_MASK	R/W	1b	Overvoltage Mask • 0b – Disabled OVP indication • 1b – Enable OVP indication (default)
4	RESERVED	R	0b	Reserved bit
3	CDC_OPTION	R/W	0b	Cable Voltage Droop Compensation Method • 0b – Internal CDC via register 05H (default) • 1b – External CDC via resistor on CDC pin
2–0	CDC	R/W	000b	Cable Droop Compensation Level (based on 50 mV at VISP - VISN): • 000b – 0.0 V rise (default) • 001b – 0.1 V rise • 010b – 0.2 V rise • 011b – 0.3 V rise • 100b – 0.4 V rise • 101b – 0.5 V rise • 110b – 0.6 V rise • 111b – 0.7 V rise

Configuration:

REG	REG Address	Mask	CDC option	Hex	Binary
CDC	5h	Default ->111b	External->1b	**0xE8h**	11101000

[Draft]

SM-39420 - MODE Register (Address = 6h) [reset = 00100000b] :

MODE controls the operating mode of the TPS55287-Q1.

Bit(s)	Field	Type	Reset Value	Description
7	OE	R/W	0b	Output Enable • 0b – Output disabled (default) • 1b – Output enabled
6	FSWDBL	R/W	0b	Switching Frequency Doubling (Buck-Boost mode) • 0b – Frequency unchanged (default) • 1b – Frequency doubled during buck-boost
5	HICCUP	R/W	1b	Hiccup Mode • 0b – Disabled during output short circuit protection • 1b – Enabled (default)
4	DISCHG	R/W	0b	Output Discharge on Shutdown • 0b – Disabled (default) • 1b – Enable 100mA discharge to GND when OE = 0
3	Force_DISCHG	R/W	0b	Force Output Discharge • 0b – Disabled (default) • 1b – Enable 100mA discharge FET
2	RESERVED	R	0b	Reserved

Bit(s)	Field	Type	Reset Value	Description
1	FPWM	R/W	0b	Light Load Operating Mode • 0b – PFM mode (default) • 1b – FPWM mode
0	RESERVED	R	0b	Reserved

Configuration:

REG	REG Address	Output EN	All other settings	Hex	Binary
MODE	6h	Enabled-> 1b	Default	**0xA0**	1010 0000

[Draft]

4.29 DDR Configuration Report

4.29.1 General Information

SM-39422 - DDR Type : *LPDDR4*
Firmware version : *2020_06_sp2* [Draft]

4.29.2 UI Configuration

SM-39415 - Device Information :

- **Clock Cycle Freq (MHz)** : *1600 MHz*
- **DDR_CLK frequency** : *800MHz*
- **Density per channel (Gb)** : *16G*
- **Number of ROW Addresses** : *17*
- **Number of Chip Selects used** : *1*
- **Number of Channels** : *2*
- **Number of COLUMN Addresses** : *10*
- **Number of BANK addresses** : *3*
- **Number of BANKS** : *8*
- **Total DRAM density (Gb)** : *32*
- **Bus Width** : *32*
- **Clock Cycle Time (ns)** : *0.625*

[Draft]

SM-39417 - Initial Board Settings :

- **PHY ODT Impedance** : *40 Ohm*
- **PHY Drive Strength** : *40 Ohm*
- **PHY Vref Quotient** : *0x13*
- **PHY Vref** : *0.163 V*
- **DRAM ODT Impedance** : *40 Ohm*
- **DRAM Drive Strength** : *40 Ohm*
- **CA Vref Training** : *yes*

[Draft]

SM-39430 - ECC options :

- **Enable Inline ECC** : *no*
- **Note** : *Enabling ECC provides SEC/DED functionality - single bit error correction and double bit error detection.*

- **Train 2D** : *yes*
- **PHY Log Level** : *Firmware complete*
- **Refresh Command Constraints** : *Legacy*
- **Static Refresh Rate [0.25x]** : *no*
- **Per Bank Refresh** : *no*

[Draft **]**

SM-39433 - DQ Swapping :

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
18	16	23	22	17	21	19	20	27	30	28	29	31	26	24	25	14	12	11	8	10	9	15	13	7	2	5	4	0	6	3	1

[Draft **]**

SM-39425 - Code Generation :

Code format : *ATF* **[** Draft **]**

SM-39411 - Advanced Settings :

Disable PMIC watchdog : *no* **[** Draft **]**

4.29.3 Mode Registers

SM-39413 - Mode Register

#	value	OP [7]	OP [6]	OP [5]	OP [4]	OP [3]	OP [2]	OP [1]	OP [0]
MR1	0x54	RPST = 0x0	nWR (for AP) = 0x5			RD-PRE = 0x0	WR-PRE = 0x1	BL = 0x0	
MR2	0x2D	WR Lev = 0x0	WLS = 0x0	WL = 0x5			RL = 0x5		
MR3	0x33	DBI-WR = 0x0	DBI-RD = 0x0	PDDS = 0x6			PPRP = 0x0	WR PST = 0x1	PU-CAL = 0x1
MR4	0x00	TUF = 0x0	Thermal Offset = 0x0		PPRE = 0x0	SR Abort = 0x0	Refresh Rate = 0x0		
MR11	0x66	Reserved = 0x0	CA ODT = 0x6			Reserved = 0x0	DQ ODT = 0x6		
MR12	0x4D	CBT Mode for Byte Mode = 0x0	VR-CA = 0x1	Vref(CA) = 0xd					
MR13	0x8	FSP-OP = 0x0	FSP-WR = 0x0	DMD = 0x0	RRO = 0x0	VRCG = 0x1	VRO = 0x0	RPT = 0x0	CBT = 0x0
MR14	0x4F	RFU = 0x0	VR(DQ) = 0x1	Vref(DQ) = 0xf					
MR16	0x00	PASR Bank Mask = 0x0							
MR17	0x00	PASR Segment Mask = 0x0							
MR22	0x6	ODTD for x8_2ch(Byte) mode = 0x0		ODTD-CA = 0x0	ODTE-CS = 0x0	ODTE-CK = 0x0	SOC ODT = 0x6		

#	value	OP [7]	OP [6]	OP [5]	OP [4]	OP [3]	OP [2]	OP [1]	OP [0]
MR24	0x00	TRR Mode = 0x0	TRR Mode BAn = 0x0			Unlimited = 0x0	MAC Value = 0x0		

[Draft]

5 EyeQ6H

SM-39407 - The boot parameters of main and surround EyeQ is given below

SuppliersManagement/EyeQ6H Boot Parameters_Primary/EyeQ6H Boot Parameters_Primary_Main_B2

SuppliersManagement/EyeQ6H Boot Parameters_Primary/EyeQ6H Boot Parameters_Primary_Surround_B2 **[** Draft **]**

6 System Power Up Sequence

SM-39410 - Comprehensive Power up and Down Sequence of a Board with MCU program steps: **[** Draft **]**

SM-39328 - If there is no number in Step column, then should ignore the row **[** Draft **]**

6.1 Main_1/Degrade Mode

SM-39330 - Step 0

- Configure all IO Expander pins as per 4.29 GPIOs Expanders Init State
- Set IO Expander Pins to output: IO_ENC_3v3_MCU_BOOTMODE3, IO_EQ_IOs_EQ0_3v3_FCMU_RST, IO_EQ_IOs_EQ1_3v3_FCMU_RST, IO_ENC_3v3_RESET_REQZ, IO_ENC_3v3_MCU_RESETZ
- Set GPIO Pins to output: PortConf_PortPin_ETH_ENC_3v3_RGMII1_RX_CLK, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RX_CTL,PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD0, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD1,PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD2, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD3
- Set GPIO Pins mode to ALT2: PortConf_PortPin_ETH_ENC_3v3_RGMII1_RX_CLK, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RX_CTL, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD0, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD1, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD2,
- Set GPIO Pins mode to ALT3: PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD3

[Draft **]**

SM-39324 - Main_1/Degraded Mode sequence

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/ Dependency	Delay
0	1	1	1	Assert WAKE.3v3_MCU_KEEPALIVE output	High		
0	3	4	4	Configure temperature sensor U127(0X49) using MCU_IF.3v3_I2C0			
0	4	5	5	Configure temperature sensor U60(0X48) using MCU_IF.3v3_I2C4			
0	5	6	6	Configure temperature sensor U62(0X49) using MCU_IF.3v3_I2C4			
0	6	7	7	Configure temperature sensor U64(0X4A) using MCU_IF.3v3_I2C4			
0	7	8	8	Configure temperature sensor U61(0X4C) using MCU_IF.3v3_I2C4			
0	8	9	9	Read DEBUG.3v3_DETECT to check the UART debug connected	High	SV62: Execute Step 9 CH63/DR64: Execute Step 10	
0	8	9	9	Read DEBUG.3v3_DETECT to check the UART debug connected	Low	SV62: Skip step 9 CH63/DR64: Skip Step 10	
0	9	10	10	Assert DEBUG.10G_PHY_RST output	High	NA for Production	
0	10	11	11	Assert MCU_CTL.SW0_RST output	High		
0	11	12	12	Assert MCU_CTL.SW1_RST output	High		
0	12	13	13	Read SAFETY_SIG.SW0_1v8_EFR	High		1ms max

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/ Dependency	Delay
0	13	14	14	Read SAFETY_SIG.SW1_1v8_EFR	High		1ms max
0	14	15	15	Read HCP5_INH	Low	Oring of these signals	
0	14	16	16	Read IGN_SENSE	Low		
0		17	17	Read B2B_CAN_INH	Low		
0		18	18	Read RADAR2_CAN_INH	Low	NA for DR64	
0	15	19	19	Read VBAT_INPUT_STAGE.3V3_DIODE_PG	High	Oring of these signals	20ms max
0		20	20	Read VBAT_INPUT_STAGE.3V3_DIODE_RED_PG	High		20ms max
0	16	21	21	Read VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON	Check ADC table	Oring of these signals	300s
0		22	22	Read VBAT_INPUT_STAGE.RED_IN_HSS_VMON	Check ADC table		
0	17	23	23	Read VBAT_INPUT_STAGE.MCU_HSS_IMON	Check ADC table		
0	18	24	24	Read SAFETY_SIG.MAIN_GND_IMON	Check ADC table	Oring of these signals	
0		25	25	Read SAFETY_SIG.RED_GND_IMON	Check ADC table		
0	19	26	26	Read VR5510_AMUX	Check ADC table		
0	20	27	27	Read 5v_AO_LDO_M	Check VM#3 table		
0	21	28	28	Read 0v8_SW0	0.8V ±5%		
0	22	29	29	Read 0v8_SW1	0.8V ±5%		
0	23	30	30	Read ETH_1v8	1.8V ±5%		
0	24	31	31	Read 1v2_SW0	1.2V ±5%		
0	25	32	32	Read 1v2_SW1	1.2V ±5%		
0	26	33	33	If DEBUG.3v3_DETECT =	High	SV62: Execute Step 28 and 27 CH63/DR64: Execute 35 and 34	
0	26	33	33	If DEBUG.3v3_DETECT =	Low	SV62: Skip step 28 and 27 CH63/DR64: Skip 35 and 34	
0	27	34		Read 0V84_10G_PHY	0.84V ±5%	NA for Production	
0	28	35		Read 1V2_10G_PHY	1.2V ±5%	NA for Production	
0	29	36	36	Assert MCU_CTL.power_EN_n output	High		10ms
0	30	37		Assert EN_RESET output	High		

[Draft]

SM-39326 - MCU_CTL.SW0_RST and MCU_CTL.SW1_RST should set to High value before setting the pin mode to push-pull and output. [Draft]

6.2 Main_2 Sequence

SM-39333 - Main_2 sequence

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/ Dependency	Delay
1	0	0	0	Assert SAFETY_SIG.DC_EQs_IN output	Low	After completion of seq 1	1ms
1	1	1	1	Read SAFETY_SIG.DC_EQ1_OUT	Low		
1	2	2	2	Assert SAFETY_SIG.DC_EQs_IN output	High		1ms
1	3	3	3	Read SAFETY_SIG.DC_EQ1_OUT	High		
1	4	4	4	Assert VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN output	High		80ms
1	5	5	5	Read VBAT_INPUT_STAGE.MAIN_HSS_VMON	Check ADC table		85ms max
1	6	6	6	Read VBAT_INPUT_STAGE.MAIN_HSS_IMON	Check ADC table		80ms max
1	7	7	7	Read VBAT_INPUT_STAGE.3v3_MAIN_HSS_STATUS	High		25ms min to 100ms max
1	8	8	8	Assert MCU_CTL.power_EN_n output	Low		50ms
1	9	9	9	Read 5V	5V±5%		20ms max
1	10	10	10	Read 3v3	3.3V±5%		2ms max
1	11	11	11	Read 1V8 using U93	1.8V±5%		2ms max
1	12	12	12	Read DES_1V2	1.2V±5%		10ms max
1	13	13	13	Read DES_0V8	0.8V ±5%		2ms max
1	14	14	14	Read EQ0_VDD_CORE	0.875V+2.5% / -3%		2ms max
1	15	15	15	Read EQ0_VDDQ	0.5V±5%		5ms max
1	16	16	16	Read EQ0_1V05	1.05V±3%		5ms max
1	17	17	17	Read EQ0_1v8_IP	1.8V±3%		16ms max
1	18	18	18	Read EQ0_1V8_PCIe	1.8V±5%		17ms max
1	19	19	19	Read EQ0_0V75	0.75V±5%		5ms max
1	20	20	20	Read EQ0_0V86	0.86V±5%		18ms max
1	21	21	21	Read EQ1_VDD_CORE	0.875V+2.5% / -3%		2ms max

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/ Dependency	Delay
1	22	22	22	Read EQ1_VDDQ	0.5V±5%		5ms max
1	23	23	23	Read EQ1_1V05	1.05V±3%		5ms max
1	24	24	24	Read EQ1_1v8_IP	1.8V±3%		16ms max
1	25	25	25	Read EQ1_1V8_PCIe	1.8V±5%		17ms max
1	26	26	26	Read EQ1_0V75	0.75V±5%		5ms max
1	27	27	27	Read EQ1_0V86	0.86V±5%		18ms max
1	28			Configure U91			
1	29		29	Read TDA CPU AVS	0.8V±5%		
1	30		30	Read TDA 1.1V	1.1V±5%		18ms max
1	31		31	Read TDA4_0v85	0.85V±5%		
1	32		32	Read TDA4_1v8_PHY_LDO	1.8V±5%		
1	33		33	Read TDA4_0v8_DLL_LDO	0.8V±5%		
1	34		34	Read TDA4_1v8_PLL_LDO	1.8v±5%		
1	35	35	35	Read 12vPoC_M	Check VM#3 table		10ms max
1	36			Read USS_12V_M	Check VM#3 table		5ms max
1	37			Assert USS.3v3_LSS_EN	High		
1	38			Read USS.3v3_LSS_STATUS	High		
1			36	Assert MCU_CTL.I2B_5V_EN output	High		
1			37	Read 5V_AUD1	5V±5%		5ms
1	39	38	38	Clear fault interrupt of 0x37 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2			
1	40	39	39	Clear fault interrupt of 0x36 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2			
1	41	40	40	Clear fault interrupt of 0x34 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2			
1	42	41	41	Clear fault interrupt of 0x35 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2			
1	43	42	42	Clear fault interrupt of 0x34 address of MCU_IF.3v3_I2C4 using INT_SCR1 and INT_SCR2			
1	44	43	43	Read SAFETY_SIG.VMs_3v3_INT	High		

[Draft]

6.3 Final

SM-39335 - Final

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/Dependency	Delay
				Power, clocks, bootstraps			
2	0		0	Read MCU_CTL.3v3_ENC_PMIC_RST_OUT	High	After completion of seq 2	100ms max
2	1	1	1	Assert MCU_CTL.3v3_EQ0_CLKs_EN output	Low		
2	2	2	2	Assert MCU_CTL.3v3_EQ1_CLKs_EN output	Low		
2	3	3	3	Assert MCU_CTL.3v3_VDIs_CLKs_EN output	Low		
2	4	4	4	Assert MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN output	Low		
2	5			Assert MCU_CTL.3v3_USS_CLK_EN output	High		
2	6	6	6	Assert EQ_IOs.EQ0_3v3_XINT output	High		
2	7	7	7	Assert EQ_IOs.EQ1_3v3_XINT output	High		
2	8	8	8	Assert EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM output	Low		
2	9	9	9	Assert EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM output	Low		
2	10	10	10	Assert EQ_IOs.EQ0_3v3_GPIO_A8_UBOOT output	Low		
2	11	11	11	Assert EQ_IOs.EQ1_3v3_GPIO_A8_UBOOT output	Low		
2	12		12	Assert ENC.3v3_MCU_BOOTMODE3 output	High		
2	13		13	Assert ENC.3v3_MCU_BOOTMODE4 output	High		
2	14		14	Assert ENC.3v3_MCU_BOOTMODE5 output	Low		
2	15		15	Assert ENC.1v8_SoC_BOOTMODE0 output	Low		
2	16		16	Assert ENC.3v3_SoC_BOOTMODE4 output	High		
2	17		17	Assert ENC.3v3_SoC_BOOTMODE5 output	Low		
2	18		18	Assert ENC.3v3_SoC_BOOTMODE6 output	High		
2	19			Assert ENC.1v8_MCU_TIMESYNC0 output High for Application mode Assert ENC.1v8_MCU_TIMESYNC0 output Low for FBL mode	High/ Low		
				Daisy chain check after EQ PWR enabled			
2	20	20	20	Assert SAFETY_SIG.DC_EQs_IN output	Low		1ms
2	21	21	21	Read SAFETY_SIG.DC_EQ1_OUT	Low		
2	22	22	22	Assert SAFETY_SIG.DC_EQs_IN output	High		1ms
2	23	23	23	Read SAFETY_SIG.DC_EQ1_OUT	High		
				Ready for deasserting reset signals			
2	24	24	24	Assert EN_RESET output	Low		1ms
2	25	25	25	Assert EQ_IOs.EQ0_3v3_POR output	High		1ms
2	26	26	26	Assert EQ_IOs.EQ0_3v3_FCMU_RST output	High		

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/Dependency	Delay
2	27	27	27	Assert EQ_IOs.EQ1_3v3_POR output	High		1ms
2	28	28	28	Assert EQ_IOs.EQ1_3v3_FCMU_RST output	High		
2	29	29	29	Assert ENC.1v8_PORZ output	High		
2	30	30	30	Assert ENC.1v8_MCU_PORZ output	High		1ms
2	31	31	31	Assert ENC.3v3_RESET_REQZ output	High		
2	32	32	32	Assert ENC.3v3_MCU_RESETZ output	High		1ms
2	33		33	Set below pins as input ENC.1v8_SoC_BOOTMODE0 ENC.3v3_SoC_BOOTMODE4 ENC.3v3_SoC_BOOTMODE5 ENC.3v3_SoC_BOOTMODE6 ENC.3v3_MCU_BOOTMODE3 ENC.3v3_MCU_BOOTMODE4 ENC.3v3_MCU_BOOTMODE5			20ms
2	34	34	34	Read TEMP_SENS_ALERT	High		
2	35	35	35	If DEBUG.3v3_DETECT =	High	SV62: Execute Step 36 to 41	
2	35	35	35	If DEBUG.3v3_DETECT =	Low	SV62: Skip step 36 to 41	
2	36	36	36	Assert ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	High	NA for Production	
2	37	37	37	Assert ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	High	NA for Production	
2	38	38	38	Assert ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	High	NA for Production	
2	39	39	39	Assert ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	Low	NA for Production	
2	40	40	40	Assert ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	Low	NA for Production	
2	41	41	41	Assert ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	Low	NA for Production	
2	42	42	42	configure pin ENC.3v3_UART_TX_R to UART			
2	43	43	43	configure pin ENC.3v3_UART_RX_R to UART			
2	44	44	44	Read SAFETY_SIG.PMIC_3v3_SS	High		
				EyeQ6_ 0&1 BIST monitoring			
2	45	45	45	Wait for the BIST occurrence			250ms max
2	46	46	46	Read SAFETY_SIG.EQ0_3v3_NCRERR	Toggle		

Sequence	Step SV62	Step CH63	Step DR64	Action	Status	Operations/Dependency	Delay
2	47	47	47	Read SAFETY_SIG.EQ1_3v3_NCRERR	Toggle		
2	48	48	48	Read SAFETY_SIG.EQ0_3v3_CRERR	Toggle		
2	49	49	49	Read SAFETY_SIG.EQ1_3v3_CRERR	Toggle		
2	50	50		Assert HEATER.3v3_HSS_DIAG_EN	High		

[Draft]

7 System Power down Sequence

7.1 Main Power down sequence

SM-39305 -

Step	IC	RefDes	MCU port	Input Net	Output Net	Status	Action	Delay	Dependency
0							assert high ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	NA for Production	
0							assert high ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	NA for Production	
0							assert high ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	NA for Production	

[Draft]

SM-39318 - Main

Step SV62	Step CH63	Step DR64	Action	Condition to go next step	wait time to go next step
		1	Assert Low MCU_CTL.I2B_5V_EN		
1	1		Assert Low HEATER.3v3_HSS_DIAG_EN		10ms
2	2		Assert Low HEATER.3v3_HSS_EN		10ms
3	3		Read HEATER.HSS_MON	read volt on HSS, read value <=0.5V	
	4	2	Assert High EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ		10ms
	5	3	Assert High EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ		10ms
	6	4	wait for EYEQ6_1 graceful shutdown complete		< 10s
	7	5	wait for EYEQ6_0 graceful shutdown complete		
	8	6	Wait for EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK	ACK == high	
	9	7	Wait for EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK	ACK == high	
4	10	8	Assert low EQ_IOs.EQ1_3v3_FCMU_RST		
5	11	9	Assert low EQ_IOs.EQ0_3v3_FCMU_RST		
6	12	10	Assert low ENC.3v3_RESET_REQZ		
7	13		Assert low MCU_CTL.3v3_USS_CLK_EN		

Step SV62	Step CH63	Step DR64	Action	Condition to go next step	wait time to go next step
				EQs are OFF, TDA4 after Shutdown Request	
8	14	11	Assert EN_RESET output Low	Low	
			Assert Low EQ0_POR_n	Internal HW step	
			Assert Low EQ_IOs.EQ1_3v3_POR	Internal HW step	
			Assert Low ENC.1v8_PORZ	Internal HW step	
			Assert Low ENC.1v8_MCU_PORZ	Internal HW step	
			Assert Low EQ_IOs.EQ0_3v3_GPIO_A4_TIMER1_CLK_PDB_0	Internal HW step	
			Assert Low PDB_DES_96793	Internal HW step	
			Assert Low ENC.3v3_SER_PDB	Internal HW step	
			Assert Low EQ_IOs.EQ0_1v8_GPIO_D4_PDB_3	Internal HW step	
			Assert Low EQ_IOs.EQ1_3v3_GPIO_A4_TIMER1_CLK_PDB_0	Internal HW step	
9	15	12	Assert high MCU_CTL.3v3_EQ1_CLKs_EN	disable = high impedance /high	10ms
10	16	13	Assert high MCU_CTL.3v3_EQ0_CLKs_EN	disable = high impedance /high	10ms
11	17	14	Assert high MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN	disable = high impedance /high	10ms
12	18	15	Assert high MCU_CTL.3v3_VDIs_CLKs_EN	disable = high impedance /high	10ms
13	19	16	Assert High MCU_CTL.power_EN_n		5ms
14	20	17	Read EQ0_1v8_IP	read value <=0.8V	
15	21	18	Read EQ0_1V8_PCIe	read value <=0.8V	
16	22	19	Read EQ0_0V86	read value <=0.3V	
17	23	20	Read EQ0_0V75	read value <=0.3V	
18	24	21	Read EQ1_1v8_IP	read value <=0.8V	
19	25	22	Read EQ1_1V8_PCIe	read value <=0.8V	
20	26	23	Read EQ1_0V86	read value <=0.3V	
21	27	24	Read EQ1_0V75	read value <=0.3V	
22	28	25	Read EQ0_VDD_CORE	read value <=0.2V	
23	29	26	Read EQ0_1V05	read value <=0.3V	
24	30	27	Read EQ0_VDDQ	read value <=0.3V	
25	31	28	Read EQ1_VDD_CORE	read value <=0.2V	
26	32	29	Read EQ1_1V05	read value <=0.3V	
27	33	30	Read EQ1_VDDQ	read value <=0.3V	
28		31	Read TDA4_0v8_VDD_CORE	read value <=0.5V	

Step SV62	Step CH63	Step DR64	Action	Condition to go next step	wait time to go next step
29		32	Read TDA4_0v8_CPU_AVS	read value <=0.5V	
30		33	Read TDA4_1v1	read value <=0.5V	
31		34	Read TDA4_0v85	read value <=0.5V	
32		35	Read TDA4_1v8_PHY_LDO	read value <=0.8V	
33		36	Read TDA4_0v8_DLL_LDO	read value <=0.5V	
34		37	Read TDA4_1v8_PLL_LDO	read value <=0.8V	
35	35	38	Read DES_1V2	read value <=0.5V	
		39	Read 12vPoC_M	read value <=1V	
36	36		Read USS_12v_M	read value <=2.2V	
			Read 5V	read value <=1V	
			Read 3v3	read value <=0.3V	
37	37	44	Read 1v8 using U93	read value <=1V	5ms max
38			Assert LOW USS.3v3_LSS_EN		
39	38	45	Assert low VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN		30ms
40	39	46	Assert low MCU_CTL.SW1_RST		
41	40	47	Assert low MCU_CTL.SW0_RST		
42	41	48	Assert low DEBUG.10G_PHY_RST		
43	42	49	Assert High DISCHARGE.ETH_MCU		0ms
44	43	53	Read 0v8_SW1	read value <=0.81V	0ms
45	44	55	Read 0v8_SW0	read value <=0.81V	0ms
46	45	57	Assert HIGH -VR5510 PMIC bit **PWRON1DIS** Register M_MODE - Address 0x01 PMIC main address : 0x20 I2C bus #4	disable wake signal	
47	46	58	Assert HIGH -VR5510 PMIC bit **GOTO_OFF** Register M_SM_CTRL1 - Address 0x02 PMIC main address : 0x20 I2C bus #4	shutdown PMIC	

[Draft]

7.2 Degraded mode power down sequence

SM-39322 - Degraded Mode

Step SV62	Step CH63	Step DR64	Action	Condition to go next step	wait time to go next step
0	0	0	Assert low MCU_CTL.SW1_RST		
1	1	1	Assert low MCU_CTL.SW0_RST		
2	2	2	Assert low DEBUG.10G_PHY_RST		
3	3	3	Assert High DISCHARGE.ETH_MCU		0ms
4	4	4	Read 0v8_SW1	read value <=0.81V	0ms
5	5	5	Read 0v8_SW0	read value <=0.81V	0ms
6	6	6	Assert HIGH -VR5510 PMIC bit **PWRON1DIS** Register M_MODE - Address 0x01 PMIC main address : 0x20 I2C bus #4	disable wake signal	
7	7	7	Assert HIGH -VR5510 PMIC bit **GOTO_OFF** Register M_SM_CTRL1 - Address 0x02 PMIC main address : 0x20 I2C bus #4	shutdown PMIC	

[Draft]

8 Reset requirement

8.1 Resetting of TDA4

SM-39311 - While performing the TDA4 reset below step are done

Step	MCU port	Output Net	Status	Action	Required delay before next step
0				Set below pin as input ENC.3v3_UART_RX_R Set below pin as input ENC.3v3_UART_TX_R	
1	I²C4 U103 P00	ENC.3v3_MCU_BOOTMODE3	High	Encoder MCU Bootmode 3 set	
2	I²C4 U103 P01	ENC.3v3_MCU_BOOTMODE4	High	Encoder MCU Bootmode 4 set	
3	I²C4 U103 P02	ENC.3v3_MCU_BOOTMODE5	Low	Encoder MCU Bootmode 5 set	
4	I²C4 U103 P03	ENC.3v3_SoC_BOOTMODE4	High	Encoder SoC Bootmode 4 set	
5	I²C4 U103 P04	ENC.3v3_SoC_BOOTMODE5	Low	Encoder SoC Bootmode 5 set	
6	PF04	ENC.1v8_SoC_BOOTMODE0	Low	Encoder SoC Bootmode 0 set	
7	I²C4 U103 P14	ENC.3v3_SoC_BOOTMODE6	High	Encoder SoC Bootmode 6 set	
8	PE01	ENC.1v8_PORZ	Low	Assert low ENC.1v8_PORZ	
9	PH02	ENC.1v8_MCU_PORZ	Low	Assert low ENC.1v8_MCU_PORZ	1ms
10	I²C4 U103 P06	ENC.3v3_RESET_REQZ	Low	Assert low ENC.3v3_RESET_REQZ	
11	I²C4 U103 P07	ENC.3v3_MCU_RESETZ	Low	Assert low ENC.3v3_MCU_RESETZ	1ms
12	PE01	ENC.1v8_PORZ	High	Assert high ENC.1v8_PORZ	
13	PH02	ENC.1v8_MCU_PORZ	High	Assert high ENC.1v8_MCU_PORZ	1ms
14	I²C4 U103 P06	ENC.3v3_RESET_REQZ	High	Assert high ENC.3v3_RESET_REQZ	
15	I²C4 U103 P07	ENC.3v3_MCU_RESETZ	High	Assert high ENC.3v3_MCU_RESETZ	1ms
16				Set below pins as input ENC.1v8_SoC_BOOTMODE0 ENC.3v3_SoC_BOOTMODE4 ENC.3v3_SoC_BOOTMODE5 ENC.3v3_MCU_BOOTMODE3 ENC.3v3_MCU_BOOTMODE4 ENC.3v3_MCU_BOOTMODE5 ENC.3v3_SoC_BOOTMODE6	20ms
17				Set below pins as UART ENC.3v3_UART_RX_R ENC.3v3_UART_TX_R	

[Draft]

8.2 Resetting of EyeQs

SM-80457 - While performing the EyeQ reset below step are done

Step	Output Net	Action	Required delay before next step
0	EQ_IOs.EQ0_3v3_FCMU_RST	Assert Low EQ_IOs.EQ0_3v3_FCMU_RST	
1	EQ_IOs.EQ0_3v3_POR	Assert Low EQ_IOs.EQ0_3v3_POR	
2	EQ_IOs.EQ1_3v3_FCMU_RST	Assert Low EQ_IOs.EQ1_3v3_FCMU_RST	1ms
3	EQ_IOs.EQ1_3v3_POR	Assert Low EQ_IOs.EQ1_3v3_POR	6ms
4	EQ_IOs.EQ0_3v3_POR	Assert High EQ_IOs.EQ0_3v3_POR	
5	EQ_IOs.EQ0_3v3_FCMU_RST	Assert High EQ_IOs.EQ0_3v3_FCMU_RST	
6	EQ_IOs.EQ1_3v3_POR	Assert High EQ_IOs.EQ1_3v3_POR	
7	EQ_IOs.EQ1_3v3_FCMU_RST	Assert High EQ_IOs.EQ1_3v3_FCMU_RST	

[Draft **]**

8.3 Resetting of Ethernet Switch

SM-80458 - While performing the Ethernet Switch reset below step are done

Step	Output Net	Action	Required delay before next step
0	MCU_CTL.SW1_RST	Assert low MCU_CTL.SW1_RST	
1	MCU_CTL.SW0_RST	Assert low MCU_CTL.SW0_RST	1ms
2	MCU_CTL.SW1_RST	Assert High MCU_CTL.SW1_RST	
3	MCU_CTL.SW0_RST	Assert High MCU_CTL.SW0_RST	

[Draft **]**

9 Debug vs. Production

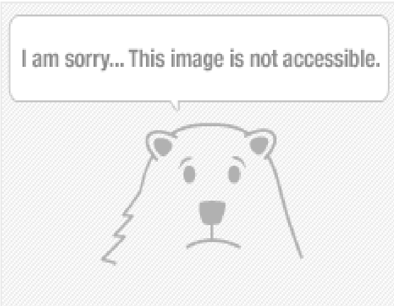
SM-39315 - Detailed MCU block changes between debug and production PCBA configurations. **[** Draft **]**

9.1 Change #1

SM-39369 - Circuit purpose: to put PMIC in debug mode while programming the MCU
Status in production: should be fully removed from BOM. Placeholders on pcb remain.
Implicated activities: **[** Draft **]**

SM-39372 - ATE: update Jig to supply 5V to PMIC while programming MCU **[** Draft **]**

SM-39363 - SW: support MCU FW updated (MCU FW should disable WD while upgrade)

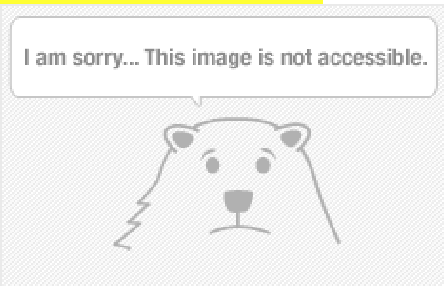


[Draft **]**

9.2 Change #2

SM-39366 - Circuit purpose: remove MCU Jtag & Aurora debug connector.
Status in production: should be fully removed from BOM. Placeholders on pcb remain.
Implicated activities:
ATE: N/A
SW: MCU FW should support SW debug (rather than HW based)

Change picture? and also add J16



[Draft **]**

9.3 Change #3

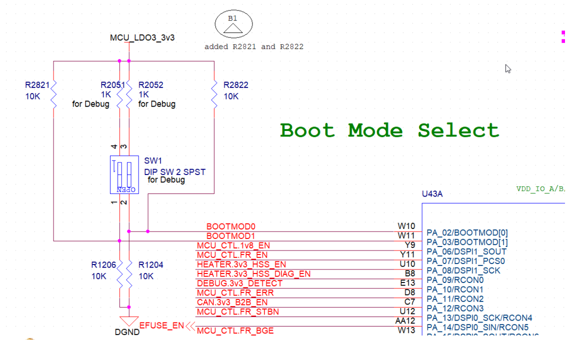
SM-39376 - Circuit purpose: MCU strapping mode
Status in production: should be fully removed from BOM. Placeholders on pcb remain.
Implicated activities: **[** Draft **]**

SM-39340 - ATE: Jig should assert Bootmode 0/1 to UART mode. MCU OTP should be fused to the required bootstrap mode
[Draft **]**

SM-39342 -

1. SW: MCU OTP fused procedure

2. Debug: can assemble & used onboard bypass resistors



[Draft]

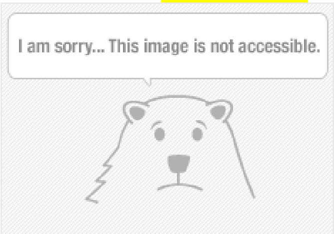
9.4 Change #6

SM-39356 - Circuit purpose: PMIC debug

Status in production: should be fully removed from BOM. Placeholders on pcb remain.

Implicated activities: None. Not in use.

PreDV direction: Can be removed



[Draft]

9.5 Change #7

Circuit purpose: SW5 debug Wake up

Status in production: should be fully removed from BOM. Placeholders on pcb remain.

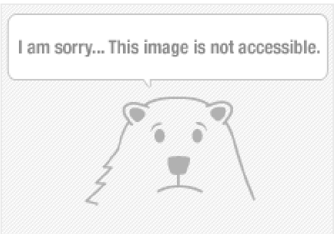
9.6 Change 11

SM-39360 - Circuit purpose: PMIC programming debug

Status in production: should be fully removed from BOM. Placeholders on pcb remain.

Implicated activities: none. Not in use.

PreDV direction: can be removed.



[Draft]

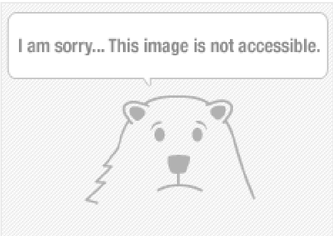
9.7 Change #15

SM-39348 - Circuit purpose: eFUSE debug

Status in production: should be fully removed from BOM. Placeholders on pcb remain.

Implicated activities: none. Not in use.

PreDV direction: can be removed.



[Draft]

10 Diagnostics

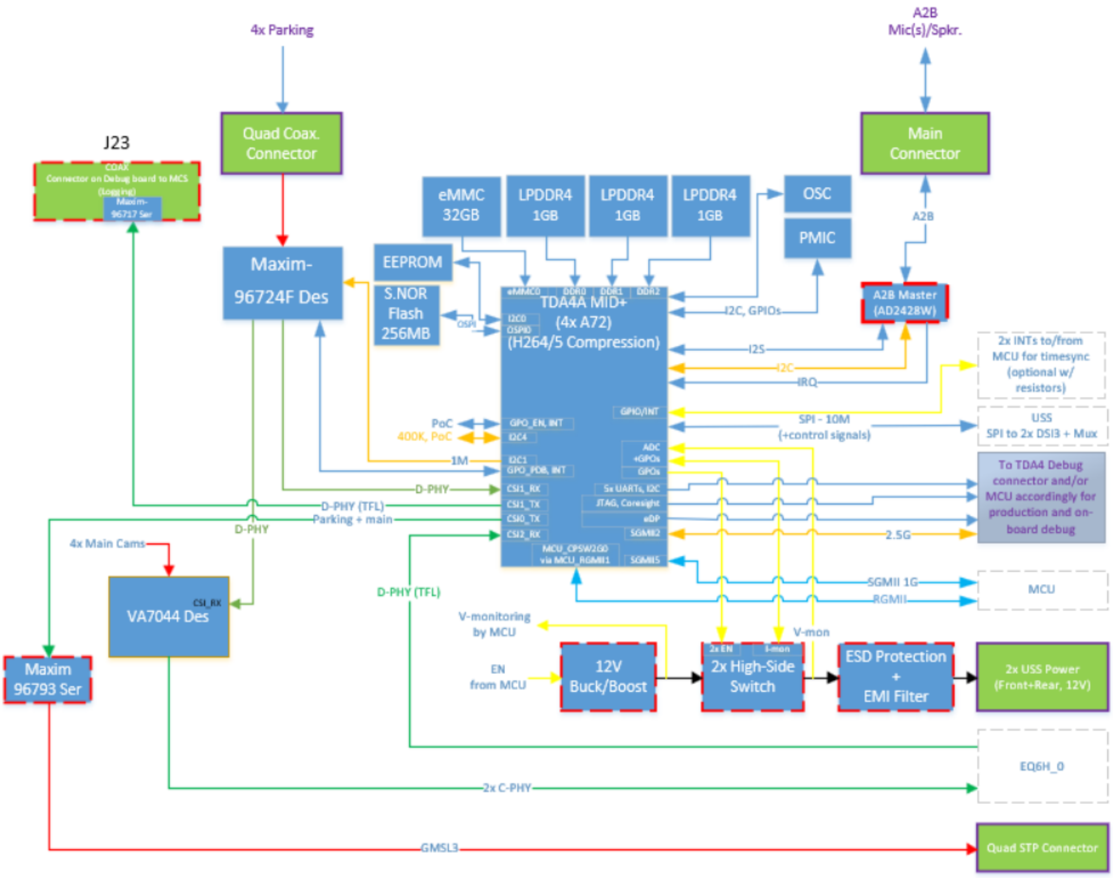
SM-39352 - All diagnostics capabilities are covered in *SuppliersManagement/Primary Board HW Diagnostics/HWDIAG* **[** Draft **]**

11 TDA4 HSI

11.1 TDA4 Block Diagram

11.1.1 TDA4 Block Diagram:

SM-39251 -



[
Draft]

11.1.2 Description

SM-39254 - TDA4 MID+ is a SOC responsible for all parking applications including:

- Video encoding for front 2MP Input of TFL.
- forwarding of raw images from front 2MP camera to debug board (AKA loop-back).
- Configuring Maxim DeSer.
- Flashing Boot Loader (FBL) Slave for SWDL flow.
- Streaming TV cameras to HMI.
- Sending vehicle USS based notification/alerts.
- Forwarding USS data to EQs.
- Run SWC in charge of forwarding video from TDA to Vehicle.
- Notify of HW and/or SW errors to TDA4.

[Draft]

11.1.3 TDA4 Peripherals Description

SM-39245 -

- 3x1GB LPDDR memory
- 32GB eMMC memory
- 256MB NOR Flash memory over OSPI
- 12 USS - SPI to 2x DSI3 + Mux
- Maxim 96724F Des of the 4 parking cameras.
- MAX96793 Ser for forwarding the parking cameras to the EQ.
- Maxim 96717 Ser for TFL debugging.
- D-PHY for providing the Top View Cameras.
- D-PHY to receive Image sharing for compression.
- ME debug connector via CSI.
- Communication with the TDA4 over 2.5G SGMII and RGMII
- PMIC – over I2C
- EEPROM
- Oscillator

[Draft]

11.1.4 Memories

11.1.4.1 P.Ns List

SM-39248 -

Device		P.N	Manuf.	Ref. Des.	Description
LPDDR4		MT53E256M32D1KS-046 AAT:L	Micron	U130, U131, U159	1GB
NOR Flash	Main	MT35XU02GCBA1G12-0AAT	Micron	U132	256MB
eMMC		MTFC32GBCAQTC-AAT	Micron	U129	32GB

[Draft]

11.1.4.2 NOR Flash

SM-39226 -

			Device ID [h]			
Device	P.N	MNF ID [h]	Byte2	Byte3	Sector Size	no. of Sectors
NOR Flash	MT35XU02GCBA1G12-0AAT	2C	5B	19	128KB	2048

[Draft]

11.1.4.3 eMMC

SM-39219 -

Device	P.N	MNF ID [h]	Byte6	Sector Size	no. of Sectors
eMMC	MTFC32GBCAQTC-AAT		4C	4KB	8M

[Draft]

11.1.4.4 LPDDR4

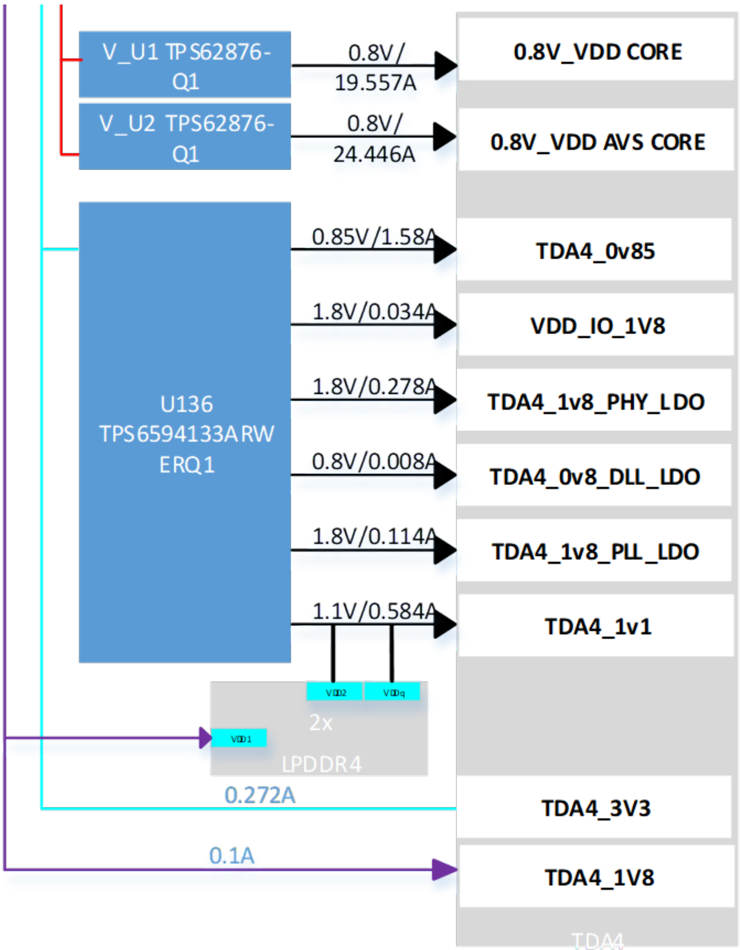
SM-39223 -

Device	P.N	MNF ID [h]	Rev ID [h]	Page Size [bytes]
LPDDR4	MT53E256M32D1KS-046 AAT:L	FF	03	2048

[Draft]

11.2 TDA4 Power

11.2.1 TDA4 Power tree diagram



SM-39238 - [Draft]

11.2.2 TDA4+ power tree description

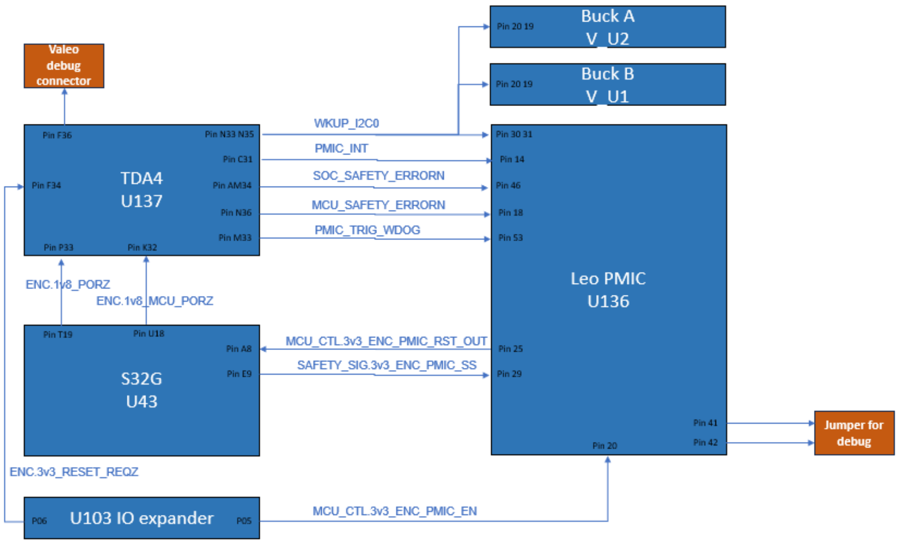
11.2.2.1 TDA4 has one power supplies:

SM-39242 -

- PMIC – IC PMIC, 4 LDO, 5 BUCK, I2C/SPI, Vin=3-5.5V, VQFN-56, AEC-Q100
    - MPN: TPS6594133ARWERQ1; Texas Instrument

[Draft]

11.2.3 TDA4 PMIC Connectivity



SM-39230 - [Draft]

11.2.4 PMIC

11.2.4.1 Configuration

SM-39234 -

- MPN: TPS6594133ARWERQ1; Texas Instrument

Mobileye shall configure the PMIC hardware through OTP based on the specified thresholds and values below.

Description	OTP Configuration Block	OTP Configuration Data
Mobileye shall configure the PMIC hardware through OTP to enable BUCK1 regulator that provides TDA4_1v1 supply to the TDA4 SOC.	BUCK1 Monitor Enable	BUCK1_CTRL.BUCK1_EN = 1 BUCK1_CTRL.BUCK1_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for BUCK1 regulator that provides TDA4_1v1 supply to the TDA4 SOC.	BUCK1 UV Configuration	BUCK1_PG_WINDOW.BUCK1_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for BUCK1 regulator that provides TDA4_1v1 supply to the TDA4 SOC.	BUCK1 OV Configuration	BUCK1_PG_WINDOW.BUCK1_OV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the BUCK1 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	BUCK1 - SOC Rail Group selection	RAIL_SEL_1.BUCK1_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable BUCK3 regulator that provides TDA4_0v85 supply to the TDA4 SOC.	BUCK3 Monitor Enable	BUCK3_CTRL.BUCK3_EN = 1 BUCK3_CTRL.BUCK3_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for BUCK3 regulator that provides TDA4_0v85 supply to the TDA4 SOC.	BUCK3 UV Configuration	BUCK3_PG_WINDOW.BUCK3_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for BUCK3 regulator that provides TDA4_0v85 supply to the TDA4 SOC.	BUCK3 OV Configuration	BUCK3_PG_WINDOW.BUCK3_OV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the BUCK3 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	BUCK3 - SOC Rail Group selection	RAIL_SEL_1.BUCK3_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable BUCK4 regulator that provides VDD_IO_1V8 supply to the TDA4 SOC.	BUCK4 Monitor Enable	BUCK4_CTRL.BUCK4_EN = 1 BUCK4_CTRL.BUCK4_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for BUCK4 regulator that provides VDD_IO_1V8 supply to the TDA4 SOC.	BUCK4 UV Configuration	BUCK4_PG_WINDOW.BUCK3_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for BUCK4 regulator that provides VDD_IO_1V8 supply to the TDA4 SOC.	BUCK4 OV Configuration	BUCK4_PG_WINDOW.BUCK3_OV_THR = 0x3

Description	OTP Configuration Block	OTP Configuration Data
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the BUCK4 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	BUCK4 - SOC Rail Group selection	RAIL_SEL_1.BUCK4_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable LDO1 regulator that provides TDA4_1v8_PHY_LDO supply to the TDA4 SOC.	LDO1 Monitor Enable	LDO1_CTRL.LDO1_EN = 1 LDO1_CTRL.LDO1_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for LDO1 regulator that provides TDA4_1v8_PHY_LDO supply to the TDA4 SOC.	LDO1 UV Configuration	LDO1_PG_WINDOW.LDO1_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for LDO1 regulator that provides TDA4_1v8_PHY_LDO supply to the TDA4 SOC.	LDO1 OV Configuration	LDO1_PG_WINDOW.LDO1_OV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the LDO1 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	LDO1 - SOC Rail Group selection	RAIL_SEL_2.LDO1_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable LDO3 regulator that provides TDA4_0v8_DLL_LDO supply to the TDA4 SOC.	LDO3 Monitor Enable	LDO3_CTRL.LDO3_EN = 1 LDO3_CTRL.LDO3_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for LDO3 regulator that provides TDA4_0v8_DLL_LDO supply to the TDA4 SOC.	LDO3 UV Configuration	LDO3_PG_WINDOW.LDO3_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for LDO3 regulator that provides TDA4_0v8_DLL_LDO supply to the TDA4 SOC.	LDO3 OV Configuration	LDO3_PG_WINDOW.LDO3_OV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the LDO3 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	LDO3 - SOC Rail Group selection	RAIL_SEL_2.LDO3_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable LDO4 regulator that provides TDA4_1v8_PLL_LDO supply to the TDA4 SOC.	LDO4 Monitor Enable	LDO4_CTRL.LDO4_EN = 1 LDO4_CTRL.LDO4_VMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the under-voltage threshold for LDO4 regulator that provides TDA4_1v8_PLL_LDO supply to the TDA4 SOC.	LDO4 UV Configuration	LDO4_PG_WINDOW.LDO4_UV_THR = 0x3
Mobileye shall configure the PMIC hardware through OTP to set the over-voltage threshold for LDO4 regulator that provides TDA4_1v8_PLL_LDO supply to the TDA4 SOC.	LDO4 OV Configuration	LDO4_PG_WINDOW.LDO4_OV_THR = 0x3

Description	OTP Configuration Block	OTP Configuration Data
Mobileye shall configure the PMIC hardware through OTP to select the SoC rail group for the LDO4 regulator to handle under-voltage and over-voltage errors. Note: for a detected voltage error on an output rail in the SoC rail group, the PMIC disables all output rails for the SoC, resets the SoC, and pulls the nINT pin low.	LDO4 - SOC Rail Group selection	RAIL_SEL_2.LDO4_GRP_SEL = 2 (SOC rail group)
Mobileye shall configure the PMIC hardware through OTP to enable the Watchdog .	Watchdog	WD_THR_CFG.WD_EN = 1
Mobileye shall configure the PMIC hardware through OTP to set the watchdog long window default value	Watchdog	WD_LONGWIN_CFG.WD_LONGWIN = 0xFF
Mobileye shall configure the PMIC hardware through OTP to select TRIGGER_MODE in the Watchdog.	Watchdog	GPIO11_CONF.GPIO11_SEL = 1 (TRIG_WDOG)
Mobileye shall configure the PMIC hardware through OTP to enable watchdog-reset function to perform Warm reset when WD_FAIL_CNT[3:0] > (WD_FAIL_TH[2:0] + WD_RST_TH[2:0]).	Watchdog	WD_THR_CFG.WD_RST_EN = 1
Mobileye shall configure the PMIC hardware through OTP to enable the BIST.	BIST	LBIST Enabled by default For ABIST, VMON_ABIST_EN= 1
Mobileye shall configure the PMIC hardware through OTP to enable the Thermal Warning and Shutdown safety mechanism	Thermal monitor	MISC_CTRL.LPM_EN = 0
Mobileye shall configure the PMIC hardware through OTP to set the Thermal Warning threshold.	Thermal monitor	CONFIG_1.TWARN_LEVEL = 0
Mobileye shall configure the PMIC hardware through OTP to set the Orderly Shutdown threshold.	Thermal monitor	CONFIG_1.TSD_ORD_LEVEL = 0
Mobileye shall configure the PMIC hardware through OTP to enable the Clock Monitoring safety mechanism.	Clock monitor	MISC_CTRL.CLKMON_EN = 1
Mobileye shall configure the PMIC hardware through OTP to enable the CRC on I2C1 interface.	CRC on I2C	I2C1_SPI_CRC_EN = 1
Mobileye shall configure the PMIC hardware through OTP to enable the CRC on I2C2 interface.	CRC on I2C	I2C2_CRC_EN = 1
Mobileye shall configure the PMIC hardware through OTP to enable the CRC on Register Map.	CRC on Register Map	REG_CRC_EN = 1
Mobileye shall configure the PMIC hardware through OTP to enable the MCU Error Signal Monitor.	ESM	ESM_MCU_MODE_CFG.ESM_MCU_EN = 1
Mobileye shall configure the PMIC hardware through OTP to select the Level mode in the MCU Error Signal Monitor.	ESM	ESM_MCU_MODE_CFG.ESM_MCU_MODE = 0
Mobileye shall configure the PMIC hardware through OTP to enable the SOC Error Signal Monitor.	ESM	ESM_SOC_MODE_CFG.ESM_SOC_EN = 1
Mobileye shall configure the PMIC hardware through OTP to select the Level mode in the SOC Error Signal Monitor.	ESM	ESM_SOC_MODE_CFG.ESM_SOC_MODE = 0
Mobileye shall configure the PMIC hardware through OTP to enable the Residual Voltage Check for BUCK1, BUCK2,BUCK3 and BUCK4 regulators	Residual Voltage Check	BUCK1_CTRL.BUCK1_RV_SEL = 1 BUCK2_CTRL.BUCK2_RV_SEL = 1 BUCK3_CTRL.BUCK3_RV_SEL = 1 BUCK4_CTRL.BUCK4_RV_SEL = 1

Description	OTP Configuration Block	OTP Configuration Data
Mobileye shall configure the PMIC hardware through OTP to enable the Residual Voltage Check for LDO1, LDO3 and LDO4 regulators	Residual Voltage Check	LDO1_CTRL.LDO1_RV_SEL = 1 LDO3_CTRL.LDO3_RV_SEL = 1 LDO4_CTRL.LDO4_RV_SEL = 1
Mobileye shall configure the PMIC hardware through OTP to set the Threshold voltage for Short Circuit and Residual Voltage Detection. Note: Information(Text) about the Threshold voltage for Short Circuit and Residual Voltage Detection available in the PMIC datasheet chapter 8.3.3 Parameter VTH_SC_RV - Threshold voltage for Short Circuit and Residual Voltage Detection Min 140mV, max 160mV	Residual Voltage Check	VTH_SC_RV = Min 140, max 160
Mobileye shall configure the PMIC hardware through OTP to enable the [Redundant Read-Back on Logic Output pins] safety mechanism for nINT pin.	Redundant Read-Back on Logic Output pins	MASK_MODERATE_ERR.NINT_READBACK_MASK = 0
Mobileye shall configure the PMIC hardware through OTP to enable the [Redundant Read-Back on Logic Output pins] safety mechanism for nRSTOUT pin.	Redundant Read-Back on Logic Output pins	MASK_MODERATE_ERR.NRSTOUT_READBACK_MASK = 0
Mobileye shall configure the PMIC hardware through OTP to enable the [Redundant Read-Back on Logic Output pins] safety mechanism for EN_DRV pin.	Redundant Read-Back on Logic Output pins	EN_DRV_READBACK_MASK = 0

[Draft]

11.2.4.2 Thresholds

11.2.4.2.1 Bucks

11.2.4.2.1.1 TDA4_1v1 (Buck1)

SM-39295 - BUCK1_PG_WINDOW Register (Offset = 18h) [Reset = 00h]

Register Name	Field Name	TPS6594	
		Value	Description
BUCK1_PG_WINDOW	BUCK1_OV_THR	0x3	+5% / +50 mV
	BUCK1_UV_THR	0x3	-5% / -50 mV

Figure 8-83. BUCK1_PG_WINDOW Register

7	6	5	4	3	2	1	0
RESERVED		BUCK1_UV_THR			BUCK1_OV_THR		
R/W-0h		R/W-0h			R/W-0h		

Table 8-48. BUCK1_PG_WINDOW Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	RESERVED	R/W	0h	
5-3	BUCK1_UV_THR	R/W	0h	Powergood low threshold level for BUCK1: (Default from NVM memory) 0h = -3% / -30mV 1h = -3.5% / -35 mV 2h = -4% / -40 mV 3h = -5% / -50 mV 4h = -6% / -60 mV 5h = -7% / -70 mV 6h = -8% / -80 mV 7h = -10% / -100mV
2-0	BUCK1_OV_THR	R/W	0h	Powergood high threshold level for BUCK1: (Default from NVM memory) 0h = +3% / +30mV 1h = +3.5% / +35 mV 2h = +4% / +40 mV 3h = +5% / +50 mV 4h = +6% / +60 mV 5h = +7% / +70 mV 6h = +8% / +80 mV 7h = +10% / +100mV

[Draft]

11.2.4.2.1.2 TDA4_0v85 (BUCK3)

SM-39298 - BUCK3_PG_WINDOW Register (Offset = 1Ah) [Reset = 00h]

Register Name	Field Name	TPS6594	
		Value	Description
BUCK3_PG_WINDOW	BUCK3_OV_THR	0x3	+5% / +50 mV
	BUCK3_UV_THR	0x3	-5% / -50 mV

Figure 8-85. BUCK3_PG_WINDOW Register

7	6	5	4	3	2	1	0
RESERVED		BUCK3_UV_THR			BUCK3_OV_THR		
R/W-0h		R/W-0h			R/W-0h		

Table 8-50. BUCK3_PG_WINDOW Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	RESERVED	R/W	0h	
5-3	BUCK3_UV_THR	R/W	0h	Powergood low threshold level for BUCK3: (Default from NVM memory) 0h = -3% / -30mV 1h = -3.5% / -35 mV 2h = -4% / -40 mV 3h = -5% / -50 mV 4h = -6% / -60 mV 5h = -7% / -70 mV 6h = -8% / -80 mV 7h = -10% / -100mV
2-0	BUCK3_OV_THR	R/W	0h	Powergood high threshold level for BUCK3: (Default from NVM memory) 0h = +3% / +30mV 1h = +3.5% / +35 mV 2h = +4% / +40 mV 3h = +5% / +50 mV 4h = +6% / +60 mV 5h = +7% / +70 mV 6h = +8% / +80 mV 7h = +10% / +100mV

[Draft]

11.2.4.2.1.3 VDD_IO_1V8 (BUCK4)

SM-39290 - BUCK4_PG_WINDOW Register (Offset = 1Bh) [Reset = 00h]

Register Name	Field Name	TPS6594	
		Value	Description
BUCK4_PG_WINDOW	BUCK4_OV_THR	0x3	+5% / +50 mV
	BUCK4_UV_THR	0x3	-5% / -50 mV

Figure 8-86. BUCK4_PG_WINDOW Register

7	6	5	4	3	2	1	0
RESERVED		BUCK4_UV_THR			BUCK4_OV_THR		
R/W-0h		R/W-0h			R/W-0h		

Table 8-51. BUCK4_PG_WINDOW Register Field Descriptions

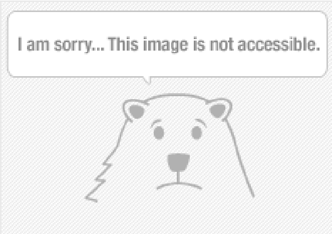
Bit	Field	Type	Reset	Description
7-6	RESERVED	R/W	0h	
5-3	BUCK4_UV_THR	R/W	0h	Powergood low threshold level for BUCK4: (Default from NVM memory) 0h = -3% / -30mV 1h = -3.5% / -35 mV 2h = -4% / -40 mV 3h = -5% / -50 mV 4h = -6% / -60 mV 5h = -7% / -70 mV 6h = -8% / -80 mV 7h = -10% / -100mV
2-0	BUCK4_OV_THR	R/W	0h	Powergood high threshold level for BUCK4: (Default from NVM memory) 0h = +3% / +30mV 1h = +3.5% / +35 mV 2h = +4% / +40 mV 3h = +5% / +50 mV 4h = +6% / +60 mV 5h = +7% / +70 mV 6h = +8% / +80 mV 7h = +10% / +100mV

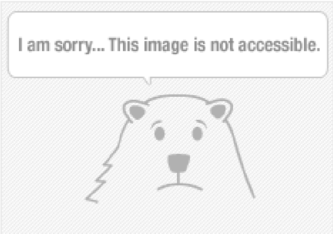
[Draft]

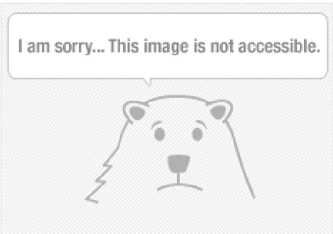
11.2.4.2.2 LDOs

11.2.4.2.2.1 TDA4_1v8_PHY_LDO (LDO1)

SM-39268 -







[Draft]

11.2.4.2.2.2 TDA4_0v8_DLL_LDO (LDO3)

SM-39271 - LDO3_PG_WINDOW Register (Offset = 29h) [Reset = 00h]

Register Name	Field Name	TPS6594	
		Value	Description
LDO3_PG_WINDOW	LDO3_OV_THR	0x3	+5% / +50 mV
	LDO3_UV_THR	0x3	-5% / -50 mV

Figure 8-99. LDO3_PG_WINDOW Register

7	6	5	4	3	2	1	0
RESERVED		LDO3_UV_THR			LDO3_OV_THR		
R/W-0h		R/W-0h			R/W-0h		

Table 8-64. LDO3_PG_WINDOW Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	RESERVED	R/W	0h	
5-3	LDO3_UV_THR	R/W	0h	Powergood low threshold level for LDO3: (Default from NVM memory) 0h = -3% / -30mV 1h = -3.5% / -35 mV 2h = -4% / -40 mV 3h = -5% / -50 mV 4h = -6% / -60 mV 5h = -7% / -70 mV 6h = -8% / -80 mV 7h = -10% / -100mV
2-0	LDO3_OV_THR	R/W	0h	Powergood high threshold level for LDO3: Default from NVM memory) 0h = +3% / +30mV 1h = +3.5% / +35 mV 2h = +4% / +40 mV 3h = +5% / +50 mV 4h = +6% / +60 mV 5h = +7% / +70 mV 6h = +8% / +80 mV 7h = +10% / +100mV

[Draft]

11.2.4.2.2.3 TDA4_1v8_PLL_LDO (LDO4)

SM-39262 - LDO4_PG_WINDOW Register (Offset = 2Ah) [Reset = 00h]

Register Name	Field Name	TPS6594	
		Value	Description
LDO4_PG_WINDOW	LDO4_OV_THR	0x3	+5% / +50 mV
	LDO4_UV_THR	0x3	-5% / -50 mV

Figure 8-100. LDO4_PG_WINDOW Register

7	6	5	4	3	2	1	0
RESERVED		LDO4_UV_THR			LDO4_OV_THR		
R/W-0h		R/W-0h			R/W-0h		

Table 8-65. LDO4_PG_WINDOW Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	RESERVED	R/W	0h	
5-3	LDO4_UV_THR	R/W	0h	Powergood low threshold level for LDO4: (Default from NVM memory) 0h = -3% / -30mV 1h = -3.5% / -35 mV 2h = -4% / -40 mV 3h = -5% / -50 mV 4h = -6% / -60 mV 5h = -7% / -70 mV 6h = -8% / -80 mV 7h = -10% / -100mV
2-0	LDO4_OV_THR	R/W	0h	Powergood high threshold level for LDO4: (Default from NVM memory) 0h = +3% / +30mV 1h = +3.5% / +35 mV 2h = +4% / +40 mV 3h = +5% / +50 mV 4h = +6% / +60 mV 5h = +7% / +70 mV 6h = +8% / +80 mV 7h = +10% / +100mV

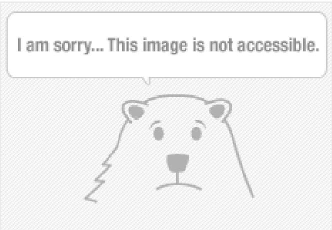
[Draft]

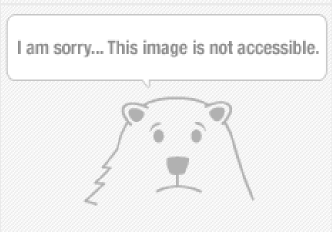
11.2.4.2.3 Watchdog

11.2.4.2.3.1 Watchdog Window Timing threshold

SM-39266 -

WD_LONGWIN_CFG Register (Offset = 405h) [Reset = FFh]





[Draft]

11.2.4.2.3.2 Watchdog Fail threshold

SM-39283 -

WD_THR_CFG Register (Offset = 409h) [Reset = FFh]

Figure 8-252. WD_THR_CFG Register

7	6	5	4	3	2	1	0
WD_RST_EN	WD_EN	WD_FAIL_TH			WD_RST_TH		
R/W-1h	R/W-1h	R/W-7h			R/W-7h		

Table 8-217. WD_THR_CFG Register Field Descriptions

Bit	Field	Type	Reset	Description
7	WD_RST_EN	R/W	1h	Watchdog reset configuration bit: This bit can be only be written when the watchdog is in the Long Window. 0h = No warm reset when WD_FAIL_CNT[3:0] > (WD_FAIL_TH[2:0] + WD_RST_TH[2:0]) 1h = Warm reset when WD_FAIL_CNT[3:0] > (WD_FAIL_TH[2:0] + WD_RST_TH[2:0]).
6	WD_EN	R/W	1h	Watchdog enable configuration bit: This bit can be only be written when the watchdog is in the Long Window. (Default from NVM memory) 0h = watchdog disabled. MCU can set ENABLE_DRV bit to 1 if all other interrupt status bits are cleared 1h = watchdog enabled. MCU can set ENABLE_DRV bit to 1 if: - watchdog is out of the Long Window - WD_FAIL_CNT[3:0] =< WD_FAIL_TH[2:0] - WD_FIRST_OK=1 - all other interrupt status bits are cleared.
5-3	WD_FAIL_TH	R/W	7h	Configuration bits for the 1st threshold of the watchdog fail counter: Device clears ENABLE_DRV bit when WD_FAIL_CNT[3:0] > WD_FAIL_TH[2:0]. These bits can be only be written when the watchdog is in the Long Window.
2-0	WD_RST_TH	R/W	7h	Configuration bits for the 2nd threshold of the watchdog fail counter: Device goes through warm reset when WD_FAIL_CNT[3:0] > (WD_FAIL_TH[2:0] + WD_RST_TH[2:0]). These bits can be only be written when the watchdog is in the Long Window.

[Draft]

11.2.4.2.4 PMIC Temperature

11.2.4.2.4.1 Thermal Warning threshold

SM-39287 -

CONFIG_1 Register (Offset = 7Dh) [Reset = C0h]

Figure 8-178. CONFIG_1 Register

7	6	5	4	3	2	1	0
NSLEEP2_MASK	NSLEEP1_MASK	EN_ILIM_FSM_CTRL	I2C2_HS	I2C1_HS	RESERVED	TSD_ORD_LEVEL	TWARN_LEVEL
R/W-1h	R/W-1h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 8-143. CONFIG_1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	NSLEEP2_MASK	R/W	1h	Masking for NSLEEP2 pin(s) and NSLEEP2B bit: (Default from NVM memory) 0h = NSLEEP2(B) affects FSM state transitions. 1h = NSLEEP2(B) does not affect FSM state transitions.
6	NSLEEP1_MASK	R/W	1h	Masking for NSLEEP1 pin(s) and NSLEEP1B bit: (Default from NVM memory) 0h = NSLEEP1(B) affects FSM state transitions. 1h = NSLEEP1(B) does not affect FSM state transitions.
5	EN_ILIM_FSM_CTRL	R/W	0h	(Default from NVM memory) 0h = Buck/LDO regulators ILIM interrupts do not affect FSM triggers. 1h = Buck/LDO regulators ILIM interrupts affect FSM triggers.
4	I2C2_HS	R/W	0h	Select I2C2 speed (input filter) (Default from NVM memory) 0h = Standard, fast or fast+ by default, can be set to Hs-mode by Hs-mode controller code. 1h = Forced to Hs-mode
3	I2C1_HS	R/W	0h	Select I2C1 speed (input filter) (Default from NVM memory) 0h = Standard, fast or fast+ by default, can be set to Hs-mode by Hs-mode controller code. 1h = Forced to Hs-mode
2	RESERVED	R/W	0h	
1	TSD_ORD_LEVEL	R/W	0h	Thermal shutdown threshold level. (Default from NVM memory) 0h = 140C 1h = 145C
0	TWARN_LEVEL	R/W	0h	Thermal warning threshold level. (Default from NVM memory) 0h = 130C 1h = 140C

[Draft]

11.2.4.2.4.2 Orderly Shutdown threshold

SM-39275 -

CONFIG_1 Register (Offset = 7Dh) [Reset = C0h]

Figure 8-178. CONFIG_1 Register

7	6	5	4	3	2	1	0
NSLEEP2_MASK	NSLEEP1_MASK	EN_ILIM_FSM_CTRL	I2C2_HS	I2C1_HS	RESERVED	TSD_ORD_LEVEL	TWARN_LEVEL
R/W-1h	R/W-1h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 8-143. CONFIG_1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	NSLEEP2_MASK	R/W	1h	Masking for NSLEEP2 pin(s) and NSLEEP2B bit: (Default from NVM memory) 0h = NSLEEP2(B) affects FSM state transitions. 1h = NSLEEP2(B) does not affect FSM state transitions.
6	NSLEEP1_MASK	R/W	1h	Masking for NSLEEP1 pin(s) and NSLEEP1B bit: (Default from NVM memory) 0h = NSLEEP1(B) affects FSM state transitions. 1h = NSLEEP1(B) does not affect FSM state transitions.
5	EN_ILIM_FSM_CTRL	R/W	0h	(Default from NVM memory) 0h = Buck/LDO regulators ILIM interrupts do not affect FSM triggers. 1h = Buck/LDO regulators ILIM interrupts affect FSM triggers.
4	I2C2_HS	R/W	0h	Select I2C2 speed (input filter) (Default from NVM memory) 0h = Standard, fast or fast+ by default, can be set to Hs-mode by Hs-mode controller code. 1h = Forced to Hs-mode
3	I2C1_HS	R/W	0h	Select I2C1 speed (input filter) (Default from NVM memory) 0h = Standard, fast or fast+ by default, can be set to Hs-mode by Hs-mode controller code. 1h = Forced to Hs-mode
2	RESERVED	R/W	0h	
1	TSD_ORD_LEVEL	R/W	0h	Thermal shutdown threshold level. (Default from NVM memory) 0h = 140C 1h = 145C
0	TWARN_LEVEL	R/W	0h	Thermal warning threshold level. (Default from NVM memory) 0h = 130C 1h = 140C

[Draft]

11.2.4.2.4.3 Immediate Shutdown threshold

SM-39279 - STAT_SEVERE_ERR Register (Offset = 76h) [Reset = 00h]

Figure 8-171. STAT_SEVERE_ERR Register

7	6	5	4	3	2	1	0
RESERVED						VCCA_OVP_STAT	TSD_IMM_STAT
R-0h						R-0h	R-0h

Table 8-136. STAT_SEVERE_ERR Register Field Descriptions

Bit	Field	Type	Reset	Description
7-2	RESERVED	R	0h	
1	VCCA_OVP_STAT	R	0h	Status bit indicating that the VCCA voltage is above overvoltage protection level.
0	TSD_IMM_STAT	R	0h	Status bit indicating that the die junction temperature is above the thermal level causing an immediate shutdown.

7.12 Thermal Monitoring and Shutdown

Over operating free-air temperature range (unless otherwise noted).

POS	PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Electrical Characteristics							
8.1a	T_{WARN_0}	TWARN_INT thermal warning threshold (no hysteresis)	TWARN_LEVEL = 0	120	130	140	°C
8.1b	T_{WARN_1}		TWARN_LEVEL = 1	130	140	150	°C
8.2a	$T_{SD\_orderly\_0}$	TSD_ORD_INT thermal shutdown rising threshold	TSD_ORD_LEVEL = 0	130	140	150	°C
8.2b	$T_{SD\_orderly\_1}$		TSD_ORD_LEVEL = 1	135	145	155	°C
8.2c	$T_{SD\_orderly\_hys\_0}$	TSD_ORD_INT thermal shutdown hysteresis	TSD_ORD_LEVEL = 0		10		°C
8.2d	$T_{SD\_orderly\_hys\_1}$		TSD_ORD_LEVEL = 1		5		°C
8.3a	T_{SD_imm}	TSD_IMM_INT thermal shutdown rising threshold		140	150	160	°C

[Draft]

11.2.4.2.5 PMIC ESM

SM-39178 - MCU hardware error delay time threshold

ESM_MCU_DELAY1_REG Register (Offset = 90h) [Reset = 00h]

Figure 8-195. ESM_MCU_DELAY1_REG Register

7	6	5	4	3	2	1	0
ESM_MCU_DELAY1							
R/W-0h							

Table 8-160. ESM_MCU_DELAY1_REG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	ESM_MCU_DELAY1	R/W	0h	These bits configure the duration of the ESM_MCU delay-1 time-interval (see Error Signal Monitor chapter). These bits can be only be written when control bit ESM_MCU_START=0.

ESM_MCU_DELAY2_REG Register (Offset = 91h) [Reset = 00h]

Figure 8-196. ESM_MCU_DELAY2_REG Register

7	6	5	4	3	2	1	0
ESM_MCU_DELAY2							
R/W-0h							

Table 8-161. ESM_MCU_DELAY2_REG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	ESM_MCU_DELAY2	R/W	0h	These bits configure the duration of the ESM_MCU delay-2 time-interval (see Error Signal Monitor chapter). These bits can be only be written when control bit ESM_MCU_START=0.

SOC hardware error delay time threshold

ESM_SOC_DELAY1_REG Register (Offset = 99h) [Reset = 00h]

Figure 8-204. ESM_SOC_DELAY1_REG Register

7	6	5	4	3	2	1	0
ESM_SOC_DELAY1							
R/W-0h							

Table 8-169. ESM_SOC_DELAY1_REG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	ESM_SOC_DELAY1	R/W	0h	These bits configure the duration of the ESM_SoC delay-1 time-interval (see Error Signal Monitor chapter). These bits can be only be written when control bit ESM_SOC_START=0.

ESM_SOC_DELAY2_REG Register (Offset = 9Ah) [Reset = 00h]

Figure 8-205. ESM_SOC_DELAY2_REG Register

7	6	5	4	3	2	1	0
ESM_SOC_DELAY2							
R/W-0h							

Table 8-170. ESM_SOC_DELAY2_REG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	ESM_SOC_DELAY2	R/W	0h	These bits configure the duration of the ESM_SoC delay-2 time-interval (see Error Signal Monitor chapter). These bits can be only be written when control bit ESM_SOC_START=0.

[Draft]

11.2.4.2.6 Residual Voltage Check

SM-39181 - Threshold voltage for Short Circuit and Residual Voltage Detection

LDOs

POS	PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
1.20	$V_{TH\_SC\_RV(LDOn)}$	Threshold voltage for Short Circuit and Residual Voltage Detection	LDOn_EN = 0 and LDOn_RV_SEL = 1	140	150	160	mV

BUCKs

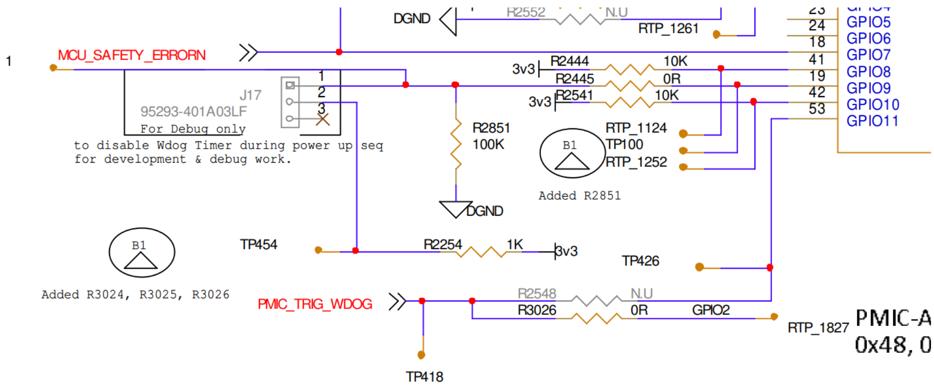
POS	PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
4.40	$V_{TH\_SC\_RV(Bn)}$	Threshold voltage for Short Circuit and Residual Voltage Detection	Bn_EN = 0 and BUCKn_RV_SEL = 1	140	150	160	mV

[Draft]

11.2.4.3 Debug Mode

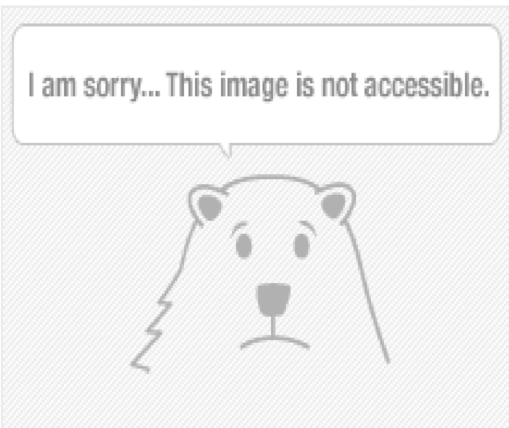
SM-39149 -

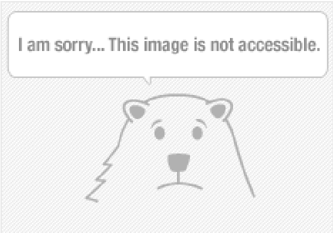
- Refer to Jumper JP:



[Draft]

11.2.5 Hardware Configuration (BOOTMOD) pins

SM-39157 - 



[Draft]

SM-39160 -

Table 4-7. BOOTMODE Pin Mapping

7	6	5	4	3	2	1	0
Backup Boot Mode Config	Primary Boot Mode Config			Backup Boot Mode			Primary Boot Mode B

[Draft]

SM-39152 -

4.3.2.3 Primary Boot Mode Configuration

When in normal boot flow (MCU Only = 0), it is possible to modify certain peripheral settings, such as chip-select, bus speed, and others, depending on the peripheral selected.

The mapping of configuration pins per boot mode is shown in Table 4-10. See the respective sections for details.

Table 4-10. Primary Boot Mode Configuration (MCU 6 = 0)

Primary Boot Mode Config Pins[(1)]			Primary Boot Mode B Pin	Primary Boot Mode A Pins			Primary Boot Mode
6	5	4	0	MCU 5	MCU 4	MCU 3	
Speed	RSVD	Csel	0	0	0	1	OSPI
0	Delay	Link stat	0	1	0	0	Ethernet RGMII
Bus reset	Reserved	Addr	0	1	1	0	I2C
		Port	0	1	1	1	UART
Port	Bus width	Voltage	1	0	0	1	eMMC
Reserved	Mode	Lane Swap	1	0	1	0	USB

(1) Shaded cells are reserved pins for that boot mode.

[Draft]

11.2.5.1 TDA4 Development Config.

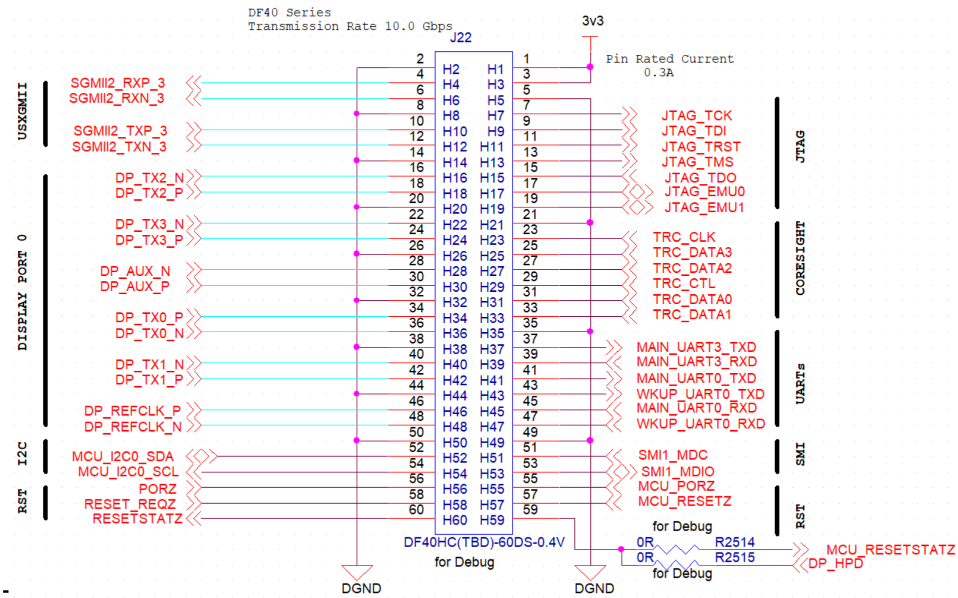
11.2.5.1.1 BOOT STRAP

SM-39155 -

Ball Num	Signal Name	Net Name	Function	Note	Bootstrap	Setting
G38	MCU_BOOTMODE00	USS.SPI_CLK	MCU_BOOTMODE00		1	PLL Reference Clock Selection: 25MHz
H36	MCU_BOOTMODE01	USS.SPI_MOSI	MCU_BOOTMODE01		1	
J38	MCU_BOOTMODE02	MCU_BOOTMODE02	MCU_BOOTMODE02		0	
H38	MCU_BOOTMODE03	MCU_BOOTMODE03	MCU_BOOTMODE03	Managed by MCU (NXP)	1	Boot Mode Selected: OSPI
J34	MCU_BOOTMODE04	MCU_BOOTMODE04	MCU_BOOTMODE04	Managed by MCU (NXP)	0	
J35	MCU_BOOTMODE05	MCU_BOOTMODE05	MCU_BOOTMODE05	Managed by MCU (NXP)	0	
H37	MCU_BOOTMODE06	MCU_BOOTMODE06	MCU_BOOTMODE06		0	Normal boot mode
K37	MCU_BOOTMODE07	MCU_BOOTMODE07	MCU_BOOTMODE07		0	OVRD
J37	MCU_BOOTMODE08	MCU_BOOTMODE08	MCU_BOOTMODE08		0	Table 4-5, p.203
K38	MCU_BOOTMODE09	MCU_BOOTMODE09	MCU_BOOTMODE09		0	
B33	BOOTMODE00	SOC_MCU_OSPI0_D0	BOOTMODE00	Managed by MCU (NXP)	1	OSPI
B32	BOOTMODE01	SOC_MCU_OSPI0_D1	BOOTMODE01		0	Not relevant
D33	BOOTMODE02	SOC_MCU_OSPI0_D4	BOOTMODE02		1	Not relevant
D34	BOOTMODE03	SOC_MCU_OSPI0_D5	BOOTMODE03		1	Not relevant
M37	BOOTMODE04	BOOTMODE04	BOOTMODE04	Managed by MCU (NXP)	0	Boot Flash is on CS0
M36	BOOTMODE05	BOOTMODE05	BOOTMODE05	Managed by MCU (NXP)	0	Not relevant
N34	BOOTMODE06	BOOTMODE06	BOOTMODE06		1	Not relevant
M34	BOOTMODE07	BOOTMODE07	BOOTMODE07		0	Not relevant

[Draft]

11.2.5.2 TDA4 debug I/F



SM-39169 - [Draft]

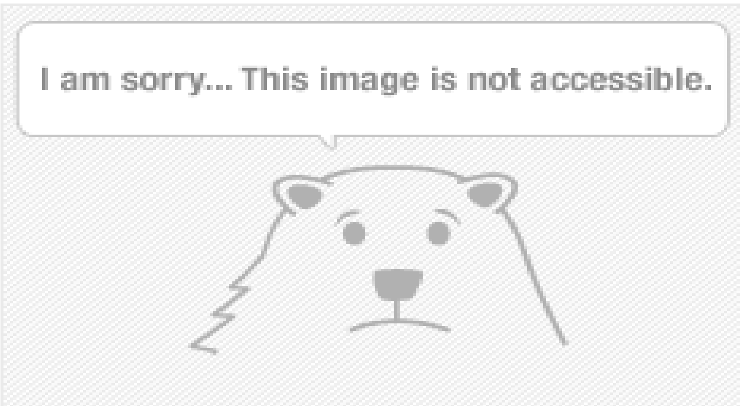
11.2.5.3 Power for DB

SM-39172 - Total: 0.3A*2pins*3.3V=1.98W [Draft]

11.2.6 TDA4 pin assignment and init state

11.2.6.1 RESETs Diagram

SM-39163 - The device incorporates four external reset pins (MCU_PORz, MCU_RESETz, PORz, and RESET_REQz) and two reset status pins (MCU_RESETSTATz and RESETSTATz).



[Draft]

11.2.6.2 RESETs Pin List

SM-39166 -

MCU (U43)/Expander pin		Dir.	Net Name	TDA4 (U137)pin	Description	Notes
U43.A8	INT GPIO	I	MCU_CTL.3v3_ENC_PMIC_RST_OUT		LEO PMIC - U136	PMIC Output

MCU (U43)/Expander pin		Dir.	Net Name	TDA4 (U137)pin	Description	Notes
U43.E9	INT GPIO	I	SAFETY_SIG.3v3_ENC_PMIC_SS		LEO PMIC - U136	PMIC Output
U103.10	I2C#4 Add. 0x74	O	ENC.3v3_RESET_REQZ	F34	Main Domain external Warm Reset request	TDA4 Input
U43.M23	GPIO	O	ENC.1v8_PORZ	P33	SoC PORz Reset Signal	TDA4 Input
U43.U18	GPIO	O	ENC.1v8_MCU_PORZ	K32	MCU Domain Cold Reset	TDA4 Input
U103.11	I2C#4 Add. 0x74	O	ENC.3v3_MCU_RESETZ	G36	MCU Domain Warm Reset	TDA4 Input
			MCU_RESETSTATZ	F36	TDA4 MCU domain Reset Status	To Valeo Debug & eMMC Reset
			RESETSTATZ	AL38	TDA4 Main domain Reset Status	To Valeo Debug & NOR Flash Reset

[Draft]

11.2.6.3 Power Supplies Enables

SM-39212 -

Net Name	Rail Name	Note
Core_Supply_EN	TDA4_0v8_VDD_CORE	Comes following PMIC power ON sequence
	TDA4_0v8_CPU_AVS	
	TDA4_0v8_DLL_LDO	

[Draft]

11.2.6.4 TDA4 Ethernet (RGMII/SGMII)

6.3.11.2 MCU Domain

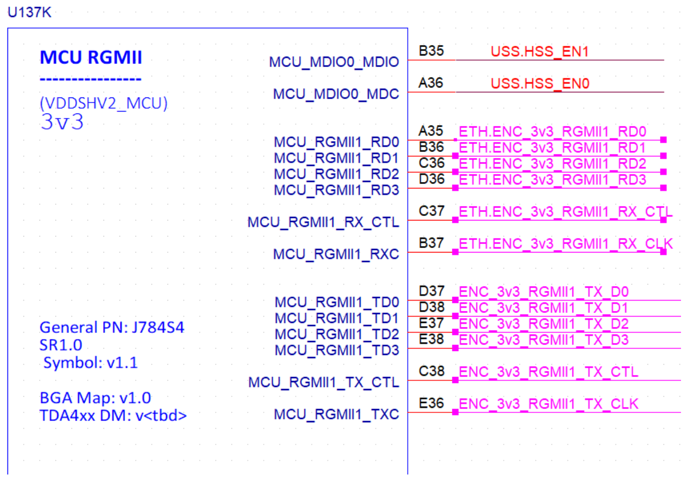
Table 6-68. MCU_CPSW2G0 Signal Descriptions

SIGNAL NAME [1]	PIN TYPE [2]	DESCRIPTION [3]	ALY PIN [4]
MCU_RGMII1_RXC	I	RGMII Receive Clock	B37
MCU_RGMII1_RX_CTL	I	RGMII Receive Control	C37
MCU_RGMII1_TXC	O	RGMII Transmit Clock	E36
MCU_RGMII1_TX_CTL	O	RGMII Transmit Control	C38
MCU_RGMII1_RD0	I	RGMII Receive Data 0	A35
MCU_RGMII1_RD1	I	RGMII Receive Data 1	B36
MCU_RGMII1_RD2	I	RGMII Receive Data 2	C36
MCU_RGMII1_RD3	I	RGMII Receive Data 3	D36
MCU_RGMII1_TD0	O	RGMII Transmit Data 0	D37
MCU_RGMII1_TD1	O	RGMII Transmit Data 1	D38
MCU_RGMII1_TD2	O	RGMII Transmit Data 2	E37
MCU_RGMII1_TD3	O	RGMII Transmit Data 3	E38
MCU_RMII1_CRS_DV	I	RMII Carrier Sense / Data Valid	C38
MCU_RMII1_REF_CLK	I	RMII Reference Clock	B37
MCU_RMII1_RX_ER	I	RMII Receive Data Error	C37
MCU_RMII1_TX_EN	O	RMII Transmit Enable	E36
MCU_RMII1_RXD0	I	RMII Receive Data 0	A35
MCU_RMII1_RXD1	I	RMII Receive Data 1	B36
MCU_RMII1_TXD0	O	RMII Transmit Data 0	D37
MCU_RMII1_TXD1	O	RMII Transmit Data 1	D38

SM-39182 - [Draft]

Figure 1: ETH Net names

SM-39184 -



[Draft]

11.2.6.5 TDA4 SerDes

11.2.6.5.1 SerDes Connectivity

6.3.12 SGMII

6.3.12.1 MAIN Domain

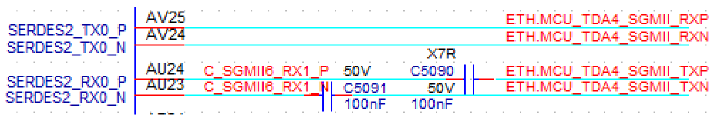
Table 6-69. CPSW9X0 Signal Descriptions

SIGNAL NAME [1]	PIN TYPE [2]	DESCRIPTION [3]	ALY PIN [4]
SGMII5_RXN0	I	SGMII Receive (negative)	AR14, AU23
SGMII5_RXP0	I	SGMIi Receive (positive)	AR15, AU24
SGMII5_TXN0	O	SGMII Transmit (negative)	AP13, AV24
SGMII5_TXP0	O	SGMII Transmit (positive)	AP14, AV25

SM-39193 -

[Draft]

SM-39196 -



[Draft]

6.3.12 SGMII

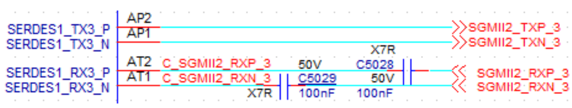
6.3.12.1 MAIN Domain

Table 6-69. CPSW9X0 Signal Descriptions

SIGNAL NAME [1]	PIN TYPE [2]	DESCRIPTION [3]	ALY PIN [4]
SGMII2_RXN0	I	SGMII Receive (negative)	AT1, AT19
SGMII2_RXP0	I	SGMIi Receive (positive)	AT2, AT20
SGMII2_TXN0	O	SGMII Transmit (negative)	AP1, AP19
SGMII2_TXP0	O	SGMII Transmit (positive)	AP2, AP20

SM-39187 - [Draft]

SM-39190 -



[Draft]

6.3.17 Display Port

6.3.17.1 MAIN Domain

Table 6-84. DP0 Signal Descriptions

SM-39205 -

SIGNAL NAME [1]	PIN TYPE [2]	DESCRIPTION [3]	ALY PIN [4]
DP0_AUXN	IO	Display Port Differential Auxiliary Data (negative)	AP22
DP0_AUXP	IO	Display Port Differential Auxiliary Data (positive)	AP23
DP0_HPD	I	Display Port Hot Plug Detection	AC34, AG33, AM37
DP0_TXN0	O	Display Port Differential Transmit (negative)	AP13
DP0_TXN1	O	Display Port Differential Transmit (negative)	AT13
DP0_TXN2	O	Display Port Differential Transmit (negative)	AT16
DP0_TXN3	O	Display Port Differential Transmit (negative)	AV18
DP0_TXP0	O	Display Port Differential Transmit (positive)	AP14
DP0_TXP1	O	Display Port Differential Transmit (positive)	AT14
DP0_TXP2	O	Display Port Differential Transmit (positive)	AT17
DP0_TXP3	O	Display Port Differential Transmit (positive)	AV19

[Draft]

Table 6-92. SERDES4 Signal Descriptions

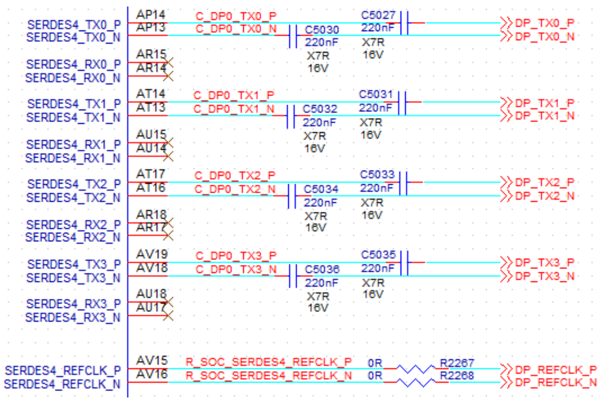
SM-39208 -

SIGNAL NAME [1]	PIN TYPE [2]	DESCRIPTION [3]	ALY PIN [4]
SERDES4_REFCLK_N	IO	Serdes Reference Clock Input/Output (negative)	AV16
SERDES4_REFCLK_P	IO	Serdes Reference Clock Input/Output (positive)	AV15

[Draft]

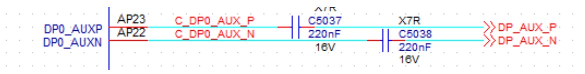
Figure 2: Main domain Net names

SM-39199 -



[Draft]

SM-39202 -



[Draft]

11.3 GPIO init state

11.3.1 TDA4 GPIOs init state

SM-39468 -

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MMC1_DAT3	AC38	VDDSHV5 - 1v8	1.8 V	ENC.1v8_MCU_TIMESYNC0	TIMER_IO2	DMTIMER	TBD	TBD	TBD	U43

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MMC1_DAT2	AA32	VDDSHV5 - 1v8	1.8 V	NC	GPIO0_61	GPIO_0_VDDSHV5	output	low	n/a	
MMC1_CLK	AB38	VDDSHV5 - 1v8	1.8 V	ENC.1v8_FSYNC	TIMER_IO6	DMTIMER	output	low	sync to DES	to U141 - max 96724F DES
MMC1_CMD	AB36	VDDSHV5 - 1v8	1.8 V	NC	GPIO0_65	GPIO_0_VDDSHV5	output	low	n/a	
MMC1_DAT0	AA33	VDDSHV5 - 1v8	1.8 V	ENC.1v8_I2C4_SCL	I2C4_SCL	I2C_4	I2C			U19 - HSS for parcking cameras PoC
MMC1_DAT1	AB34	VDDSHV5 - 1v8	1.8 V	ENC.1v8_I2C4_SDA	I2C4_SDA	I2C_4				

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_RGMII1_RD0	A35	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RD0	MCU_RGMII1_RD0	MCU_CPSW2G_0	RGMII input			from U43
MCU_RGMII1_RD1	B36	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RD1	MCU_RGMII1_RD1	MCU_CPSW2G_0				
MCU_RGMII1_RD2	C36	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RD2	MCU_RGMII1_RD2	MCU_CPSW2G_0				
MCU_RGMII1_RD3	D36	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RD3	MCU_RGMII1_RD3	MCU_CPSW2G_0				
MCU_RGMII1_RXC	B37	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RX_CLK	MCU_RGMII1_RXC	MCU_CPSW2G_0				
MCU_RGMII1_RX_CTL	C37	VDDSHV2_MCU - 3v3	3.3 V	ETH.ENC_3v3_RGMII1_RX_CTL	MCU_RGMII1_RX_CTL	MCU_CPSW2G_0				

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_RGMII1_TD0	D37	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_D0	MCU_RGMII1_TD0	MCU_CPSW2G_0	RGMII output			to U43
MCU_RGMII1_TD1	D38	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_D1	MCU_RGMII1_TD1	MCU_CPSW2G_0				
MCU_RGMII1_TD2	E37	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_D2	MCU_RGMII1_TD2	MCU_CPSW2G_0				
MCU_RGMII1_TD3	E38	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_D3	MCU_RGMII1_TD3	MCU_CPSW2G_0				
MCU_RGMII1_TXC	E36	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_CLK	MCU_RGMII1_TXC	MCU_CPSW2G_0				
MCU_RGMII1_TX_CTL	C38	VDDSHV2_MCU - 3v3	3.3 V	ENC_3v3_RGMII1_TX_CTL	MCU_RGMII1_TX_CTL	MCU_CPSW2G_0				
MCU_MDIO0_MDIO	B35	VDDSHV2_MCU - 3v3	3.3 V	USS.HSS_EN1	WKUP_GPIO0_52	WKUP_GPIO_0_VDDSHV2_MCU	output	low	0 - out 1 HSS disabled1 - out 1 HSS enabled	to U74

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_MDIO0_MDC	A36	VDDSHV2_MCU - 3v3	3.3 V	USS.HSS_EN0	WKUP_GPIO0_53	WKUP_GPIO_0_VDDSHV2_MCU	output	low	0 - out 0 HSS disabled 1 - out 0 HSS enabled	to U74
MCASP2_AXR0	AC36	VDDSHV2 - 1v8	1.8 V	SMI1_MDIO	MDIO1_MDIO	CPSW9X_0	MDC/IO			J22
MCASP2_AFSX	AE37	VDDSHV2 - 1v8	1.8 V	SMI1_MDC	MDIO1_MDC	CPSW9X_0				
MCAN12_TX	AG36	VDDSHV2 - 1v8	1.8 V	TRC_CLK	TRC_CLK	TRACE_0	TRACE			to J22
MCAN12_RX	AJ33	VDDSHV2 - 1v8	1.8 V	TRC_CTL	TRC_CTL	TRACE_0				
MCAN13_TX	AF33	VDDSHV2 - 1v8	1.8 V	TRC_DATA0	TRC_DATA0	TRACE_0				
MCAN13_RX	AH33	VDDSHV2 - 1v8	1.8 V	TRC_DATA1	TRC_DATA1	TRACE_0				
MCAN14_RX	AK36	VDDSHV2 - 1v8	1.8 V	TRC_DATA3	TRC_DATA3	TRACE_0				
GPIO0_12	AK37	VDDSHV2 - 1v8	1.8 V	TRC_DATA2	TRC_DATA2	TRACE_0				
MCAN14_TX	AG33	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_5	GPIO_0_VDDSHV2	output	low	n/a	

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCAN15_TX	AG34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_7	GPIO_0_VDDSHV2	output	low	n/a	
MCAN15_RX	AJ35	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_8	GPIO_0_VDDSHV2	output	low	n/a	
MCAN16_TX	AH34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_9	GPIO_0_VDDSHV2	output	low	n/a	
MCAN16_RX	AE33	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_10	GPIO_0_VDDSHV2	output	low	n/a	
GPIO0_11	AL32	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_11	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_ACLKX	AK35	VDDSHV2 - 1v8	1.8 V	MCASP0_ACLKX	GPIO0_14	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AFSX	AK38	VDDSHV2 - 1v8	1.8 V	MCASP0_AFSX	GPIO0_15	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR0	AF37	VDDSHV2 - 1v8	1.8 V	BIT_Data	GPIO0_16	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR1	AG37	VDDSHV2 - 1v8	1.8 V	Audio_In	GPIO0_17	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR14	AF34	VDDSHV2 - 1v8	1.8 V	MCASP0_ACLKR	GPIO0_42	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR15	AE34	VDDSHV2 - 1v8	1.8 V	MCASP0_AFSR	GPIO0_43	GPIO_0_VDDSHV2	output	low	n/a	

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCASP0_AXR2	AK33	VDDSHV2 - 1v8	1.8 V	ENC.1v8_SYS_CHECK_2	GPIO0_18	GPIO_0_VDDSHV2	input	n/a	0 - no error 1 - error	from U141 - max 96724F DES
MCASP0_AXR3	AJ38	VDDSHV2 - 1v8	1.8 V	ENC.1v8_SYS_CHECK_3	GPIO0_31	GPIO_0_VDDSHV2	input	n/a	0 - no error 1 - error	from U141 - max 96724F DES
MCASP0_AXR4	AK34	VDDSHV2 - 1v8	1.8 V	ENC.1v8_INTb	GPIO0_32	GPIO_0_VDDSHV2	input	n/a	0 - no error 1 - error	from U141 - max 96724F DES
MCASP0_AXR5	AG38	VDDSHV2 - 1v8	1.8 V	ENC.1v8_HSS_INT	GPIO0_33	GPIO_0_VDDSHV2	input	n/a	0 - error 1 - no error	from HSS for parcking cameras PoC
MCASP0_AXR6	AF36	VDDSHV2 - 1v8	1.8 V	ENC.1v8_PDB	GPIO0_34	GPIO_0_VDDSHV2	output	low	0 - DES is reset 1 - DES out of reset	to U141 - max 96724F DES
MCASP0_AXR7	AE35	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_35	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR8	AC35	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_36	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR9	AG35	VDDSHV2 - 1v8	1.8 V	ENC.1v8_LOCK	GPIO0_37	GPIO_0_VDDSHV2	input	n/a	0 - no lock 1 - lock	from U141 - max 96724F DES
MCASP1_AXR1	AC32	VDDSHV2 - 1v8	1.8 V	I2B_I2C2_SCL	GPIO0_19	GPIO_0_VDDSHV2	output	low	n/a	
MCASP1_AXR2	AC37	VDDSHV2 - 1v8	1.8 V	I2B_I2C2_SDA	GPIO0_20	GPIO_0_VDDSHV2	output	low	n/a	

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCASP2_ACLKX	AD37	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_21	GPIO_0_VDDSHV2	output	low	n/a	
MCASP2_AXR1	AE36	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_24	GPIO_0_VDDSHV2	output	low	n/a	
MCAN0_TX	AF38	VDDSHV2 - 1v8	1.8 V	ENC.1v8_SYS_CHECK_0	GPIO0_25	GPIO_0_VDDSHV2	input	n/a	0 - no error 1 - error	from U141 - max 96724F DES
MCAN0_RX	AE38	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_26	GPIO_0_VDDSHV2	output	low	n/a	
MCAN1_RX	AH38	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_28	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR10	AH36	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_38	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR11	AF35	VDDSHV2 - 1v8	1.8 V	ENC.1v8_SYS_CHECK_1	GPIO0_39	GPIO_0_VDDSHV2	input	n/a	0 - no error 1 - error	from U141 - max 96724F DES
MCASP0_AXR12	AD34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_40	GPIO_0_VDDSHV2	output	low	n/a	
MCASP0_AXR13	AJ36	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_41	GPIO_0_VDDSHV2	output	low	n/a	
MCASP1_AXR3	AL33	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_44	GPIO_0_VDDSHV2	output	low	n/a	

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCASP1_AXR4	AL34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_45	GPIO_0_VDDSHV2	output	low	n/a	
MCASP1_ACLKX	AC34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_46	GPIO_0_VDDSHV2	output	low	n/a	
ECAP0_IN_APWM_OUT	AD36	VDDSHV2 - 1v8	1.8 V	ENC.1v8_I2C1_SCL	I2C1_SCL	I2C_1	I2C			to U141 - max 96724F DES
EXT_REFCLK1	AJ32	VDDSHV2 - 1v8	1.8 V	ENC.1v8_I2C1_SDA	I2C1_SDA	I2C_1				
MCAN1_TX	AJ37	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_27	GPIO_0_VDDSHV2	output	low	n/a	
PMIC_WAKE0n	AJ34	VDDSHV2 - 1v8	1.8 V	NC	GPIO0_13	GPIO_0_VDDSHV2	output	high	n/a	n/a
MCASP1_AFSX	AD33	VDDSHV2 - 1v8	1.8 V	MAIN_UART0_RXD	UART0_RXD	USART_0	UART			to J22
MCASP1_AXR0	AD38	VDDSHV2 - 1v8	1.8 V	MAIN_UART0_TXD	UART0_TXD	USART_0				

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCAN2_TX	AC33	VDDSHV2 - 1v8	1.8 V	MAIN_UART3_RXD	UART3_RXD	USART_3	UART			to J22
MCAN2_RX	AH37	VDDSHV2 - 1v8	1.8 V	MAIN_UART3_TXD	UART3_TXD	USART_3				
MCU_OSPI1_CSn1	G33	VDDSHV1_MCU - 1v8	1.8 V	ENC.1v8_MCU_TIMESYNC1	MCU_TIMER_IO0	MCU_DMTIMER	TBD	TBD	TBD	U43

MCU_OSPI0_CLK	E32	VDDSHV1_MCU - 1v8	1.8 V	SOC_MCU_OSPI0_CLK_R	MCU_OSPI0_CLK	MCU_OSPI_0	OSPI	U132 - OSPI memory
MCU_OSPI0_CSn0	A32	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_CS0	MCU_OSPI0_CSn0	MCU_OSPI_0		
MCU_OSPI0_D0	B33	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D0	MCU_OSPI0_D0	MCU_OSPI_0		
MCU_OSPI0_D1	B32	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D1	MCU_OSPI0_D1	MCU_OSPI_0		
MCU_OSPI0_D2	C33	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D2	MCU_OSPI0_D2	MCU_OSPI_0		
MCU_OSPI0_D3	C35	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D3	MCU_OSPI0_D3	MCU_OSPI_0		
MCU_OSPI0_D4	D33	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D4	MCU_OSPI0_D4	MCU_OSPI_0		
MCU_OSPI0_D5	D34	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D5	MCU_OSPI0_D5	MCU_OSPI_0		
MCU_OSPI0_D6	E34	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D6	MCU_OSPI0_D6	MCU_OSPI_0		
MCU_OSPI0_D7	E33	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_D7	MCU_OSPI0_D7	MCU_OSPI_0		
MCU_OSPI0_DQS	C34	VDDSHV1_MCU - 1v8	1.8 V	OSPI0_DQS	MCU_OSPI0_DQS	MCU_OSPI_0		

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_OSPI0_CSn3	C32	VDDSHV1_MCU - 1v8	1.8 V	NC	MCU_OSPI0_ECC_FAIL	MCU_OSPI_0	output	low	n/a	
MCU_OSPI0_LBCLKO	D32	VDDSHV1_MCU - 1v8	1.8 V	NC	MCU_OSPI0_LBCLKO	MCU_OSPI_0	output	low	n/a	
MCU_OSPI0_CSn2	B34	VDDSHV1_MCU - 1v8	1.8 V	NC	MCU_OSPI0_RESET_OUT0	MCU_OSPI_0	output	low	n/a	
MCU_OSPI0_CSn1	A33	VDDSHV1_MCU - 1v8	1.8 V	NC	WKUP_GPIO0_28	WKUP_GPIO_0_VDDSHV1_MCU	output	low	n/a	
MCU_OSPI1_CLK	F32	VDDSHV1_MCU - 1v8	1.8 V	1v8_USS_HSS_DEN	WKUP_GPIO0_31	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - current measurment disabled 1 - current measurment enabled	to U166
MCU_OSPI1_LBCLKO	C31	VDDSHV1_MCU - 1v8	1.8 V	PMIC_INT	WKUP_GPIO0_32	WKUP_GPIO_0_VDDSHV1_MCU	input	n/a	0 - error 1 - no error	from U136 - PMIC
MCU_OSPI1_DQS	F31	VDDSHV1_MCU - 1v8	1.8 V	NC	WKUP_GPIO0_33	WKUP_GPIO_0_VDDSHV1_MCU	output	low	n/a	
MCU_OSPI1_D0	E35	VDDSHV1_MCU - 1v8	1.8 V	USS_FRONT_EN_2	WKUP_GPIO0_34	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - transistor close 1 - transistor open	Q30-1
MCU_OSPI1_D1	D31	VDDSHV1_MCU - 1v8	1.8 V	USS_FRONT_EN_3	WKUP_GPIO0_35	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - transistor close 1 - transistor open	Q33-1

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_OSPI1_D2	G31	VDDSHV1_MCU - 1v8	1.8 V	1v8_USS_HSS_DSEL	WKUP_GPIO0_36	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - sense output 0 1 - sense output 1	to U166
MCU_OSPI1_D3	F33	VDDSHV1_MCU - 1v8	1.8 V	USS_FRONT_EN_1	WKUP_GPIO0_37	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - transistor close 1 - transistor open	Q27-1
MCU_OSPI1_CSn0	G32	VDDSHV1_MCU - 1v8	1.8 V	USS_FRONT_EN_0	WKUP_GPIO0_38	WKUP_GPIO_0_VDDSHV1_MCU	output	low	0 - transistor close 1 - transistor open	Q24-1
MCU_I2C0_SCL	M35	VDDSHV0_MCU - 3v3	3.3 V	MCU_I2C0_SCL	MCU_I2C0_SCL	MCU_I2C_0	I2C			to U128 - EEPROM
MCU_I2C0_SDA	G34	VDDSHV0_MCU - 3v3	3.3 V	MCU_I2C0_SDA	MCU_I2C0_SDA	MCU_I2C_0				
MCU_MCAN0_TX	K33	VDDSHV0_MCU - 3v3	3.3 V	USS.INT	WKUP_GPIO0_60	WKUP_GPIO_0_VDDSHV0_MCU	input	PU	0 - interrupt 1 - no interrupt	from U75 - USS ASIC
MCU_MCAN0_RX	F38	VDDSHV0_MCU - 3v3	3.3 V	USS.RST	WKUP_GPIO0_61	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - U75 in reset 1 - U75 out of reset	to U75 - USS ASIC
MCU_SPI0_CS0	F37	VDDSHV0_MCU - 3v3	3.3 V	USS.SPI_CS	MCU_SPI0_CS0	MCU_SPI_0	output	low	1 - Chip select	to U75 - USS ASIC

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_SPI0_CLK	G38	VDDSHV0_MCU - 3v3	3.3 V	USS.SPI_CLK	MCU_SPI0_CLK	MCU_SPI_0	SPI			to U75 - USS ASIC
MCU_SPI0_D0	H36	VDDSHV0_MCU - 3v3	3.3 V	USS.SPI_MOSI	MCU_SPI0_D0	MCU_SPI_0				
MCU_SPI0_D1	J38	VDDSHV0_MCU - 3v3	3.3 V	USS.SPI_MISO	MCU_SPI0_D1	MCU_SPI_0				
WKUP_GPIO0_11	M38	VDDSHV0_MCU - 3v3	3.3 V	3v3_EEPROM_WP	WKUP_GPIO0_11	WKUP_GPIO_0_VDDSHV0_MCU	output	high	0 - WP disabled 1 - WP enabled	to U128 - EEPROM
WKUP_GPIO0_10	L33	VDDSHV0_MCU - 3v3	3.3 V	NC	WKUP_GPIO0_10	WKUP_GPIO_0_VDDSHV0_MCU	output	low	n/a	
WKUP_GPIO0_0	H38	VDDSHV0_MCU - 3v3	3.3 V	MCU_BOOTMODE03	WKUP_GPIO0_0	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	from U43
WKUP_GPIO0_1	J34	VDDSHV0_MCU - 3v3	3.3 V	MCU_BOOTMODE04	WKUP_GPIO0_1	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	from U43
WKUP_GPIO0_2	J35	VDDSHV0_MCU - 3v3	3.3 V	MCU_BOOTMODE05	WKUP_GPIO0_2	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	from U43

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
WKUP_GPIO0_3	J36	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_0	WKUP_GPIO0_3	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q24-2
WKUP_GPIO0_4	H35	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_1	WKUP_GPIO0_4	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close1 - transistor open	Q27-2
WKUP_GPIO0_5	K36	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_2	WKUP_GPIO0_5	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q30-2
WKUP_GPIO0_6	L37	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_3	WKUP_GPIO0_6	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q33-2
WKUP_GPIO0_7	L36	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_4	WKUP_GPIO0_7	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q36-2
WKUP_GPIO0_8	L35	VDDSHV0_MCU - 3v3	3.3 V	USS.FRONT_EN_4	WKUP_GPIO0_8	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q36-1
WKUP_GPIO0_9	L34	VDDSHV0_MCU - 3v3	3.3 V	USS.FRONT_EN_5	WKUP_GPIO0_9	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q39-1

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
WKUP_GPIO0_12	J37	VDDSHV0_MCU - 3v3	3.3 V	ENC.3v3_MCU_UART0_TXD	MCU_UART0_TXD	MCU_USART_0	UART			to J19
WKUP_GPIO0_13	K38	VDDSHV0_MCU - 3v3	3.3 V	ENC.3v3_MCU_UART0_RXD	MCU_UART0_RXD	MCU_USART_0				
WKUP_GPIO0_14	H37	VDDSHV0_MCU - 3v3	3.3 V	MCU_BOOTMODE06	WKUP_GPIO0_14	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	
WKUP_GPIO0_15	K37	VDDSHV0_MCU - 3v3	3.3 V	MCU_BOOTMODE07	WKUP_GPIO0_15	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	
WKUP_GPIO0_49	M33	VDDSHV0_MCU - 3v3	3.3 V	PMIC_TRIG_WDOG	WKUP_GPIO0_49	WKUP_GPIO_0_VDDSHV0_MCU	output	low	n/a	to U136 - PMIC
WKUP_GPIO0_56	M37	VDDSHV0_MCU - 3v3	3.3 V	BOOTMODE04	WKUP_GPIO0_56	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	from U43
WKUP_GPIO0_57	M36	VDDSHV0_MCU - 3v3	3.3 V	BOOTMODE05	WKUP_GPIO0_57	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	from U43
WKUP_GPIO0_66	N34	VDDSHV0_MCU - 3v3	3.3 V	BOOTMODE06	WKUP_GPIO0_66	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
WKUP_GPIO0_67	M34	VDDSHV0_MCU - 3v3	3.3 V	BOOTMODE07	WKUP_GPIO0_67	WKUP_GPIO_0_VDDSHV0_MCU	input	n/a	n/a	
WKUP_I2C0_SCL	N33	VDDSHV0_MCU - 3v3	3.3 V	WKUP_I2C0_SCL	WKUP_I2C0_SCL	WKUP_I2C_0	I2C			U137 - PMIC
WKUP_I2C0_SDA	N35	VDDSHV0_MCU - 3v3	3.3 V	WKUP_I2C0_SDA	WKUP_I2C0_SDA	WKUP_I2C_0				
MCU_RESETSTATz	F36	VDDSHV0_MCU - 3v3	3.3 V	MCU_RESETSTATZ	MCU_RESETSTATz	WKUP_SYSTEM_0_MCU	input	n/a	0 - TDA4 in reset 1 - TDA4 active	from U43 and J22
MCU_RESETz	G36	VDDSHV0_MCU - 3v3	3.3 V	MCU_RESETZ	MCU_RESETz	WKUP_SYSTEM_0_MCU	input	n/a	0 - TDA4 in reset 1 - TDA4 active	from U43 and J22
PMIC_POWER_EN1	L38	VDDSHV0_MCU - 3v3	3.3 V	USS.REAR_EN_5	WKUP_GPIO0_88	WKUP_GPIO_0_VDDSHV0_MCU	output	low	0 - transistor close 1 - transistor open	Q39-2
RESET_REQz	F34	VDDSHV0_MCU - 3v3	3.3 V	RESET_REQZ	RESET_REQz	WKUP_SYSTEM_0_MCU	input	n/a	0 - TDA4 in reset1 - TDA4 active	from U43 and J22
WKUP_UART0_RXD	K35	VDDSHV0_MCU - 3v3	3.3 V	WKUP_UART0_RXD	WKUP_UART0_RXD	WKUP_USART_0	UART			J22
WKUP_UART0_TXD	K34	VDDSHV0_MCU - 3v3	3.3 V	WKUP_UART0_TXD	WKUP_UART0_TXD	WKUP_USART_0				

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
EMU0	F35	VDDSHV0_MCU - 3v3	3.3 V	JTAG_EMU0	EMU0	WKUP_JTAG	JTAG bus			to J22
EMU1	H34	VDDSHV0_MCU - 3v3	3.3 V	JTAG_EMU1	EMU1	WKUP_JTAG				
TRSTn	G37	VDDSHV0_MCU - 3v3	3.3 V	JTAG_TRST	TRSTn	WKUP_JTAG				
TCK	G35	VDDSHV0_MCU - 3v3	3.3 V	JTAG_TCK	TCK	WKUP_JTAG				
TDI	AL37	VDDSHV0 - 3v3	3.3 V	JTAG_TDI	TDI	JTAG_VDDSHV0				
TDO	AL35	VDDSHV0 - 3v3	3.3 V	JTAG_TDO	TDO	JTAG_VDDSHV0				
TMS	AL36	VDDSHV0 - 3v3	3.3 V	JTAG_TMS	TMS	JTAG_VDDSHV0				
SPI0_CS0	AM37	VDDSHV0 - 3v3	3.3 V	DP_HPD	DP0_HPD	DP_0	per valeo use for DP debug			to J22
EXTINTn	AN35	VDDSHV0 - 3v3	3.3 V	pull up resistor	GPIO0_0	GPIO_0_VDDSHV0	output	high	n/a	
SPI0_CS1	AP38	VDDSHV0 - 3v3	3.3 V	ENC.1v8_HSS_EN	GPIO0_52	GPIO_0_VDDSHV0	output	low	0 - HSS output disabled 1 - HSS output enabled	U19 - HSS for parcking cameras PoC

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
SPI0_CLK	AN38	VDDSHV0 - 3v3	3.3 V	ENC.3v3_DES_CLK_EN	GPIO0_53	GPIO_0_VDDSHV0	output	low	0 - clock active 1 - clock disabled	U88 - clock generator for DES (U141) clock
I2C0_SCL	AN36	VDDSHV0 - 3v3	3.3 V	pull up resistor	GPIO0_56	GPIO_0_VDDSHV0	output	low		
I2C0_SDA	AP37	VDDSHV0 - 3v3	3.3 V	pull up resistor	GPIO0_57	GPIO_0_VDDSHV0	output	low		
TIMER_IO0	AR38	VDDSHV0 - 3v3	3.3 V	ENC.3v3_SER_PDB	GPIO0_58	GPIO_0_VDDSHV0	output	low		
TIMER_IO1	AN37	VDDSHV0 - 3v3	3.3 V	pull down resistor	OBSCLK0	SYSTEM_0_VDDSHV0	output	low	n/a	n/a
RESETSTATz	AL38	VDDSHV0 - 3v3	3.3 V	RESETSTATZ	RESETSTATz	SYSTEM_0_VDDSHV0	output	high	0 - eMMC in reset 1 - eMMC out of reset	U129 - eMMC J22
SOC_SAFETY_ERRORn	AM34	VDDSHV0 - 3v3	3.3 V	SOC_SAFETY_ERRORN	SOC_SAFETY_ERRORn	SYSTEM_0_VDDSHV0	output	high	0 - error 1 - no error	to U136 - PMIC
SPI0_D0	AM35	VDDSHV0 - 3v3	3.3 V	ENC.3v3_UART2_RXD	UART2_RXD	USART_2	UART			J19
SPI0_D1	AM36	VDDSHV0 - 3v3	3.3 V	ENC.3v3_UART2_TXD	UART2_TXD	USART_2				

MMC0_CLK	AK5	VDDS_MMC0 - 1v8	1.8 V	eMMC0_CLK	MMC0_CLK	MMC_0	eMMC	U129 - eMMC
MMC0_CMD	AL8	VDDS_MMC0 - 1v8	1.8 V	eMMC0_CMD	MMC0_CMD	MMC_0		
MMC0_DAT0	AK9	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D0	MMC0_DAT0	MMC_0		
MMC0_DAT1	AL6	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D1	MMC0_DAT1	MMC_0		
MMC0_DAT2	AK8	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D2	MMC0_DAT2	MMC_0		
MMC0_DAT3	AK6	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D3	MMC0_DAT3	MMC_0		
MMC0_DAT4	AK7	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D4	MMC0_DAT4	MMC_0		
MMC0_DAT5	AL7	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D5	MMC0_DAT5	MMC_0		
MMC0_DAT6	AL5	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D6	MMC0_DAT6	MMC_0		
MMC0_DAT7	AK3	VDDS_MMC0 - 1v8	1.8 V	eMMC0_D7	MMC0_DAT7	MMC_0		
MMC0_DS	AK4	VDDS_MMC0 - 1v8	1.8 V	eMMC0_DS	MMC0_DS	MMC_0		

Design Signal Pad Name	Ball Name	IO Power Supply Name	IO Power Setting	Net Name	Mode Signal Name	Mode Interface Name	init state	Init Value	Active Level description / Comments	connected to
MCU_PORz	K32	VDDA_WKUP-TDA4_1v8_PLL_LDO	1.8 V	MCU_PORZ	MCU_PORz	WKUP_SYSTEM_0_VDDA_WKUP	input	n/a	0 - TDA4 in reset 1 - TDA4 active	from U43 and J22
MCU_SAFETY_ERRORn	N36	VDDA_WKUP-TDA4_1v8_PLL_LDO	1.8 V	MCU_SAFETY_ERRORN	MCU_SAFETY_ERRORn	WKUP_SYSTEM_0_VDDA_WKUP	output	high	0 - error 1 - no error	to U136 - PMIC
PORz	P33	VDDA_WKUP-TDA4_1v8_PLL_LDO	1.8 V	PORZ	PORz	WKUP_SYSTEM_0_VDDA_WKUP	input	n/a	0 - TDA4 in reset 1 - TDA4 active	from U43 and J22

[Draft]

SM-39469 - The external circuit controlling the mode pins must be disabled once MCU_PORz rises. [Draft]

11.3.2 SerDes

SM-39466 -

Mode Interface Name	bus name on SCH	direction	connected to
DSI0	ENC.VDI2_PARKING_MIPI_OUT	out	U184 - max 96793 SER
DSI1	ENC.VDI1_LOGGING_MIPI_OUT	out	J23
CSI1	ENC.VDI2_MIPI	in	U145 - max 96724F DES
CSI2	EQs_MIPI.EQ0_MIPI_OUT	in	A-U1 - main EQ
SGMII2	SGMII2	bi-dir	J22
SGMII5	ETH.MCU_TDA4_SGMII	bi-dir	U43
DP0	DP	out	J22
UFS	not used		
USB0	not used		

[Draft]

11.3.3 Timers

11.3.3.1 MCU Domain

SM-39467 -

Timer unit	Net Name	Timer channel	Timer mode	Timer usage	Comments
MCU_Timer0					
MCU_Timer1					
MCU_Timer2					
MCU_Timer3					
MCU_Timer4					
MCU_Timer5					
MCU_Timer6					
MCU_Timer7					
MCU_Timer8					
MCU_Timer9					
GTC					

[Draft]

11.3.3.2 MAIN Domain

SM-39472 -

Timer unit	Net Name	Timer channel	Timer mode	Timer usage	Comments
Timer_IO0					
Timer_IO1					
Timer_IO2	ENC.1v8_MCU_TIMESYNC0				
Timer_IO3					
Timer_IO4					
Timer_IO5					
Timer_IO6	ENC.1v8_FSYNC				
Timer_IO7					

[Draft]

12 Revision history

Ver.	Date	Updated by	Section	Essence of change	Approved by	
0.1	11-11-2025	Rohith Krishnan		Updated the VSUP1 and PWRON1 to 1.35V so that the PMIC will be operational at 27V SM-31879 the limit updated from 0.88V to 0.937V(17V to 18.1V)		
0.2	24-03-2026	Chaithra Shetty	7.1	Main power down -> Added step 1-Assert Low MCU_CL.I2B_5V_EN		
0.3	10-08-2026	Chaithra Shetty	4.26 4.27	ENC.1v8_MCU_TIMESYNC1 made NA ENC.1v8_MCU_TIMESYNC1 - GPIO init defined as HIGH-Z ,NA ,PD as not in use.	Frank Bergen Moshe Shalev	